For Burnside High School Use in 2010 Only

Calculus for Scholarship Students

- **This is a write on workbook for the Scholarship Mathematics with Calculus course.**

- **Designed to support and extend the Level Three Mathematics with Calculus course.**

- **Contains previous scholarship questions with worked solutions plus additional exercises designed to extend scholarship students**

- **For use in extension classes for scholarship students or as a text/workbook for self guided scholarship students.**

Alan Coulston

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Alan Coulston

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Preface

This book is a result of my own frustrations over several years about the paucity of material for the Scholarship Mathematics with Calculus course.

With the advent of the examination in 2004 the approach of most teachers (me included) was to go through previous examination papers, which initially meant Scholarship and NZEST papers. The problem with this approach, particularly once the first NCEA scholarship examinations became available, was that the average scholarship student did not know how to even get the questions started. The reason for this is that the level at which the scholarship course is pitched is considerably beyond that of the Level Three Mathematics with Calculus course.

To achieve success at scholarship level students must be used to working beyond Level 3. In some centres this is achieved by students attending university courses which stretch them beyond Level 3. The material they cover does not necessarily have to be that of the NCEA Level 3 syllabus but simply needs to extend the student in similar areas.

This book contains notes, examples and exercises at the level appropriate for the Level 3 course but contains additional work which is not in the Level 3 syllabus and allows the student to extend themselves beyond the requirements of Level 3. As is appropriate for such a book it also contains exercises from previous years' papers (with worked and annotated solutions) included with the appropriate topics.

This book is a workbook and is not intended to be a textbook (although it does straddle the gap between workbook and textbook a little). As such it is intended as a supplement to the Level 3 textbook, not a replacement for it. Success, for most schools at this level, comes from additional classes (often before school) to the normal Level 3 class. The primary use of this book is as a workbook for such classes. It could, however, be used as a text/work book for individual students who do not have the option of additional classes at their school.

I wish you the best of luck with scholarship in the upcoming year.

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Contents

Chapter 1 Algebra

1.1	Logarithms and Exponents	1
1.2	Surd Equations	7
1.3	Sum and Product of the Roots of a Quadratic Equation	11
1.4	Rational Expressions	16
1.5	Partial Fractions	20
1.6	Infinite Series	27
1.7	General Complex Numbers	34
1.8	The Geometry of Complex Numbers	45

Chapter 2 Trigonometry

2.1	Trigonometric Graphs and Modelling	63
2.2	Identities	68
2.3	Equations	75
2.4	Triangles	86

Chapter 3 Differentiation

3.1	Common derivatives	95
3.2	Limits	96
3.3	First principles	100
3.4	The chain, product and quotient rules	103
3.5	Implicit and parametric differentiation	107

3.6 Inverse trigonometric functions 117

3.7 Properties of curves 121

3.8 Related rates of change 125

3.9 Optimisation problems 131

Chapter 4 Integration

4.1 Substitutions 141

4.2 Integration by Parts 150

4.3 Further trigonometric integration methods 155

4.4 Solids of revolution: Disc and Shell methods 166

4.5 Arc length and Surface of revolution 181

4.6 Differential equations 188

Chapter 5 Conic Sections

5.1 The Cartesian and Parametric Forms of Conic Sections 219

5.2 Focus, Eccentricity and Directrix 221

5.3 Tangents and Normals 226

5.4 An alternative approach to tangents 227

5.5 Translation of Conic Sections 229

Answers and worked solutions 255

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1. Algebra

This chapter covers a mix of basic algebraic skills and complex numbers. Some of the skills covered here extend beyond the Level 3 Mathematics with Calculus syllabus. This is in order to cover knowledge which is presumed to be held by scholarship students, or simply to provide good manipulation practice.

1.1 Logarithms and Exponents

Properties of Indices

1. $a^x a^y = a^{x+y}$
2. $\frac{a^x}{a^y} = a^{x-y}$
3. $(a^x)^n = a^{nx}$
4. $\left(\frac{a}{b}\right)^x = \frac{a^x}{b^x}$
5. $(ab)^x = a^x b^x$

Properties of logarithms

1. $\log_a b = c \Leftrightarrow b = a^c$
2. $\log_a x - \log_a y = \log_a \left(\frac{x}{y}\right)$
3. $\log_a x + \log_a y = \log_a xy$
4. $\log_a x^n = n\log_a x$

Proof of the logarithm rules

Let $p = \log_a x$ and $q = \log_a y \Leftrightarrow a^p = x$ and $a^q = y$

1.

$xy = a^p a^q$

$xy = a^{p+q}$

$\log_a xy = p + q$

$\log_a xy = \log_a x + \log_a y$

2.

$\frac{x}{y} = \frac{a^p}{a^q}$

$\frac{x}{y} = a^{p-q}$

$\log_a \left(\frac{x}{y}\right) = p - q$

$\log_a \left(\frac{x}{y}\right) = \log_a x - \log_a y$

3.

$x^n = (a^p)^n$

$x^n = a^{np}$

$\log_a x^n = np$

$\log_a x^n = n\log_a x$

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Examples

1. Simplify:

$$\left(\frac{3^{2n-1}}{(3^n)^3}\right)\div\left(\frac{27^{n+1}}{3^{4n+1}}\right)$$

$$=\frac{3^{2n-1}}{3^{3n}}\times\frac{3^{4n+1}}{3^{3n+3}}$$

$$=\frac{3^{6n}}{3^{6n+3}}$$

$$=\frac{1}{27}$$

2. Solve:

$4^x-2^{x+1}-8=0$

$(2^x)^2-2.2^x-8=0$

$(2^x-4)(2^x+2)=0$

$2^x=4$ or $2^x=-2$

So $x=2$ is the only solution.

3. Solve the simultaneous equations:

$y=\log_3 9x$ and $y=\log_9 x$

$9x=3^y$ and $x=9^y$

$x=\frac{3^y}{9}=3^{y-2}$ and $x=3^{2y}$

So:

$3^{y-2}=3^{2y}$

$2y=y-2$

$y=-2$

$x=3^{-4}=\frac{1}{81}$

4. Solve the simultaneous equations:

$\log_3(x+2y+1)=0$ (1)

$\log_3(1-x)-2\log_3 y=1$ (2)

Rearranging (1):

$x+2y+1=3^0=1$

$x=-2y$

Sub in (2):

$\log_3(1+2y)-\log_3 y^2=1$

$\log_3\left(\frac{1+2y}{y^2}\right)=1$

$\frac{1+2y}{y^2}=3^1=3$

$3y^2-2y-1=0$

$(3y+1)(y-1)=0$

$y=-\frac{1}{3},1$

$\left(\frac{2}{3},-\frac{1}{3}\right)$ and $(-2,1)$ are solutions?

But y cannot be negative in (2).

Hence $(-2,1)$ is the only solution.

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Ex 1.1 Logarithms and Exponents

1. Solve the equation $\log_4(\log_2 x) = 1$

2. If x is a real number find the minimum value of $3^{x^2 - 4x}$

3. Solve the equation $3^x - 3^{x-1} = 162$

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4. Solve the equation $\sqrt{3^x} - \frac{18}{\sqrt{3^x}} = 3$

5. If $a^2 + b^2 = 14ab$, prove $\log\left(\frac{a+b}{4}\right) = \frac{1}{2}(\log a + \log b)$

6. Solve $\log_x x + \log_x\left(4x^2 + x - 4\right) = 4$

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7. Solve the system of equations $y = \log_4 16x$ and $y = \log_{16} x$

8. Determine **all** values of k for which the equation $k3^x + 3^{-x} = 4$ has a single root.

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1.2 Surd Equations

Surd equations have a radical (root) sign.

Although not necessary, it helps to look at the graphs of these equations to understand what is happening with the solutions.

Examples

1. $x - 2 = \sqrt{x+4}$

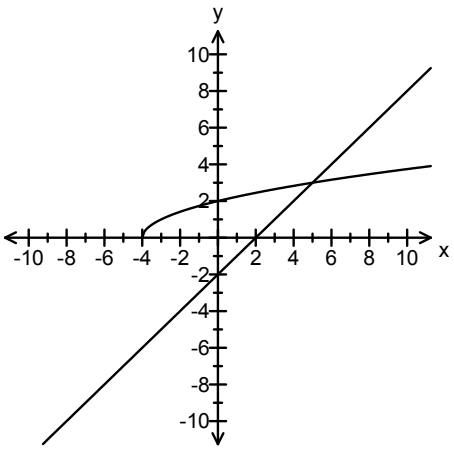
Square both sides.

$x^2 - 4x + 4 = x + 4$

$x^2 - 5x = 0$

$x(x-5) = 0$

$x = 0, 5$



Solutions of surd equations must be tested to ensure they work.

$x = 0$

$-2 \neq \sqrt{0+4}$

$x = 5$

$5 - 2 = \sqrt{5+4}$

$3 = 3$

So, x = 5 is the only solution.
This is confirmed by the graph where there is only one intersection.

2. $\sqrt{x-3} + \sqrt{x} = 3$

Separate the radical signs.
If the radical signs are not separated manipulation towards the solution becomes more difficult.

$\sqrt{x-3} = 3 - \sqrt{x}$

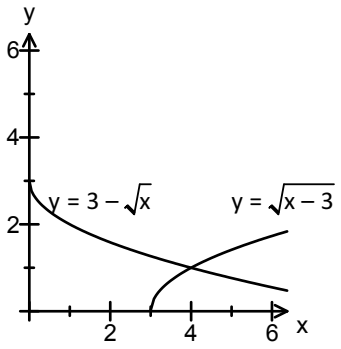
$x - 3 = 9 - 6\sqrt{x} + x$

$-12 = -6\sqrt{x}$

$2 = \sqrt{x}$

$x = 4$

In this case there is only the one solution.



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3. $\sqrt{x-2}-\sqrt{x}=2$

Separate the radical signs.

$\sqrt{x-2}=\sqrt{x}+2$

$x-2=x+4\sqrt{x}+4$

$-6=4\sqrt{x}$

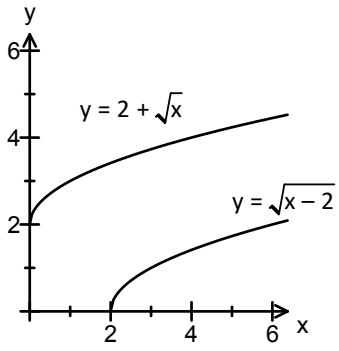
$36=16x$

$x=\frac{9}{4}$

Testing:

$\sqrt{\frac{9}{4}-2}-\sqrt{\frac{9}{4}}=-1$

$-1\neq 2$



There is no solution. The graphs do not intersect.

Ex 1.2 Surd Equations

1. $x-1=\sqrt{5x-9}$

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2. $x-3=\sqrt{30-2x}$

3. $\sqrt{2x+6}-\sqrt{x+4}=1$

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4. $\sqrt{2x+3} - \sqrt{x+2} = 2$

5. $\sqrt{2x+9} - \sqrt{x+1} = \sqrt{x+4}$

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1.3 Sum and Product of the Roots of a Quadratic

Consider the general quadratic equation $ax^2 + bx + c = 0$ which has two roots α and β.

Rewriting as:

$$x^2 + \frac{b}{a}x + \frac{c}{a} = 0 \qquad (1)$$

If α and β are the roots then:

$$(x - \alpha)(x - \beta) = 0$$

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0 \qquad (2)$$

Comparing coefficients in (1) and (2):

$$\alpha + \beta = -\frac{b}{a} \text{ and } \alpha\beta = \frac{c}{a}$$

Examples

(1) Find the sum and product of the roots of $3x^2 + 2x + 5 = 0$

$$\alpha + \beta = -\frac{2}{3}, \ \alpha\beta = \frac{5}{3}$$

Note: although $b^2 - 4ac = -56$, the sum and product do exist because there are roots, but they are not real.

(2) $3x^2 + (p + 1)x + 24 = 0$ has one root equal to twice the other find the value(s) of p.

Let the roots be $\alpha = r$ and $\beta = 2r$:

$$\alpha + \beta = 3r = -\frac{p+1}{3} \qquad (1) \quad \text{and} \quad \alpha\beta = 2r^2 = \frac{24}{3} \qquad (2)$$

Solving (2):

$r = \pm 2$

Substituting in (1) :

p = -19, 17 respectively

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(3) Form the quadratic with roots α and β where $\alpha + \beta = \frac{3}{4}$, $\alpha\beta = \frac{1}{2}$

$x^2 - \frac{3}{4}x + \frac{1}{2} = 0$

$4x^2 - 3x + 2 = 0$

(4) The roots of the equation $2x^2 - 7x + 4 = 0$ are α and β.

Write down the equation whose roots are $\frac{1}{\alpha}$ and $\frac{1}{\beta}$.

$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta}$

Now, $\alpha + \beta = \frac{7}{2}$ and $\alpha\beta = 2$

Hence, $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{7}{4}$ and $\frac{1}{\alpha\beta} = \frac{1}{2}$

So, the equation is:

$x^2 - \frac{7}{4}x + \frac{1}{2} = 0$

$4x^2 - 7x + 2 = 0$

(5) If α and β are the roots of $x^2 + 3x - 2 = 0$, find the equation whose roots are α^3 and β^3

$\alpha + \beta = -3$ and $\alpha\beta = -2$

Now,

$$(\alpha+\beta)^3 = \alpha^3 + 3\alpha^2\beta + 3\alpha\beta^2 + \beta^3$$
$$= \alpha^3 + \beta^3 + 3\alpha\beta(\alpha+\beta)$$

Rearranging,

$$\alpha^3 + \beta^3 = (\alpha+\beta)^3 - 3\alpha\beta(\alpha+\beta)$$
$$= (-3)^3 - 3.-2.-3$$
$$= -45$$
$$\alpha^3\beta^3 = (-2)^3 = -8$$

So the equation is $x^2 + 45x - 8 = 0$

For an example of the Sum and Product formulae in a scholarship question see 2004 Question Six in the Integration Chapter Ex 4.6.5

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Ex 1.3 Sum and Product of the Roots of a Quadratic

1. Form the quadratic equation with roots α and β for the following sums and products:

(a) $\alpha + \beta = 5, \alpha\beta = 3$

(b) $\alpha + \beta = -\frac{4}{5}, \alpha\beta = \frac{1}{10}$

(c) $\alpha + \beta = 5a, \alpha\beta = 3a^2$

2. Find the roots of each quadratic equation subject to the given conditions :

(a) $(3p - 1)x^2 + (2p + 5)x + p = 0$ where the roots are reciprocals of each other.
Verify the answer.

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(b) $2px^2 - px + p + 1 = 0$ where the sum of the roots are equal to twice the product of the roots.

3. The roots of $3x^2 - 11x + 6 = 0$ are α and β. Find the equation with roots α^2 and β^2

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4. The roots of $x^2 - 5x - 9 = 0$ are α and β. Find the equation with roots $\frac{1}{\alpha}$ and $\frac{1}{\beta}$

5. Form the equation with roots $2+\sqrt{3}$ and $2-\sqrt{3}$

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1.4 Rational Expressions

The ability to quickly and accurately manipulate rational functions is an essential skill for this course.

Examples

1. $\dfrac{3}{x+1}+\dfrac{4}{x^2-1}$

$=\dfrac{3}{x+1}+\dfrac{4}{(x-1)(x+1)}$

$=\dfrac{3(x-1)}{(x+1)(x-1)}+\dfrac{4}{(x-1)(x+1)}$

$=\dfrac{3x+1}{(x+1)(x-1)}$

2. $\dfrac{x+6}{x^2-4}-\dfrac{4}{x+2}-\dfrac{2}{x-2}$

$=\dfrac{x+6}{(x+2)(x-2)}-\dfrac{4}{x+2}-\dfrac{2}{x-2}$

$=\dfrac{x+6}{(x+2)(x-2)}-\dfrac{4(x-2)}{(x+2)(x-2)}-\dfrac{2(x+2)}{(x+2)(x-2)}$

$=\dfrac{x+6-4x+8-2x-4}{(x+2)(x-2)}$

$=\dfrac{-5x+10}{(x+2)(x-2)}$

$=\dfrac{-5(x-2)}{(x+2)(x-2)}$

$=\dfrac{-5}{(x+2)},\ x\neq 2$

3. $\dfrac{-5}{x+4}+\dfrac{5}{x^2+8x+16}-\dfrac{1}{x^2-16}$

$=\dfrac{-5}{x+4}+\dfrac{5}{(x+4)^2}-\dfrac{1}{(x+4)(x-4)}$

$=\dfrac{-5(x+4)(x-4)+5(x-4)-(x+4)}{(x+4)^2(x-4)}$

$=\dfrac{-5x^2+80+5x-20-x-4}{(x+4)^2(x-4)}$

$=\dfrac{-5x^2+4x+56}{(x+4)^2(x-4)}$

4. $\dfrac{5}{4(x+1)}-\dfrac{13}{8(x-1)}+\dfrac{11}{8(x+3)}$

$=\dfrac{10(x-1)(x+3)-13(x+1)(x+3)+11\left(x^2-1\right)}{8(x+1)(x-1)(x+3)}$

$=\dfrac{10\left(x^2+2x-3\right)-13\left(x^2+4x+3\right)+11x^2-11}{8(x+1)(x-1)(x+3)}$

$=\dfrac{10x^2+20x-30-13x^2-52x-39+11x^2-11}{8(x+1)(x-1)(x+3)}$

$=\dfrac{8x^2-32x-80}{8(x+1)(x-1)(x+3)}$

$=\dfrac{x^2-4x-10}{(x+1)(x-1)(x+3)}$

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Ex 1.4 Rational Expressions

Simplify these expressions.

1. $\dfrac{5}{2x+3}+\dfrac{7}{x-1}$

2. $\dfrac{2x-2}{x^2+4x+3}-\dfrac{x-1}{x^2+5x+6}$

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3. $\dfrac{2}{x^2+5x+6} - \dfrac{4}{x^2+4x+3} + \dfrac{3}{x^2+3x+2}$

4. $\dfrac{2x-8}{3x^2+8x+4} + \dfrac{x+3}{3x^2+5x+2}$

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5. $\dfrac{-5}{x^2+3x-4}+\dfrac{5}{x^2+2x-3}+\dfrac{1}{x^2+7x+12}$

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1.5 Partial Fractions

Partial fractions involve the decomposition of a rational function $\frac{P(x)}{Q(x)}$ into the sums (or differences) of simpler rational fractions, where the degree of polynomial P(x) is less than the degree of polynomial Q(x).

Example:

$$\frac{8x-42}{(x+6)(x-3)} = \frac{10}{x+6} - \frac{2}{x-3}$$

The terms of the partial fractions (LHS in the example above) vary according to the type of term(s) on the denominator as shown in the table:

Type of term on the denominator	Example	Term(s) in the partial fraction
Linear	$ax + b$	$\frac{A}{ax+b}$
Linear to a power	$(ax + b)^n$	$\frac{A_1}{ax+b} + \frac{A_2}{(ax+b)^2} + ... + \frac{A_n}{(ax+b)^n}$
Quadratic which cannot be factorised	$ax^2 + bx + c$	$\frac{Ax+B}{ax^2+bx+c}$
Quadratic to a power	$(ax^2 + bx + c)^n$	$\frac{A_1x+B_1}{ax^2+bx+c} + \frac{A_2x+B_2}{(ax^2+bx+c)^2} + ... + \frac{A_nx+B_n}{(ax^2+bx+c)^n}$

Linear term on the denominator

To express $\frac{8x-42}{x^2+3x-18}$ as partial fractions, first factorise the denominator:

$$\frac{8x-42}{x^2+3x-18} = \frac{8x-42}{(x+6)(x-3)}$$

Now express as the sum of individual fractions with the linear factors on the denominator:

$$\frac{8x-42}{(x+6)(x-3)} = \frac{A}{x+6} + \frac{B}{x-3}$$

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Express with a common denominator:

$$\frac{8x-42}{(x+6)(x-3)} = \frac{A(x-3)+B(x+6)}{(x+6)(x-3)}$$

There are two methods to proceed from here:

Method One: Comparing coefficients on the numerators

$$\frac{8x-42}{(x+6)(x-3)} = \frac{A(x-3)+B(x+6)}{(x+6)(x-3)}$$

Comparing the coefficients of x:

$A + B = 8$

Comparing the constant terms:

$-3A + 6B = -42$

Solving the simultaneous equations:

$A = 10, B = -2,$ hence:

$$\frac{8x-42}{(x+6)(x-3)} = \frac{10}{x+6} - \frac{2}{x-3}$$

Method Two: Substitution

$$\frac{8x-42}{(x+6)(x-3)} = \frac{A(x-3)+B(x+6)}{(x+6)(x-3)}$$

Any value can be substituted for x but values which make the brackets equal zero are good choices. We need only concern ourselves with the numerators.

x = 3	**x = -6**
$24 - 42 = 9B$	$-48 - 42 = -9A$
$B = -2$	$A = 10$

Hence:

$$\frac{8x-42}{(x+6)(x-3)} = \frac{10}{x+6} - \frac{2}{x-3}$$

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Linear term to a power on the denominator

To express $\frac{8x-42}{(x-3)^2}$ as partial fractions, express as individual fractions with the factors on the denominator being descending powers of the repeated factor:

$$\frac{8x-42}{(x-3)^2}=\frac{A}{(x-3)^2}+\frac{B}{x-3}$$

Express with a common denominator:

$$\frac{8x-42}{(x-3)^2}=\frac{A+B(x-3)}{(x-3)^2}$$

Using substitution:

x = 3	**x = 0**
24 – 42 = A	-42 = A – 3B
A = -18	B = 8

Hence:

$$\frac{8x-42}{(x-3)^2}=\frac{8}{x-3}-\frac{18}{(x-3)^2}$$

Quadratic term or factor on the denominator

If the quadratic can be factorised then factorise and proceed as before. Otherwise, to express $\frac{x^2+x}{(x-4)(x^2+4)}$ as partial fractions, express as individual fractions with the factors on the denominator being the factors of the given rational expression:

$$\frac{x^2+x}{(x-4)(x^2+4)}=\frac{A}{x-4}+\frac{Bx+C}{x^2+4}$$

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Express with a common denominator and group the terms:

$$\frac{x^2+x}{(x-4)(x^2+4)} = \frac{A(x^2+4)+Bx(x-4)+C(x-4)}{(x-4)(x^2+4)}$$

$$\frac{x^2+x}{(x-4)(x^2+4)} = \frac{(A+B)x^2-(4B-C)x+4(A-C)}{(x-4)(x^2+4)}$$

Using equating coefficients:

$A + B = 1$

$-4B + C = 1$

$4(A - C) = 0$

Solving gives $A = 1$, $B = 0$, $C = 1$

Hence:

$$\frac{x^2+x}{(x-4)(x^2+4)} = \frac{1}{x-4} + \frac{1}{x^2+4}$$

Ex 1.5 Partial Fractions

Split the following rational functions into partial fractions:

1. $\dfrac{2x+3}{(x+2)(x+1)}$

_____

_____

_____

_____

_____

_____

_____

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2. $\dfrac{5x+3}{(x+3)(x-1)}$

3. $\dfrac{12x}{x^2+x-2}$

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4. $\dfrac{5}{2x^2 + x}$

5. $\dfrac{9 - x}{(x - 1)(x - 3)^2}$

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6. $\dfrac{3x^2 + 3x}{(x-2)(x^2+2)}$

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1.6 Infinite Series

There are two important series with which students should already be familiar viz: the Geometric and the Binomial series.

1.61 The Geometric Series

The sum to infinity of the geometric series $1 - x + x^2 - x^3 + x^4 + \ldots$ where, a = 1 and r = -x, is given by $\frac{1}{1+x}$ but, as is the case with infinite series there are conditions on this.

In this case: $-1 < x < +1$.

Example

$$1 - \tfrac{1}{2} + \tfrac{1}{4} - \tfrac{1}{8} + \ldots\ldots = \frac{1}{1 - \frac{-1}{2}}$$

$$= \frac{2}{3}$$

1.62 The Binomial Series

For the particular case where the first term in the bracket is one:

$$(1+x)^n = 1 + \binom{n}{1}x + \binom{n}{2}x^2 + \binom{n}{3}x^3 + \ldots\ldots + \binom{n}{r}x^r + \ldots\ldots + x^n$$

$$= 1 + nx + n(n-1)\frac{x^2}{2!} + n(n-1)(n-2)\frac{x^3}{3!} + \ldots\ldots + n(n-1)(n-2)\ldots(n-r+1)\frac{x^r}{r!} + \ldots\ldots + x^n$$

For positive values of n the polynomial expression is closed ie: has a finite number of terms.

Example

$$(1+x)^4 = 1 + 4x + 6x^2 + 4x^3 + x^4$$

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Consider the Geometric and Binomial series together:

$$\frac{1}{1+x} = 1 - x + x^2 - x^3 + x^4 + \ldots\ldots \qquad \text{[Geometrics Series]}$$

$$(1+x)^{-1} = 1 + {}^{-}1.x + {}^{-}1.{}^{-}2.\frac{x^2}{2!} + {}^{-}1.{}^{-}2.{}^{-}3.\frac{x^3}{3!} + {}^{-}1.{}^{-}2.{}^{-}3.{}^{-}4.\frac{x^4}{4!} + \ldots\ldots \qquad \text{[Binomial Series]}$$

$$= 1 - x + x^2 - x^3 + x^4 + \ldots\ldots$$

While not a proof, this indicates that the binomial expansion holds for negative values of n.

But:

(i) the series is infinite and

(ii) its convergence is subject to conditions ie: $-1 < x < 1$.

Example

Expand $\frac{1}{(1-2x)}$ as a series of ascending powers of x giving:

(a) the first four terms

(b) the range of value of x for which the expansion is valid

Now using the binomial expansion with n = -1

(a) $(1 - 2x)^{-1} = 1 + 2x + 4x^2 + 8x^3 + 16x^4 + \ldots.$

(b) This is true for $-1 < 2x < 1$ ie: $-\frac{1}{2} < x < \frac{1}{2}$

1.63 The Maclaurin Series

In the previous examples expressions of the type $(1+x)^n$ can be represented by polynomial expansions, subject to conditions.

It remains to be seen whether functions such as sine, cosine, log etc can also be represented by polynomial expansions.

The Maclaurin series is used to represent these functions as polynomial expansions:

$$f(x) = f(0) + f'(0).x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!} + \ldots\ldots.$$

The validity of the series is subject to strict the conditions that:

(i) f(x) is infinitely differentiable

(ii) the value of the function and all its derivatives exist at x = 0.

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Examples

1. The Mclaurin series of sinx:

$f(x) = \sin x$ $\quad f(0) = 0$

$f'(x) = \cos x$ $\quad f'(0) = 1$

$f''(x) = -\sin x$ $\quad f''(0) = 0$

$f'''(x) = -\cos x$ $\quad f'''(0) = -1$

$f^{(4)}(x) = \sin x$ $\quad f^{(4)}(0) = 0$ $\quad$ etc...

$$\sin(x) = 0 + 1x + 0 - \frac{x^3}{3!} + 0 + \frac{x^5}{5!} + 0 - \frac{x^7}{7!} + \ldots.$$

$$= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \ldots\ldots\ldots,$$ where x is in radians.

2. Now let us look at $\log_e x$:

$f(x) = \log_e x$ $\quad f(0) = \log_e 0$ which is undefined.

$f'(x) = \frac{1}{x}$ $\quad f'(0) = \frac{1}{0}$ which is undefined.

$f''(x) = -\frac{1}{x^2}$ $\quad f''(0) = -\frac{1}{0^2}$ which is undefined.

So, there is no polynomial expansion for $\log_e x$, instead look at $\log_e(1+x)$:

$f(x) = \log_e(1+x\ \ )$ $\quad f(0) = \log_e 1 = 0$

$f'(x) = \frac{1}{1+x}$ $\quad f'(0) = \frac{1}{(1+0)} = 1$

$f''(x) = -\frac{1}{(1+x)^2}$ $\quad f''(0) = -\frac{1}{(1+0)^2} = -1$

$f'''(x) = \frac{2}{(1+x)^3}$ $\quad f'''(0) = \frac{2}{(1+0)^3} = 2$

$f^{(4)}(x) = -\frac{3\times 2}{(1+x)^4}$ $\quad f^{(4)}(0) = -\frac{3\times 2}{(1+0)^4} = -3\times 2$ $\quad$ etc...

$$\log_e(1+x) = 0 + 1.x + {}^{-}1.\frac{x^2}{2!} + 2.\frac{x^3}{3!} + {}^{-}3\times 2.\frac{x^4}{4!} + 4\times 3\times 2.\frac{x^5}{5!} + \ldots..$$

$$= x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} - \ldots..$$

In this expansion we should note that:

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(i) there are no factorial expressions on the denominator, which actually contributes to the condition that $x \leq 1$ in order for the series to converge.

(ii) $x > -1$

Hence for the expansion to be valid $-1 < x \leq 1$

1.64 Combining Partial Fractions and Series

Consider the series $\sum_{r=2}^{n} \frac{1}{(r-1)r} = \frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \ldots\ldots + \frac{1}{(n-1)n}$.

In this form it is not easy to evaluate the sum.

But if $\frac{1}{(r-1)r}$ is expressed as partial fractions:

$$\frac{1}{(r-1)r} = \frac{A}{r-1} + \frac{B}{r}$$

$$= \frac{(A+B)r - B}{(r-1)r}$$

Either by substitution or equating coefficients it can be shown that A = 1 and B = -1

Hence:

$$\sum_{r=2}^{n} \frac{1}{(r-1)r} = \sum_{r=2}^{n} \frac{1}{r-1} - \frac{1}{r}$$

$$= \sum_{r=2}^{n} \frac{1}{r-1} - \sum_{r=2}^{n} \frac{1}{r}$$

List enough terms of each to see what is happening:

$$= (1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \ldots\ldots \frac{1}{n-1}) - (\frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \ldots\ldots \frac{1}{n-1} + \frac{1}{n})$$

$$= 1 - \frac{1}{n}$$

$$= \frac{n-1}{n}$$

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Ex 1.6 Infinite Series

1. Use the binomial series to expand the following as far as x^4, stating the conditions for which it is valid:

(a) $\dfrac{1}{4x+1}$

(b) $\dfrac{1}{x^2+3}$

2. Use the Maclaurin series to expand the following as far as x^4:

(a) $f(x) = \cos x$

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(b) $f(x) = e^x$

3. Use partial fractions to find the sum of each of the following series:

(a) $\sum_{r=1}^{n} \frac{1}{r(r+2)}$

(b) $\sum_{r=3}^{n} \frac{1}{(r+1)(r-2)}$

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4. The function f(n) is defined for $n \in \{1, 2, 3, ...\}$ by: :

$$f(n)=\sum_{k=1}^{k=n}\frac{2k-1}{k(k+1)(k+2)}=\frac{1}{1\times2\times3}+\frac{3}{2\times3\times4}+\frac{5}{3\times4\times5}+.....+\frac{2n-1}{n(n+1)(n+2)}$$

Show that f(n) may also be written in the form:

$$f(n)=\sum_{k=1}^{k=n}\left(-\frac{1}{2k}+\frac{3}{k+1}-\frac{5}{2(k+2)}\right)$$

and hence find f(n) in terms of n and show that $\lim_{n\to\infty} f(n)=\frac{3}{4}$

(Scholarship 2005 Question Five (b))

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1.7 General Complex Numbers

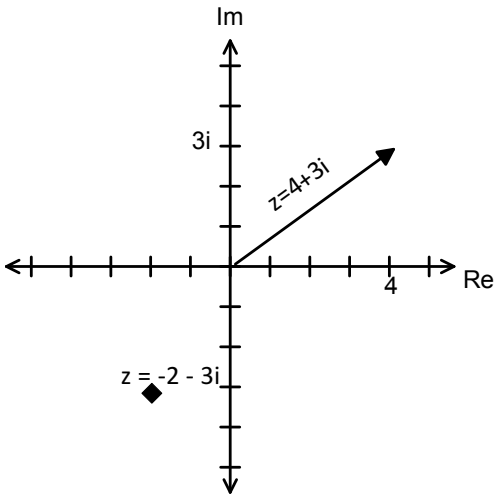
1.71 Basic Skills Reference

Rectangular Form	**z = x + iy**
Addition/Subtraction	Add/Subtract the real and complex parts **Example:** 2 + 3i + 3 - 4i = 5 - i
Multiplication	Expand the brackets, remembering that $i^2 = -1$ **Example:** (2 + i)(3 – 4i) = 6 – 8i + 3i + 4 = 10 – 5i
Complex Conjugate	The complex conjugate of z = a + ib is $\overline{z} = a - ib$ NB: $z\overline{z} = a^2 + b^2$
Division	Multiply top and bottom by the complex conjugate of the denominator. **Example:** $\frac{2+i}{1+2i}$ $= \frac{2+i}{1+2i} \times \frac{1-2i}{1-2i}$ $= \frac{4-3i}{5} = \frac{4}{5} - \frac{3}{5}i$

1.72 The Argand Diagram

The complex number z = x + iy can be represented on the Argand plane where the x axis is the ***Real*** axis and the y axis the ***Imaginary*** axis.

The complex number is represented as a point on the Argand plane or as a vector.



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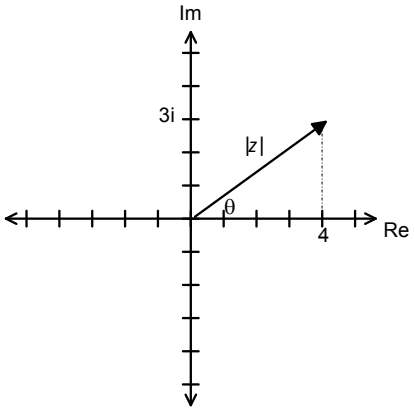
1.73 The Modulus and Argument

The modulus, $|z|$, of the complex number z is the length (magnitude) of the vector ie:

$$|z| = \sqrt{x^2 + y^2}$$

NB:

$$|z|^2 = z\bar{z}$$



The argument, θ, of the complex number z is the angle the vector makes with the positive direction of the x axis ie:

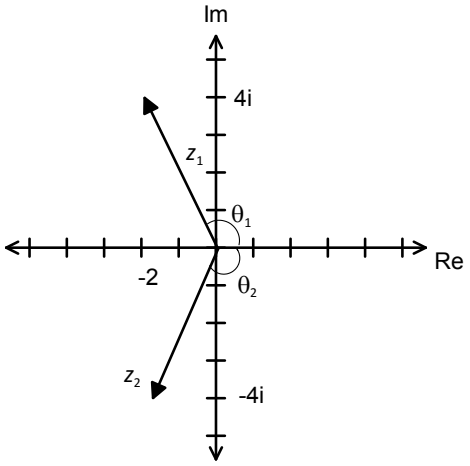
$$\tan\theta = \frac{y}{x}$$

As there are two such values of θ that will satisfy the expression, a diagram is necessary to determine the particular value of θ required.

Example:

1. $$\text{Arg}(z_1) = \pi - \tan^{-1}\left(\frac{4}{2}\right)$$
$$= 2.03$$

2. $$\text{Arg}(z_2) = -\left(\pi - \tan^{-1}\left(\frac{-4}{-2}\right)\right)$$
$$= -2.03$$



NB: the argument is defined on the domain $-\pi \le \theta \le \pi$, hence $\text{Arg}(z_2)$ is -2.03 NOT 4.25

Properties of the modulus

1. $|z_1 z_2| = |z_1||z_2|$
2. $\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}$
3. $|z_1 + z_2| \le |z_1| + |z_2|$
4. $|z_1 - z_2| \ge |z_1| - |z_2|$

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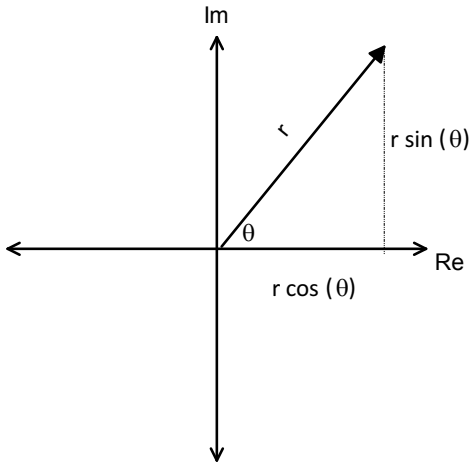
1.74 Polar Form of a Complex Number

In polar form the complex number is expressed in terms of the modulus and argument.
The complex number **z** with modulus **r** can be split into horizontal and vertical components of rcosθ and rsinθ respectively.
Hence:

$z = r\cos\theta + r\sin\theta i$
$z = r(\cos\theta + i\sin\theta)$

$\cos\theta + i\sin\theta$ is written as $\text{cis}\theta$

So, $z = r\text{cis}\theta$



Multiplication	Let $z_1 = r_1\text{cis}\theta_1$ and $z_2 = r_2\text{cis}\theta_2$ $z_1z_2 = r_1r_2\ \text{cis}\theta_1\ \text{cis}\theta_2$ $z_1z_2 = r_1r_2\ (\cos\theta_1 + i\sin\theta_1)(\cos\theta_2 + i\sin\theta_2)$ $z_1z_2 = r_1r_2(\cos\theta_1\cos\theta_2 + i(\cos\theta_1\sin\theta_2 + \sin\theta_1\cos\theta_2) - \sin\theta_1\sin\theta)$ $z_1z_2 = r_1r_2(\cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2))$ $z_1z_2 = r_1r_2\text{cis}(\theta_1 + \theta_2)$ ie: multiply the moduli and add the arguments.
Division	Similarly, $\frac{z_1}{z_2} = \frac{r_1}{r_2}\text{cis}(\theta_1 - \theta_2)$
De Moivre's Theorem	$z^n = r^n\text{cis}(n\theta)$, where n is an integer

Examples:

1. Solve $z^4 = 1 + i$

Writing in polar form:

$z^4 = \sqrt{2}\text{cis}\left(\frac{\pi}{4} + 2\pi n\right)$ - the $2\pi n$ is added to allow all four solutions to be obtained

Let $z = r\text{cis}\theta$

$z^4 = r^4\text{cis}4\theta$

de Moivre's theorem is defined for integer values of n, hence it is used in this way:

Comparing the RHS of the two expressions:

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$r^4 = \sqrt{2}$ and $4\theta = \frac{\pi}{4} + 2\pi n$

$r = \sqrt[8]{2}$ $\theta = \frac{\pi}{16} + \frac{\pi n}{2}$

So, $z = \sqrt[8]{2}\text{cis}\left(\frac{\pi}{16} + \frac{\pi n}{2}\right)$

Listing the solutions in polar form and ensuring that, $-\pi \le \theta \le \pi$:

$\sqrt[8]{2}\text{cis}\left(\frac{\pi}{16}\right), \sqrt[8]{2}\text{cis}\left(\frac{9\pi}{16}\right), \sqrt[8]{2}\text{cis}\left(-\frac{15\pi}{16}\right), \sqrt[8]{2}\text{cis}\left(-\frac{7\pi}{16}\right)$

2. α is one of the complex roots of $z^3 = -1$.

By expressing the other complex root in terms of α show that $\alpha^2 + 1 = \alpha$

$z^3 = \text{cis}(\pi + 2\pi n)$

Let $z = r\text{cis}\theta$

$z^3 = r^3\text{cis}3\theta$

Comparing the RHS of both expressions :

$r = 1$ and $\theta = \frac{\pi}{3} + \frac{2\pi n}{3}$

$z = \text{cis}\left(\frac{\pi}{3} + \frac{2\pi n}{3}\right)$

The solutions are:

$\text{cis}\frac{\pi}{3}, \text{cis}\pi, \text{cis}-\frac{\pi}{3}$

The two complex roots are the 1st and the 3rd which, in rectangular form, are:

$\frac{1}{2} + \frac{\sqrt{3}}{2}i, \frac{1}{2} - \frac{\sqrt{3}}{2}i$

The sum of these two roots is one

Let $\text{cis}\frac{\pi}{3} = \alpha$, then $\text{cis}-\frac{\pi}{3} = \alpha^{-1}$

$\alpha + \alpha^{-1}$

$= \alpha + \frac{1}{\alpha}$

$= \frac{\alpha^2 + 1}{\alpha}$

But the sum of the two complex roots is one.

Hence:

$\frac{\alpha^2 + 1}{\alpha} = 1$

$\alpha^2 + 1 = \alpha$

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Ex 1.7 General Complex Numbers

1. Using the roots of $z^5 = 1$, or otherwise, write $z^4 + z^3 + z^2 + z + 1$ as the product of two quadratic expressions with real coefficients.

   Hence find the exact value of the product $\cos\left(\frac{2\pi}{5}\right)\cos\left(\frac{4\pi}{5}\right)$.

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(Scholarship 2004 Question 4 (b))

2. If α is a complex root of the equation $z^5 = 1$, show that $\alpha + \alpha^2 + \alpha^3 + \alpha^4 = -1$.

Hence find the equation with real coefficients whose roots are $\alpha + \alpha^4$ and $\alpha^2 + \alpha^3$.

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(Scholarship 2005 Question 1 (a))

3. Show that $\cos\left(\frac{\pi}{8}\right)$ can be written exactly as $\frac{\sqrt{2+\sqrt{2}}}{2}$.

Hence, or otherwise, solve the equation $z^2 = \frac{1+7i}{-3+4i}$, where z is a complex number, giving exact solutions in the form a + ib where a and b are constants in surd form.

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(Scholarship 2006 Question 2 (a))

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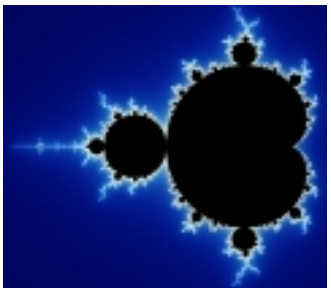
4. (a) If $z=\sqrt{\frac{1}{2}\left(a+\sqrt{\left(a^2+b^2\right)}\right)}+i\sqrt{\frac{1}{2}\left(-a+\sqrt{\left(a^2+b^2\right)}\right)}$ is a complex number, with $i=\sqrt{-1}$ and a, b real numbers, find z^2 in the form p + iq.

The Mandelbrot set (see diagram) is constructed by plotting in black all the complex numbers, c = a + ib, such that:

$$\text{if } f(z)=z^2+c \text{ then } \left|f^n(0)\right|<2 \text{ for all } n\in\{1,2,3,....\}$$

where $f^n(z)$ represents the modulus of the complex number $\underbrace{f(f(f(...f(z)))))}_{n}$

That is, $f^2(z)=f(f(z))=(z^2+c)^2+c$. The sequence starts with z = 0, so f(0) = c, $f^2(0) = c^2 + c$, etc



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(b) Use the definition of the Mandelbrot set above for $f(z) = z^2 + c$, beginning with $z = 0$, so $f(0) = c$.

(i) Show that $c = 1 + \frac{i}{2}$ is not part of the Mandelbrot set, but that c = I is.

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(ii) For $z = z^2 + c$, with $c = \sqrt{\frac{1}{2}\left(a + \sqrt{\left(a^2 + b^2\right)}\right)} + i\sqrt{\frac{1}{2}\left(-a + \sqrt{\left(a^2 + b^2\right)}\right)}$

and $b^2 = 3a^2$ find $f^2(0)$ in terms of a.

Hence show that when $a = \frac{1}{8}$, $\left|f^2(0)\right| = \frac{1}{4}\sqrt{\left(5 + 2\sqrt{3}\right)}$.

(Scholarship 2007 Question Four)

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1.8 The Geometry of Complex Numbers

1.81 Addition of Vectors and the triangle inequality

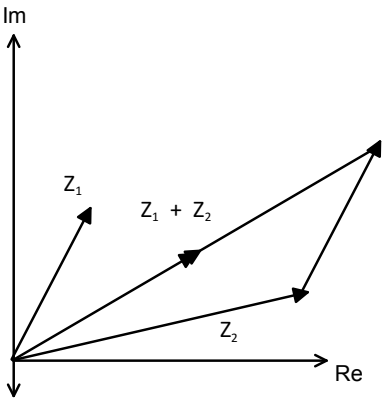
When adding two vectors the longer side can be no longer than the sum of the two shorter sides:

$$|z_1 + z_2| < |z_1| + |z_2|$$

OR

$$|z_1 + z_2| \leq |z_1| + |z_2|$$, if the vectors are parallel.

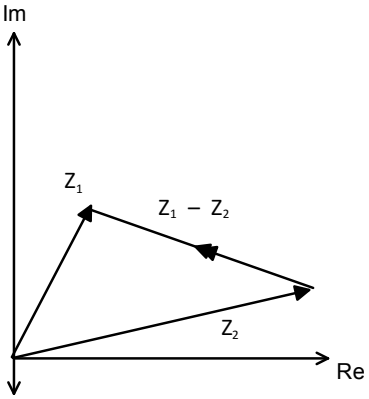
This is the triangle inequality.



1.82 Subtraction of Vectors

It is harder to show but the equivalent expression for subtraction is:

$$|z_1 - z_2| \geq |z_1| - |z_2|$$



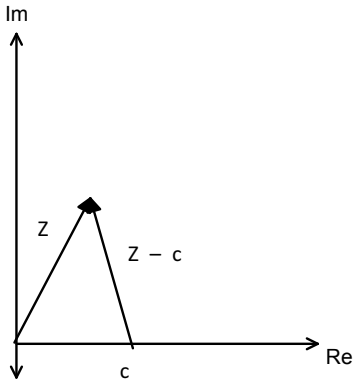
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1.83 Translation

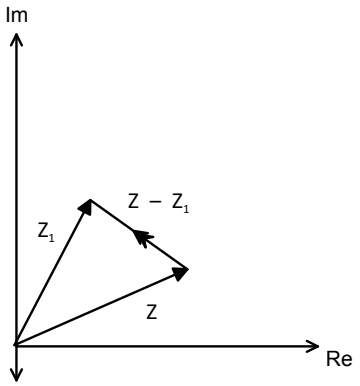
Consider z – c, where z is a complex number and c is real.

z – c starts at c and closes the triangle.

NB: c – z is in the opposite direction.



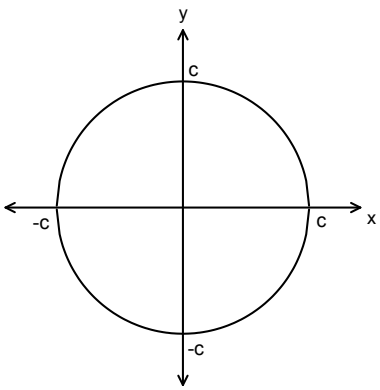
Consider $z - z_1$, where z and z_1 are complex numbers.



1.84 The modulus and geometry

1. $|z| = c$, where c is real

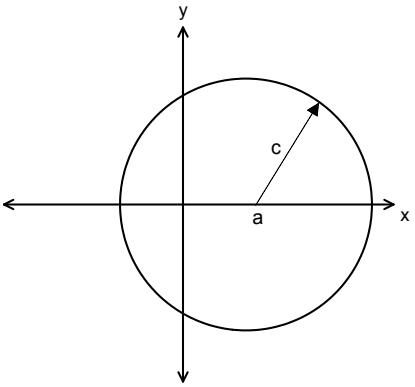
Circle centre (0, 0), radius c



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2. $|z-a|=c$, where a, c are real

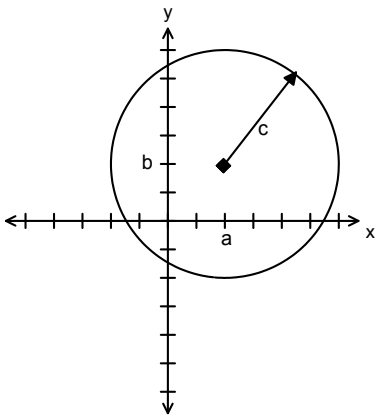
Circle centre (a, 0), radius c



3. $|z-z_1|=c$, where z_1 is complex and c is real

Let $z_1 = a + ib$

Circle centre (a, b), radius c



4. $|z-z_1|=|z-a|$

The above three cases are reasonably obvious.
This one is not as obvious, so we will follow an example through.

Example:

$|z-2i|=|z-4|$

Let $z = x + iy$

$|z-2i|=|z-4|$

$|x+iy-2i|=|x+iy-4|$

$|x+(y-2)i|^2=|x-4+iy|^2$

$x^2+(y-2)^2=(x-4)^2+y^2$

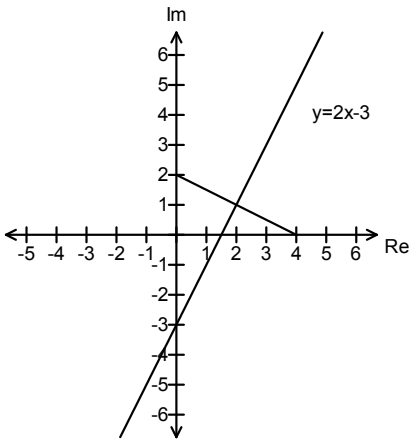
$x^2+y^2-4y+4=x^2-8x+16+y^2$

$y=2x-3$

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This is clearly a straight line, but looking at it more carefully:

- the gradients of the two lines are 2 and – ½ , hence they are perpendicular
- Solving y = 2x + 3 and $y = -\frac{1}{2}x + 2$, gives the intersection at the midpoint of the line between 2i and 4.

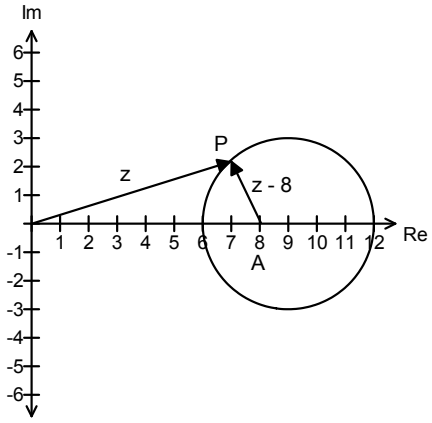


Hence y = 2x + 3 is the perpendicular bisector of the line between 2i and 4

Another way of looking at this is "the distance between z and 2i is the same as the distance between z and 4", which is the perpendicular bisector.

5. On an Argand diagram the point A represents 8.

Show that the points P such that OP = 3AP lie on a circle.



$|z| = 3|z-8|$

$|z|^2 = 9|z-8|^2$

Let $z = x + iy$

$x^2 + y^2 = 9[(x-8)^2 + y^2]$

$x^2 + y^2 = 9[x^2 - 16x + 64 + y^2]$

$x^2 + y^2 = 9x^2 - 144x + 576 + 9y^2$

$8x^2 + 8y^2 - 144x + 576 = 0$

$x^2 + y^2 - 18x + 72 = 0$

$(x-9)^2 + y^2 = 9$

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Ex 1.8 The Geometry of Complex Numbers

1. Identify, on an Argand diagram, the points representing the following (working not required):

   (a) $|z| = 3$

   (b) $|z - 3| = 3$

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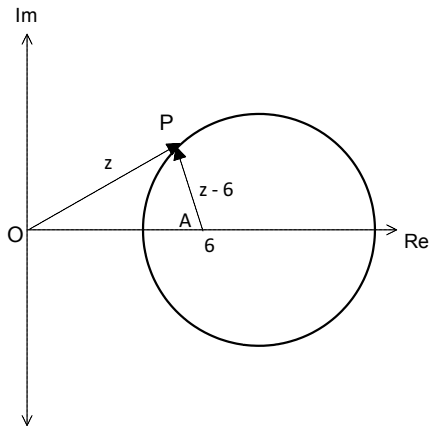
(c) $|z+3i| = |z+2|$

(d) $z - \overline{z} = 4i$

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2. On an Argand diagram the point A represents 6.

Show that the points P such that OP = 2AP lie on a circle.



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3. P is the set of points on an Argand diagram corresponding to the complex number z which satisfies the equation:

$$|6 + z| - |6 - z| = 8 .$$

Explain why the points P lie on a hyperbola.

Work out the equation of the hyperbola.

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4. P is the set of points on an Argand diagram corresponding to the complex number z which satisfies the equation:

$$|z + 5| + |z - 5| = 26 .$$

Explain why the points P lie on an ellipse.
Work out the equation of the ellipse.

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5. If v = 1 + i and z = x + iy, for any real numbers x and y:

(i) Show that the equation

$$|z-v|=|zv|$$

represents a circle, and find its radius and centre

(ii) Find the intersection of the circle in part (i) and the straight line

$$|z-v|=|z+v|$$

Scholarship 2004 Question Four (a))

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6. The roots $z_i (i = 1,2,3,4,5)$ of $z^5 - 32 = 0$ are represented in an Argand diagram by the points A, B, C, D and E, respectively, taken clockwise, with

$|\arg z_2| = |\arg z_5|$ and $|\arg z_3| = |\arg z_4|$ and A = (2,0)

(i) Sketch an Argand diagram, showing clearly the points A, B, C, D and E.

Find the distance between the points represented by C and E in terms of b, the y coordinate of B.

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(ii) A variable complex number w is given by $w = t(1+i)$, $t \in R$, and w is represented in the Argand diagram by the variable point W.

If F is the point (1, -1) find the exact value of the minimum distance AW + FW.

(You do not need to prove that it is a minimum).

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(Scholarship 2005 Question One (b))

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7. (a) Solve the equation $(z + 1)^3 = 8$, where z is a complex number.

Give your answers in terms of w, where w is a complex number and $w^3 = 1$.

Hence, or otherwise, show that the sum of the roots of $(z + 1)^3 = 8$ is -3.

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(b) Solve the equation $(z + 1)^3 = 8(z – 1)^3$, giving exact answers in the form a + ib, where i $= \sqrt{-1}$.

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(c) Plot the three solutions to the equation in part (b) in the complex plane as the points A, B and C.

Plot the three solutions to the equation

$$\left(\frac{1}{z}+1\right)^3 = 8\left(\frac{1}{z}-1\right)^3$$

in the complex plane as the points E, F and G.

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Show that the ratio of the areas of the two triangles ABC and EFG is given by:

$$\frac{\Delta ABC}{\Delta EFG} = \frac{81}{49}$$

(Scholarship 2008 Question Two)

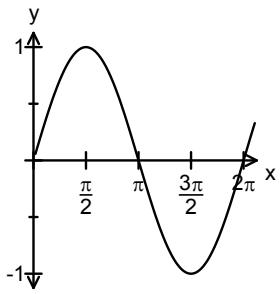
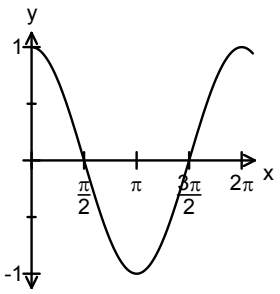
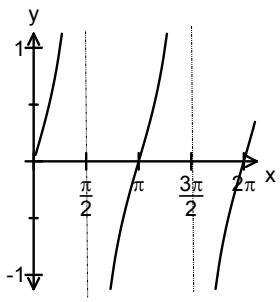
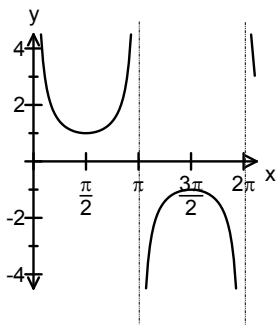
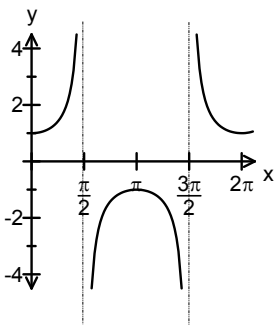
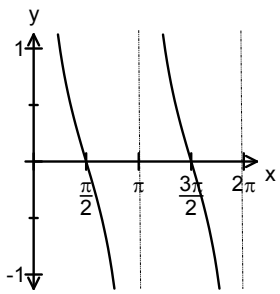
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2. Trigonometry

Trigonometry has only explicitly, played a small part in the scholarship examinations to date but a familiarity with identities and equations is necessary in other topics. As a result this chapter is a short one, briefly covering the topics with just a few examples where necessary.

2.1 Trigonometric Graphs and Modelling

Although the Graphic Calculator can be used to draw trigonometric graphs, it is still not as quick as having an understanding of the basic graphs, particularly when it comes to the solution of modelling questions.

Sine (sin)	Cosine (cos)	Tangent (tan)
		
Cosecant (cosec)	**Secant (sec)**	**Cotangent (cot)**
		

Where $\text{cosec}\theta = \frac{1}{\sin\theta}$, $\sec\theta = \frac{1}{\cos\theta}$, $\cot\theta = \frac{1}{\tan\theta}$

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Example

The depth of water in the middle of a tidal channel changes as the tide comes in and out.
This depth can be modelled by a trigonometric function
Successive high tides occur every 12 hours. The maximum depth of water at high tide is 3.5m while the minimum depth at low tide is 0.5m.

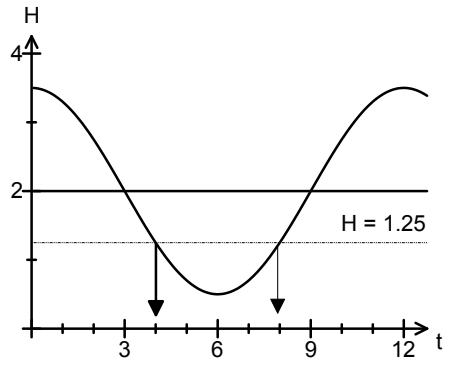
(a) Write down an equation to model this situation

It is safe to walk across the inlet provided the depth of water is not more than 1.25 metre.
High tide on a particular day occurs at 11:00am.

(b) Between what times in the afternoon of that day will it be safe to walk across the inlet?

Solution

(a) Draw a graph:



The situation can be modelled by a cosine function of the form H = AcosBt + C

Evaluate the constants:

$$A = \frac{3.5 - 0.5}{2} = 1.5$$

$$B = \frac{2\pi}{12} = \frac{\pi}{6}$$

$$C = \frac{3.5 + 0.5}{2} = 2$$

Hence the equation is $H = 1.5\cos\left(\frac{\pi t}{6}\right) + 2$

From the graph H = 1.25 when t = 4 and t = 8 ie: between 3 and 7pm

This can be solved algebraically but we will leave such solutions to section 2.3.

Ex 2.1 Graphs and Modelling

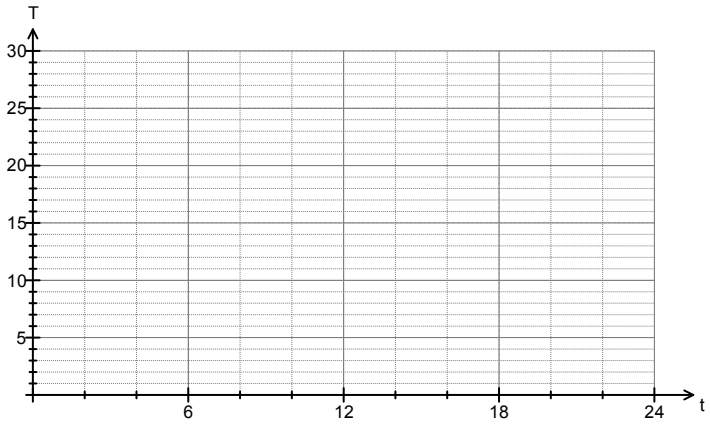
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1. James' teacher has asked him to create a mathematical model for the temperatures for a cloudless and windless day in February.

   He has been told that the lowest temperature was 7°C at 3am and the highest temperature was 29°C at 3pm.

   James decides to use a trigonometric model for the temperature.

   (a) Sketch the graph and use it to determine a trigonometric equation that models the given data from 3am that day to 3am the next day ie over a 24 hour period.



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   (b) Use the model to find the temperature at midnight.

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   (c) Give one time when the temperature is 18° .

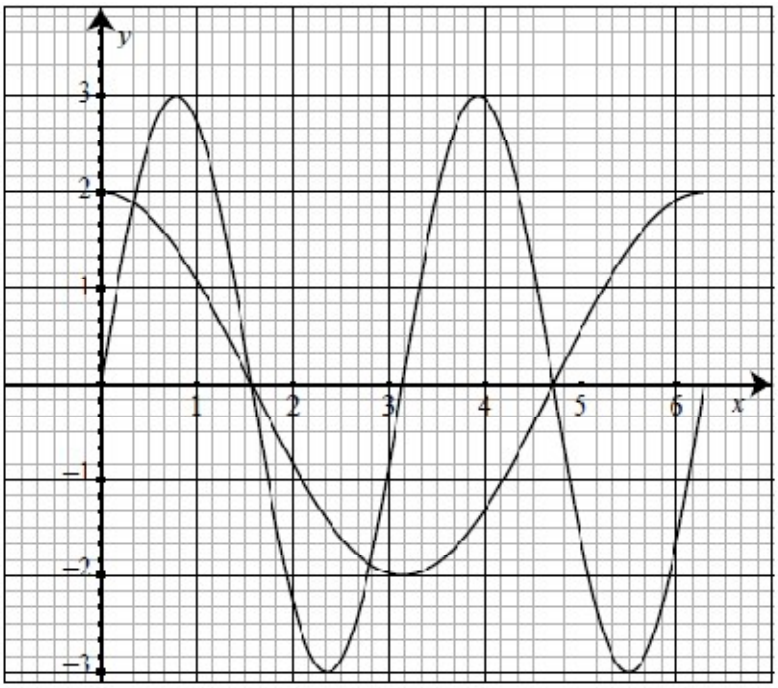
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2. The graph of the two functions

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$f(x) = 3\sin 2x,\ 0 \le x \le 2\pi$ and $g(x) = 2\cos x,\ 0 \le x \prec 2\pi$ are shown



(a) $h(x) = k + 2\cos x,\ 0 \le x \le 2\pi$, where k is a positive constant.

Find the x-coordinate of all the points where f(x) and h(x) have the same gradient.

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(b) The graph of $p(x) = -a(x-b)^2 - c$, $0 \le x \le 2\pi$, where $a > 0$, $b > 0$ and $c > 0$, intersects each of the graphs of f(x) and g(x) at exactly one point only.

The x-coordinate of the point of intersection with f(x) is 2.575, and the vertex of p(x) lies on g(x). Find the values of a, b and c.

(c) Find the range of values of k for which f(x) and h(x) intersect at exactly two points.

Note: k is greater than zero and both f(x) and g(x) have the domain $0 \le x \le 2\pi$.

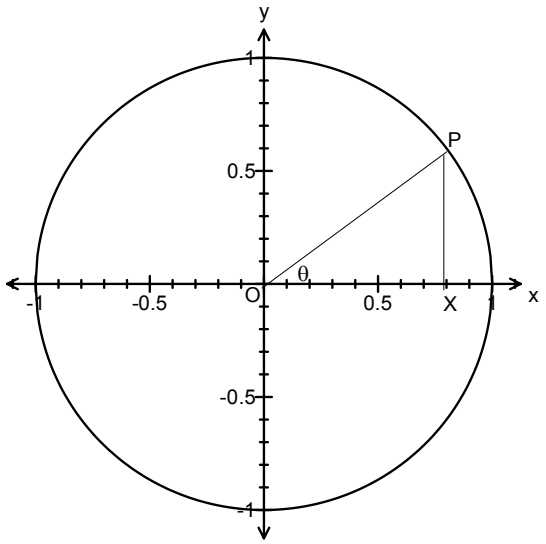
(Scholarship 2007 Question Three)

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2.2 Identities

The Pythagorean Identities

These identities are based on the theorem of Pythagoras:



P is any point on a circle of radius 1.

From right angled triangle OPX:

$OX = \cos\theta$ and $PX = \sin\theta$

By Pythagoras:

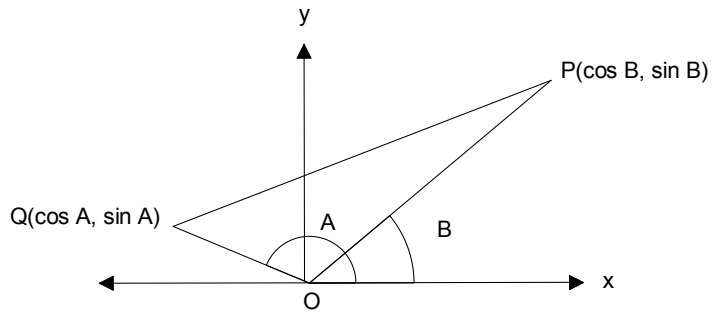
$$OX^2 + PX^2 = 1$$
$$\cos^2\theta + \sin^2\theta = 1$$

This is the first identity. Dividing each side first by $\cos^2\theta$ then $\sin^2\theta$ yields the next two identities:

$$\tan^2\theta + 1 = \sec^2\theta$$
$$\cot^2\theta + 1 = \text{cosec}^2\theta$$

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The Compound Angle Formulae



Using Pythagoras (or length of a line formula):

$$PQ^2 = (\cos B - \cos A)^2 + (\sin B - \sin A)^2$$
$$OQ^2 = \sin^2 A + \cos^2 A = 1$$
$$OP^2 = \sin^2 B + \cos^2 B = 1$$

Using the Cosine Rule in triangle OPQ:

$$PQ^2 = OQ^2 + OP^2 - 2 \times OP \times OQ \times \cos(A - B)$$
$$(\cos B - \cos A)^2 + (\sin B - \sin A)^2 = 1 + 1 - 2 \times 1 \times 1 \times \cos(A - B)$$
$$\cos^2 B - 2\cos B\cos A + \cos^2 A + \sin^2 B - 2\sin B\sin A + \sin^2 A = 2 - 2 \times \cos(A - B)$$
$$2 - 2\cos B\cos A - 2\sin B\sin A = 2 - 2 \times \cos(A - B)$$
$$\cos B\cos A + \sin B\sin A = \cos(A - B)$$

OR

$$\cos(A - B) = \cos A\cos B + \sin A\sin B$$

The other compound angle formulae can be derived directly from this and their derivation can be found in standard Level Three texts. Here I will simply derive the formula for $\cos(A + B)$ because it utilises a simple but very important idea:

$$\cos(A - B) = \cos A\cos B + \sin A\sin B$$
$$\cos(A + B) = \cos(A - \bar{B})$$
$$\cos(A + B) = \cos A\cos \bar{B} + \sin A\sin \bar{B}$$
$$\cos(A + B) = \cos A\cos B - \sin A\sin B$$

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The last step of the proof is based on the fact that cosine is an even function ie: $\cos(-A) = \cos A$ and sine is an odd function ie: $\sin(-A) = -\sin A$. Similarly tangent and cosecant are also odd and secant even. This simple idea occurs frequently in the work to follow.

In summary the compound angle formulae are:

$$\sin(A \pm B) = \sin A\cos B \pm \cos A\sin B$$
$$\cos(A \pm B) = \cos A\cos B \mp \sin A\sin B$$
$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A\tan B}$$

The remaining formulae follow. Some of the proofs will be included as examples at the end.

Double Angle Formulae

$$\sin 2A = 2\sin A\cos A$$
$$\cos 2A = \cos^2 A - \sin^2 A$$
$$= 2\cos^2 A - 1$$
$$= 1 - 2\sin^2 A$$
$$\tan 2A = \frac{2\tan A}{1 - \tan^2 A}$$

Product Formulae

$$2\sin A\cos A = \sin(A + B) + \sin(A - B)$$
$$2\cos A\sin B = \sin(A + B) - \sin(A - B)$$
$$2\cos A\cos B = \cos(A + B) + \cos(A - B)$$
$$2\sin A\sin B = \cos(A - B) - \cos(A + B)$$

Sum Formulae

$$\sin C + \sin D = 2\sin\left(\frac{C + D}{2}\right)\cos\left(\frac{C - D}{2}\right)$$
$$\sin C - \sin D = 2\cos\left(\frac{C + D}{2}\right)\sin\left(\frac{C - D}{2}\right)$$
$$\cos C + \cos D = 2\cos\left(\frac{C + D}{2}\right)\cos\left(\frac{C - D}{2}\right)$$
$$\cos C - \cos D = -2\sin\left(\frac{C + D}{2}\right)\sin\left(\frac{C - D}{2}\right)$$

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Proofs

1. Work on one side of the proof at a time

2. The best side to work on is usually the more complicated side eg: one with two fractions to be added, but there is no hard and fast rule.

3. Be familiar with the "shape" of the identities/formulae as this will make it more likely that you will open up the formula sheet for a helpful formula.

4. At each step be aware of where you wish to go as this will help in the appropriate selection of formulae.

5. Sometimes it might be necessary to switch to the other side and bring it towards the side you have been previously working on. This is not usually necessary.

6. If all else fails start again on the other side.

Examples

1. Prove that $\tan(A+B) = \dfrac{\tan A + \tan B}{1 - \tan A \tan B}$

In this example we will do the proof twice as it exhibits two different "tricks":

$$\text{RHS} = \frac{\dfrac{\sin A}{\cos A} + \dfrac{\sin B}{\text{Cos}B}}{1 - \dfrac{\sin A}{\cos A} \times \dfrac{\sin B}{\cos B}}$$

$$= \frac{\dfrac{\sin A \cos B + \cos A \sin B}{\cos A \cos B}}{\dfrac{\cos A \cos B - \sin A \sin B}{\cos A \cos B}}$$

$$= \frac{\sin A \cos B + \cos A \sin B}{\cos A \cos B - \sin A \sin B}$$

$$= \frac{\sin(A+B)}{\cos(A+B)}$$

$$= \tan(A+B)$$

$$= \text{LHS}$$

This is an example of starting on the "more complicated side" and manipulating fractions.

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$$\text{LHS} = \tan(A+B)$$

$$= \frac{\sin(A+B)}{\cos(A+B)}$$

$$= \frac{\sin A\cos B + \cos A\sin B}{\cos A\cos B - \sin A\sin B}$$

Now, sneaking a look at the RHS we see that not only is it necessary to get simple tangent expressions but there must also be a 1 on the lower left hand side of the fraction. To achieve this 1 it is necessary to divide by the term that is currently there viz: cosAcosB.
A quick check of the other terms verifies that this will produce the desired result all round.

$$\text{LHS} = \frac{\frac{\sin A\cos B}{\cos A\cos B} + \frac{\cos A\sin B}{\cos A\cos B}}{1 - \frac{\sin A\sin B}{\cos A\cos B}}$$

$$= \frac{\frac{\sin A}{\cos A} + \frac{\sin B}{\cos B}}{1 - \frac{\sin A}{\cos A} \times \frac{\sin B}{\cos B}}$$

$$= \frac{\tan A + \tan B}{1 - \tan A\tan B}$$

$$= \text{RHS}$$

2. If A, B, C are the angles of a triangle show that:

$$\sin A + \sin B + \sin C = 4\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}$$

$$\text{LHS} = (\sin A + \sin B) + \sin C$$

$$= 2\sin\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right) + 2\sin\frac{C}{2}\cos\frac{C}{2}$$

$$\text{But } A + B + C = 180^o \Rightarrow \frac{A+B}{2} + \frac{C}{2} = 90^o$$

$$\text{So, } \sin\left(\frac{A+B}{2}\right) = \sin\left(90^o - \frac{C}{2}\right) = \cos\left(\frac{C}{2}\right)$$

$$\text{and, } \sin\left(\frac{C}{2}\right) = \sin\left(90^o - \frac{A+B}{2}\right) = \cos\left(\frac{A+B}{2}\right)$$

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$$\sin A + \sin B + \sin C = 2\cos\left(\frac{C}{2}\right)\cos\left(\frac{A-B}{2}\right) + 2\cos\left(\frac{A+B}{2}\right)\cos\left(\frac{C}{2}\right)$$

$$= 2\cos\left(\frac{C}{2}\right)\left[\cos\left(\frac{A-B}{2}\right) + \cos\left(\frac{A+B}{2}\right)\right]$$

$$= 2\cos\left(\frac{C}{2}\right)\left[2\cos\left(\frac{1}{2}\left(\frac{A-B}{2} + \frac{A+B}{2}\right)\right)\cos\left(\frac{1}{2}\left(\frac{A-B}{2} - \frac{A+B}{2}\right)\right)\right]$$

$$= 4\cos\left(\frac{C}{2}\right)\cos\left(\frac{A}{2}\right)\cos\left(-\frac{B}{2}\right)$$

$$= 4\cos\left(\frac{A}{2}\right)\cos\left(\frac{B}{2}\right)\cos\left(\frac{C}{2}\right)$$

Converting a sum into a product should be obvious.

The trick here is when to use the angle sum of a triangle.
Because the RHS has only terms in cosine then using the property $\sin\left(90^o - \theta\right) = \cos\theta$ should also have been an obvious tactic.

Ex 2.2 Identities

1. Prove that $\dfrac{\sin A - \sin B}{\sin A + \sin B} = \cot\left(\dfrac{A+B}{2}\right)\tan\left(\dfrac{A-B}{2}\right)$

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2. If A, B and C are the angles of a triangle, show that:

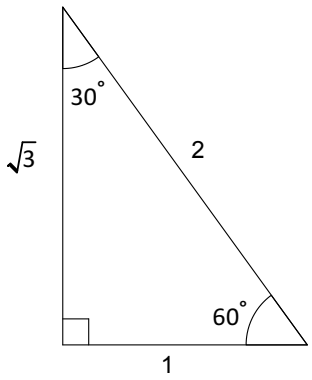
$$\frac{\sin A + \sin B}{\cos A + \cos B} = \cot\left(\frac{C}{2}\right)$$

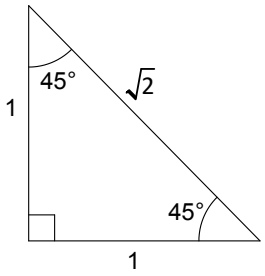
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2.3 Trigonometric Equations

There are several methods for the solution of trigonometric equations.
Being familiar with them all allows for the adoption of the most appropriate method for the problem in hand.

But first, the special triangles:

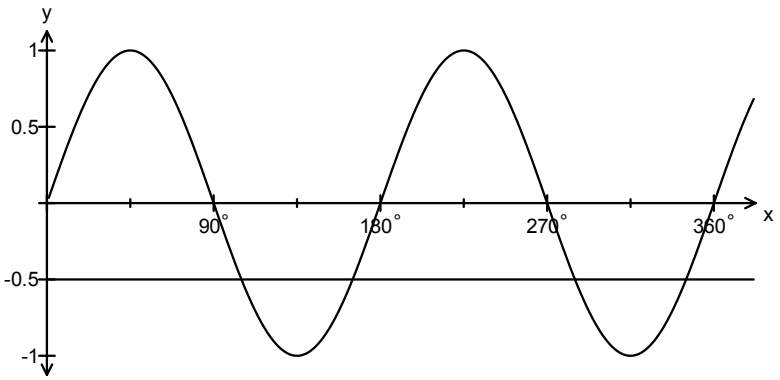




The values of the trigonometric functions for 30°, 45°, 60° can easily be obtained from these triangles eg: $\sin 60^o = \frac{\sqrt{3}}{2}$

Graphical Method

To solve $\sin 2x = -0.5,\ \ 0^o \le x \le 360^o$, graph:



The calculator, or special triangles, will give the first solution as 2x = -30° or x= -15° (345°).

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Observing the graph the remaining solutions are thus 270 + 15, 180 – 15, 90 + 15 ie: x = 105°, 165°, 285°, 345°

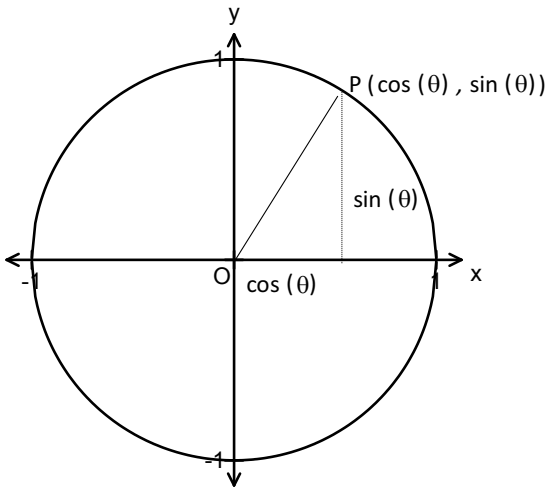
As the equations become more complicated the graphical solution becomes more cumbersome, although the Graphics Calculator is reasonably easy to use, albeit a little slow in setting it up.

The Quadrant Method

This is based on the trigonometric functions as circular functions.

As P moves in an anticlockwise direction around the unit circle, sin θ is the vertical distance above the x-axis and cos θ the horizontal distance from the y-axis.

The tangent $\left(\tan\theta = \frac{\sin\theta}{\cos\theta}\right)$ is the gradient of OP.



As it moves through the 1st, 2nd, 3rd, 4th quadrants the sign of each changes:

- **A**ll positive in the 1st quadrant
- **S**ine positive in the 2nd quadrant
- **T**angent positive in the 3rd quadrant
- **C**osine positive in the 4th quadrant

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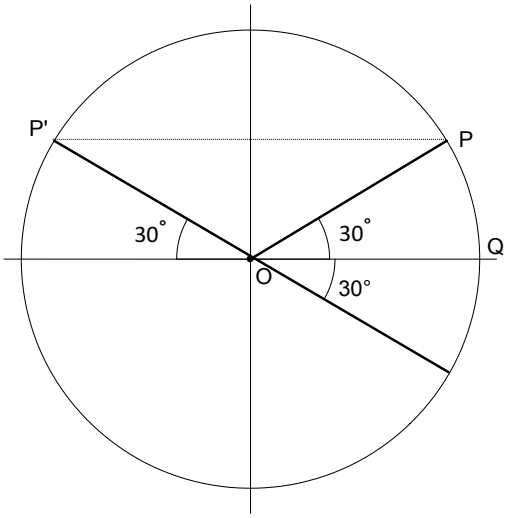
The examples on the following page show how to use this method.

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From the special triangles we know:

$\sin 30^o = \frac{1}{2},\ \cos 30^o = \frac{\sqrt{3}}{2}$

and $\tan 30^\circ = \frac{1}{\sqrt{3}}$



<QOP = 150° but from the symmetry of the diagram:

$\sin 150^o = \frac{1}{2},\ \cos 150^o = -\frac{\sqrt{3}}{2}$

and $\tan 150^\circ = -\frac{1}{\sqrt{3}}$

Similarly:

$\sin 330^o = -\frac{1}{2},\ \cos 330^o = \frac{\sqrt{3}}{2}$

and $\tan 330^\circ = -\frac{1}{\sqrt{3}}$

Example:

Solve $\sin 2x = -0.5,\ \ 0^o \le x \le 360^o$

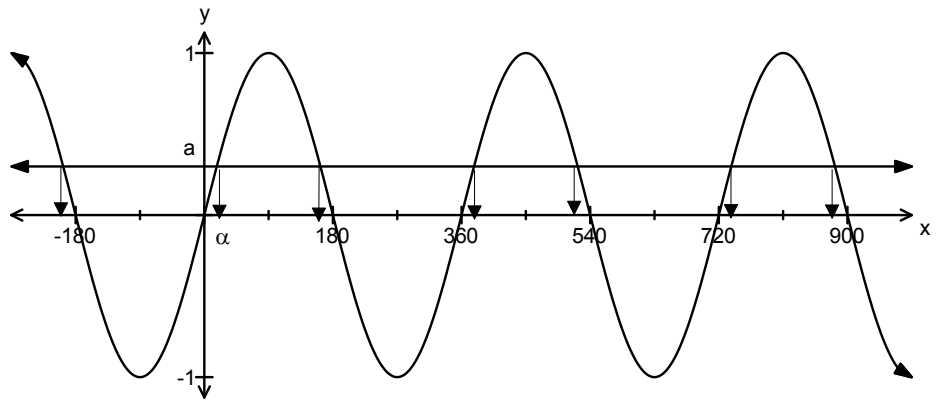
- Sine is negative in the 3rd & 4th quadrant
- 2x = 210° and 2x = 330° yield solutions of 105° and 165°
- But 2x = 210° + 360° and 2x = 330° + 360° yield solutions of 285° and 345° which are also in the required domain.
- So, x = 105°, 165°, 285°, 345°

This method is much quicker than drawing a graph with or without a graphics calculator

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The Formulae

The formulae are derived from the graphs:



The graph shows the multiple solutions of $\sin x = a$, the first of which is α

By symmetry, the other solutions shown are:

$-180-\alpha,\ 180-\alpha,\ 360+\alpha,\ 540-\alpha,\ 720+\alpha,\ 900-\alpha$

OR

$-1\times180-\alpha,\ 1\times180-\alpha,\ 2\times180+\alpha,\ 3\times180-\alpha,\ 4\times180+\alpha,\ 5\times180-\alpha$

Generalising this pattern:

$$x = 180n + (-1)^n \times \alpha$$

Similarly formulae can be derived for cosine and tangent:

Sine	Cosine	Tangent
$x = 180n + (-1)^n \times \alpha$ OR $x = \pi n + (-1)^n \times \alpha$	$x = 360n \pm \alpha$ OR $x = 2\pi n \pm \alpha$	$x = 180n + \alpha$ OR $x = \pi n + \alpha$

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Examples

1. Solve $2\sin\left(2x-90^o\right)=\sqrt{2},\ 0\le x\le 360^o$

$2\sin\left(2x-90^o\right)=\sqrt{2}$

$\sin\left(2x-90^o\right)=\dfrac{\sqrt{2}}{2}$

$\alpha=\sin^{-1}\left(\dfrac{\sqrt{2}}{2}\right)$

$\alpha=45^\circ$ (Using the calculator or special triangles)

Using the formula:

$2x-90=180n+(-1)^n\times 45$

$x=90n+(-1)^n\times 22.5+45$

$n=0\quad x=67.5$

$n=1\quad x=112.5$

$n=2\quad x=245$

$n=3\quad x=292.5$

2. Solve $\cos\left(2x+\pi\right)=-0.5,\ 0\le x\le 2\pi$

$\cos\left(2x+\pi\right)=-0.5$

$\alpha=\dfrac{2\pi}{3}$ (Special triangles + quadrant method)

$2x+\pi=2\pi n\pm\dfrac{2\pi}{3}$

The next step has the potential for error viz: the + and – options must be separated.

$2x+\pi=2\pi n+\dfrac{2\pi}{3}$ OR $2x+\pi=2\pi n-\dfrac{2\pi}{3}$

$2x=2\pi n-\dfrac{\pi}{3}$ OR $2x=2\pi n-\dfrac{5\pi}{3}$

$x=\pi n-\dfrac{\pi}{6}$ OR $x=\pi n-\dfrac{5\pi}{6}$

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$n = 0$ gives solutions outside the domain

$n = 1 \quad x = \frac{5\pi}{6}$ OR $\frac{\pi}{6}$

$n = 2 \quad x = \frac{11\pi}{6}$ OR $\frac{7\pi}{6}$

The formulae are useful for complicated examples or for those who like the mechanical nature of formulae. The approach is quite long and the other two methods are usually best for modelling questions where the equations can be quite simple.

Familiarity with all three approaches allows for flexibility. The quickest approach for the particular problem at hand is generally the one which should be used.

A Modelling Example

The depth of water in the middle of a tidal channel changes as the tide comes in and out.

This depth can be modelled by the trigonometric function $H = 1.5\cos\left(\frac{\pi t}{6}\right) + 2$

It is safe to walk across the channel when the depth of water is less than 1.5 metres.
If high tide is at 11am, between what times in the afternoon is it safe to cross?

Drawing a **sketch** graph is always helpful:

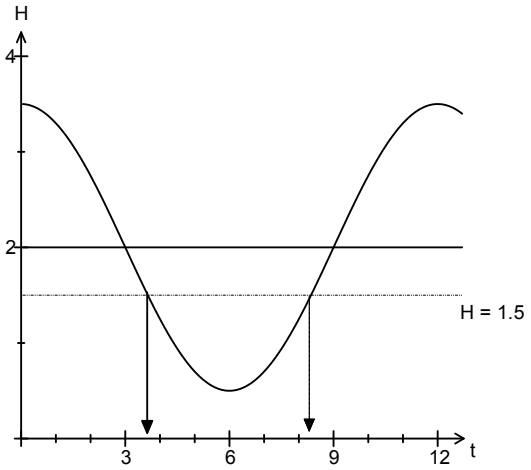
$1.5 = 1.5\cos\left(\frac{\pi t}{6}\right) + 2$

$-0.5 = 1.5\cos\left(\frac{\pi t}{6}\right)$

$-\frac{1}{3} = \cos\left(\frac{\pi t}{6}\right)$

$\frac{\pi t}{6} = 1.9106$

$t = 3.649$



Observing the graph, this is clearly the first solution.
By symmetry, the second is 9 – 0.649 = 8.351.
These correspond to times of 2:39pm and 7:21pm.

This question is clearly easiest done by direct reference to the graph but the second solution could be found using the formula where $\alpha = 1.9106$

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Ex 2.3 Equations

1. Expand $\cos(2A+B)$ and hence prove that

$$\frac{1}{4}\cos 3\theta = \cos^3\theta - \frac{3}{4}\cos\theta$$

Hence, by putting $x = \frac{2}{3}\cos\theta$, find the exact roots of the equation

$$27x^3 - 9x = 1$$

in terms of π.

Hence, without using the calculator and clearly showing your method, write down the exact value of the product

$$\cos\frac{\pi}{9}\cos\frac{3\pi}{9}\cos\frac{5\pi}{9}\cos\frac{7\pi}{9}$$

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(Scholarship 2005 Question Five (a))

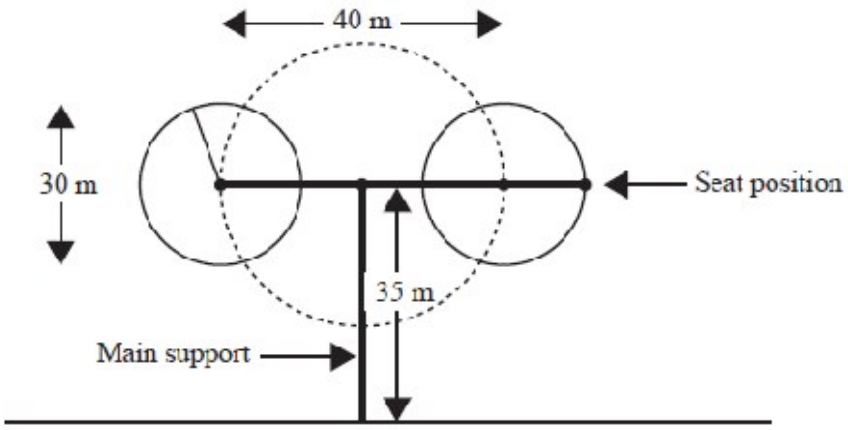
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2.

(a) Solve for $0 \le x \le 2\pi$

$$10\pi\cos\left(\frac{\pi x}{2}\right)+10\pi\cos\left(\frac{2\pi x}{3}\right)=0$$

(b) An amusement park has a giant double Ferris wheel as shown



The double Ferris wheel has a rotating arm 40 metres long attached at its centre to a main support 35 metres above the ground. At each end of the rotating arm is attached a Ferris wheel measuring 30 metres in diameter as shown in the diagram. The rotating arm takes 4 minutes to complete one full revolution, and each wheel takes 3 minutes to complete a

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revolution about the wheel's hub.

All revolutions are anti clockwise in a vertical plane.

At time t = 0 the rotating arm is parallel to the ground and your seat is at the 3-o'clock position of the rightmost wheel.

(i) Find a formula for h(t), your height above the ground in metres, as a function of time in minutes.

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(ii) Hence show your maximum height above the ground, clearly showing that it is a maximum.

(Scholarship 2007 Question Two)

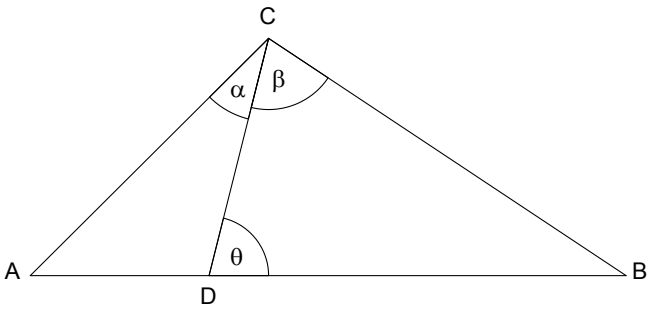
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2.4 Triangles

Triangle problems with numerical answers are too simplistic at this level but triangles do provide a possible context for algebraic manipulation.

Following is an example where the cotangent formula, useful in some equilibrium problems in mechanics, is proved.

In the triangle below prove that if D divides AB in the ratio m:n then:
$(m+n)\cot\theta = m\cot\alpha - n\cot\beta$



In ΔACD:

$$\frac{CD}{\sin A} = \frac{AD}{\sin\alpha}$$

But:

$$A = \theta - \alpha \quad (\text{ext} < \Delta)$$

$$CD = \frac{AD\sin(\theta-\alpha)}{\sin\alpha}$$

In ΔBCD:

$$\frac{CD}{\sin B} = \frac{BD}{\sin\beta}$$

$$CD = \frac{BD\sin(180-(\theta+\beta))}{\sin\beta}$$

$$CD = \frac{BD\sin(\theta+\beta)}{\sin\beta}$$

Hence:

$$\frac{AD\sin(\theta-\alpha)}{\sin\alpha} = \frac{BD\sin(\theta+\beta)}{\sin\beta}$$

But:

$$AD:BD = m:n$$

$$\frac{m\sin(\theta-\alpha)}{\sin\alpha} = \frac{n\sin(\theta+\beta)}{\sin\beta}$$

$$\frac{m(\sin\theta\cos\alpha - \cos\theta\sin\alpha)}{\sin\alpha} = \frac{n(\sin\theta\cos\beta + \cos\theta\sin\beta)}{\sin\beta}$$

$$\frac{m\sin\theta\left(\cos\alpha - \frac{\cos\theta}{\sin\theta}\sin\alpha\right)}{\sin\alpha} = \frac{n\sin\theta\left(\cos\beta + \frac{\cos\theta}{\sin\theta}\sin\beta\right)}{\sin\beta}$$

$$m\left(\frac{\cos\alpha}{\sin\alpha} - \frac{\cos\theta}{\sin\theta}\right) = n\left(\frac{\cos\beta}{\sin\beta} + \frac{\cos\theta}{\sin\theta}\right)$$

$$m(\cot\alpha - \cot\theta) = n(\cot\beta + \cot\theta)$$

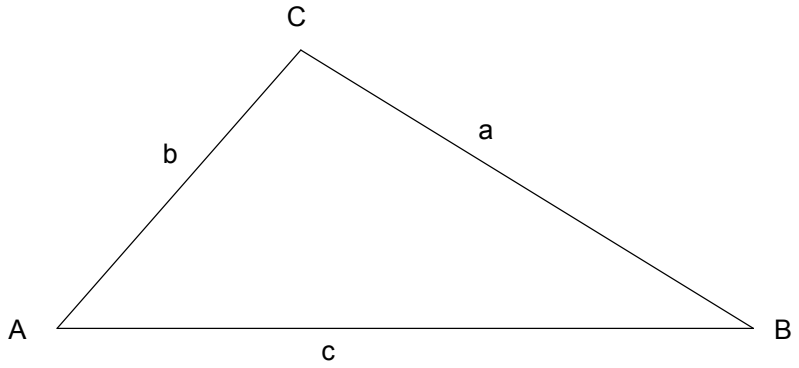
$$m\cot\alpha - m\cot\theta = n\cot\beta + n\cot\theta$$

$$(m+n)\cot\theta = m\cot\alpha - n\cot\beta$$

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Ex 2.4 Triangles

1.

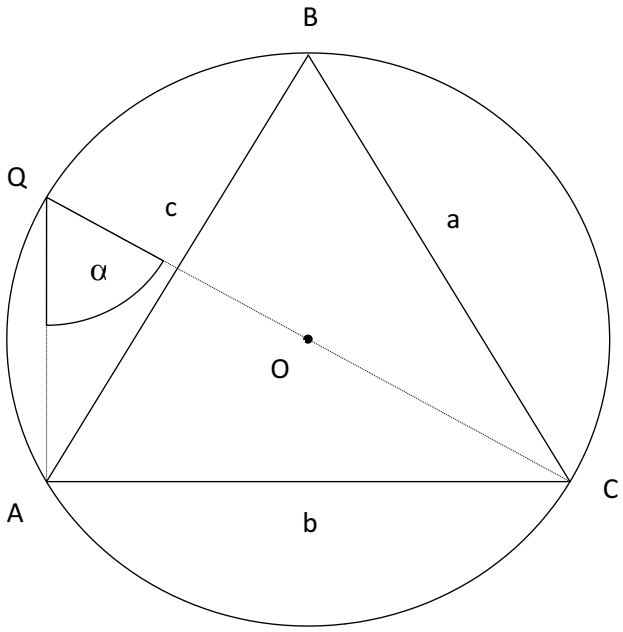


Prove that $\frac{b-c}{b+c} = \tan\left(\frac{B-C}{2}\right)\tan\left(\frac{A}{2}\right)$

` Hint: Let $\frac{c}{\sin C} = \frac{b}{\sin B} = k$

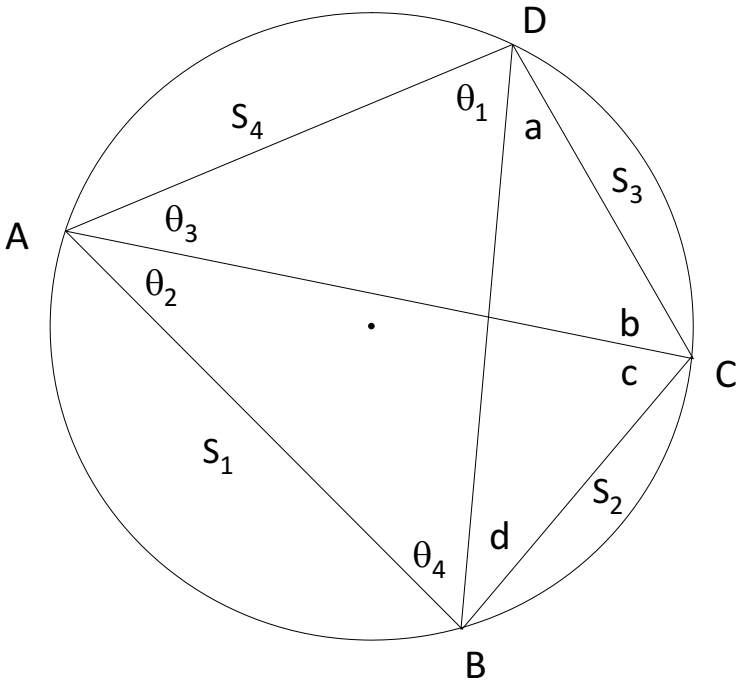
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2. (a) Assuming the Sine Rule $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$, use the diagram to show that $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$, where R is the radius of the circle and O is the centre.



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(b)



ABCD is a cyclic quadrilateral and O is the centre of the circle with radius R.

(i) Match a, b, c, d with the appropriate values of $\theta_1, \theta_2, \theta_3, \theta_4$

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Ptolemy's Theorem states that $S_1S_3 + S_2S_4 = AC.DB$

(ii) Using Ptolemy's Theorem and the result of (a) show that:

$$\sin\theta_1\sin\theta_3 + \sin\theta_2\sin\theta_4 = \sin(\theta_3 + \theta_2)\sin(\theta_3 + \theta_4)$$

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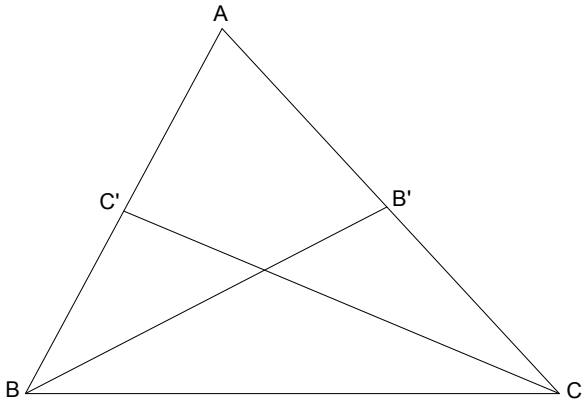
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3. The Steiner-Lehmus theorem states that if two of the angle bisectors of a triangle are equal then the triangle is isosceles. This question presents a proof of the theorem.

BB' and CC' are the angle bisectors of angles ABC and ACB respectively and BB' = CC'.



(a) From triangles BB'C and BC'C respectively show that:

$$\frac{BB'}{\sin C} = \frac{BC}{\sin\left(C + \frac{B}{2}\right)} \text{ and } \frac{CC'}{\sin B} = \frac{BC}{\sin\left(B + \frac{C}{2}\right)}$$

Hence show that, given BB' = CC':

$$\cos\left(\frac{C}{2}\right)\cos\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right) = \cos\left(\frac{B}{2}\right)\cos\left(\frac{A+C}{2}\right)\cos\left(\frac{A-C}{2}\right)$$

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(b) Show that $\cos(x+y).\cos(x-y) = \cos^2 x + \cos^2 y - 1$ and apply it to the result in (a) to show that triangle ABC is isosceles.

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3. Differentiation

This chapter contains some revision of Level Three differentiation but does not revise Level Two skills.

It also covers a great deal of work which either extends those ideas or covers ideas which go beyond the Level Three syllabus.

Topics included are:

(i) Common derivatives
(ii) Limits
(iii) First Principles
(iv) Chain, Product and Quotient Rules
(v) Implicit and Parametric Differentiation
(vi) Derivatives of Inverse Trigonometric functions
(vii) Properties of Curves
(viii) Related Rates of Change
(ix) Optimisation Problems

3.1 Common derivatives

The table below summarises the derivatives which should be known at Level Three.

y = f(x)	$\frac{dy}{dx} = f'(x)$
e^x	e^x
$\ln x$	$\frac{1}{x}$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$
$\sec x$	$\sec x \tan x$
$\operatorname{cosec} x$	$-\operatorname{cosec} x \cot x$
$\cot x$	$-\operatorname{cosec}^2 x$

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3.2 Limits

Some important properties of limits

1. $\lim_{x\to a}(f(x)+g(x)) = \lim_{x\to a} f(x) + \lim_{x\to a} g(x)$

   The limit of the sums equals the sum of the limits

2. $\lim_{x\to a}(f(x) - g(x)) = \lim_{x\to a} f(x) - \lim_{x\to a} g(x)$

   The limit of the differences equals the difference of the limits

3. $\lim_{x\to a}(f(x) \times g(x)) = \lim_{x\to a} f(x) \times \lim_{x\to a} g(x)$

   The limit of the product equals the product of the limits

4. $\lim_{x\to a}\left(\frac{f(x)}{g(x)}\right) = \frac{\lim_{x\to a} f(x)}{\lim_{x\to a} g(x)}$

   The limit of the quotients equals the quotient of the limits

The Sandwich Theorem

If $g(x) \le f(x) \le h(x)$ for all $x \ne c$ in some interval about c and $\lim_{x\to c} g(x) = \lim_{x\to c} h(x) = L$, then $\lim_{x\to c} f(x) = L$. ie:

if f(x) is sandwiched between two functions with limit L then the limit of f(x) is also L

Example One

Show $\lim_{x\to 0} x^2 \sin\left(\frac{1}{x}\right) = 0$

Now, $\lim_{x\to 0} x^2 \sin\left(\frac{1}{x}\right) = \lim_{x\to 0} x^2 \times \lim_{x\to 0} \sin\left(\frac{1}{x}\right)$

And, $-1 \le \lim_{x\to 0} \sin\left(\frac{1}{x}\right) \le 1$

$-1 \times \lim_{x\to 0} x^2 \le \lim_{x\to 0} x^2 \sin\left(\frac{1}{x}\right) \le +1 \times \lim_{x\to 0} x^2$

$-1 \times 0 \le \lim_{x\to 0} x^2 \sin\left(\frac{1}{x}\right) \le +1 \times 0$

$0 \le \lim_{x\to 0} x^2 \sin\left(\frac{1}{x}\right) \le +1$

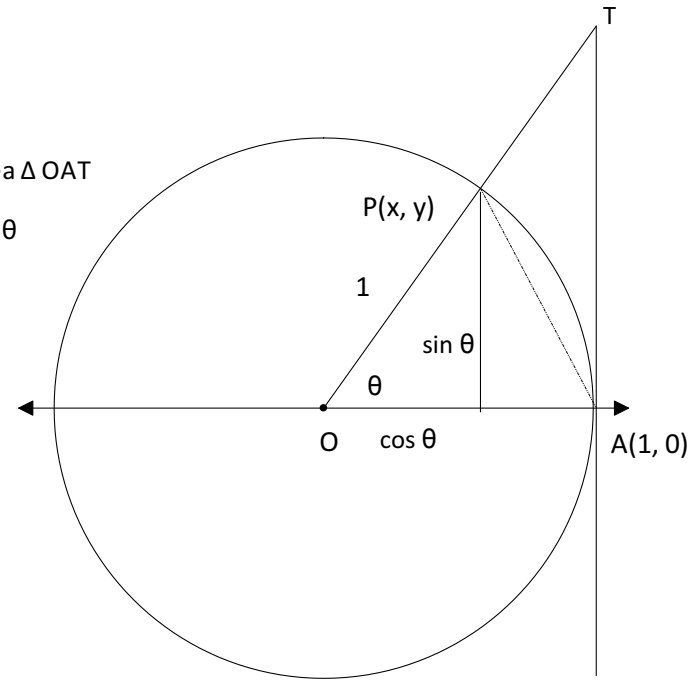
and hence $\lim_{x\to 0} x^2 \sin\left(\frac{1}{x}\right) = 0$

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Example Two

Show $\lim\limits_{x\to 0}\dfrac{\sin x}{x}=1$

Consider the diagram:



$\text{Area }\Delta AOP \le \text{Area sector } AOP \le \text{Area }\Delta OAT$

$\dfrac{1}{2}\times 1\times\sin\theta \le \dfrac{1}{2}\times 1^2\times\theta \le \dfrac{1}{2}\times 1\times\tan\theta$

$\sin\theta \le \theta \le \tan\theta$

$\sin\theta \le \theta \le \dfrac{\sin\theta}{\cos\theta}$

$1 \le \dfrac{\theta}{\sin\theta} \le \dfrac{1}{\cos\theta}$

$1 \ge \dfrac{\sin\theta}{\theta} \ge \cos\theta$

$\cos\theta \le \dfrac{\sin\theta}{\theta} \le 1$

$\lim\limits_{\theta\to 0}\cos\theta \le \lim\limits_{\theta\to 0}\dfrac{\sin\theta}{\theta} \le \lim\limits_{\theta\to 0}1$

$1 \le \lim\limits_{\theta\to 0}\dfrac{\sin\theta}{\theta} \le 1$

So, by the sandwich theorem $\lim\limits_{x\to 0}\dfrac{\sin x}{x}=1$

Example Three

Show $\lim\limits_{x\to 0}\dfrac{1-\cos x}{x}=0$

$\dfrac{1-\cos x}{x}$

$=\dfrac{1-\cos x}{x}\times\dfrac{1+\cos x}{1+\cos x}$

$=\dfrac{1-\cos^2 x}{x(1+\cos x)}$

$=\dfrac{\sin^2 x}{x(1+\cos x)}$

$\lim\limits_{x\to 0}\dfrac{1-\cos x}{x}$

$=\lim\limits_{x\to 0}\dfrac{\sin^2 x}{x(1+\cos x)}$

$=\lim\limits_{x\to 0}\dfrac{\sin x}{x}\times\dfrac{\sin x}{1+\cos x}$

$=\lim\limits_{x\to 0}\dfrac{\sin x}{x}\times\lim\limits_{x\to 0}\dfrac{\sin x}{1+\cos x}$

$=1\times\dfrac{0}{2}$

$=0$

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Example Four

$$\lim_{x\to 0}\left(\frac{\sin 3x}{\sin 5x}\right)$$

$$= \lim_{x\to 0}\left(\frac{3x}{5x}\times\frac{5x}{3x}\times\frac{\sin 3x}{\sin 5x}\right)$$

$$= \lim_{x\to 0}\left(\frac{3x}{5x}\times\frac{\sin 3x}{3x}\times\frac{5x}{\sin 5x}\right)$$

$$= \lim_{x\to 0}\left(\frac{3x}{5x}\right)\times\lim_{x\to 0}\left(\frac{\sin 3x}{3x}\right)\times\lim_{x\to 0}\left(\frac{5x}{\sin 5x}\right)$$

$$= \frac{3}{5}$$

Ex 3.2 Limits

1. Find $\lim_{x\to 0}\left(\frac{1-\cos 3x}{3x^2}\right)$ by first multiplying by $\frac{1+\cos 3x}{1+\cos 3x}$

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2. Find $\lim_{x \to 0} \left(\frac{\tan x}{x^2 + x} \right)$

3. Find $\lim_{x \to 25} \frac{\sqrt{x-9} - 4}{\sqrt{x} - 5}$

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3.3 First Principles

The derived function of f(x) is defined by $f'(x) = \lim_{h\to 0}\left(\frac{f(x+h)-f(x)}{h}\right)$

Examples:

1. $f(x) = x^2 - 1$

$$f'(x) = \lim_{h\to 0}\left(\frac{(x+h)^2 - 1 - x^2 + 1}{h}\right)$$

$$f'(x) = \lim_{h\to 0}\left(\frac{x^2 + 2xh + h^2 - 1 - x^2 + 1}{h}\right)$$

$$f'(x) = \lim_{h\to 0}\left(\frac{2xh + h^2}{h}\right)$$

$$f'(x) = \lim_{h\to 0}\left(\frac{h(2x+h)}{h}\right)$$

$$f'(x) = \lim_{h\to 0}(2x+h)$$

$$f'(x) = 2x$$

2. $f(x) = \sin x$

$$f'(x) = \lim_{h\to 0}\left(\frac{\sin(x+h) - \sin x}{h}\right)$$

$$f'(x) = \lim_{h\to 0}\left(\frac{2\cos\left(\frac{x+h+x}{2}\right)\sin\left(\frac{x+h-x}{2}\right)}{h}\right)$$

$$f'(x) = \lim_{h\to 0}\left(\frac{\cos\left(x+\frac{h}{2}\right)\sin\left(\frac{h}{2}\right)}{h/2}\right)$$

$$f'(x) = \lim_{h\to 0}\left(\frac{\sin\left(\frac{h}{2}\right)}{h/2}\right)\lim_{h\to 0}\cos\left(x+\frac{h}{2}\right)$$

Using the result of 3.2 Example 2 :

$$f'(x) = 1\times\cos x$$

$$f'(x) = \cos x$$

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Ex 3.3 First Principles

1. Show that the derivative of $\operatorname{cosec} x$ is $-\operatorname{cosec} x \cot x$ by first principles.

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2. Show that the derivative of $\frac{1}{\sqrt{x}}$ is $\frac{-1}{2\sqrt{x^3}}$ by first principles.

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3.4 The Chain, Product and Quotient Rules

The Chain Rule (for composite functions)

If $y = f(g(x))$ then $\left[f(g(x))\right]' = g'(x).f'(g(x))$

– the derivative of the "inside" × the derivative of the "whole lot"

OR

$$\frac{dy}{dx} = \frac{du}{dx} \times \frac{dy}{du}$$

– the "du's cancel just like fractions" to give $\frac{dy}{dx}$

Examples

1. $y = \ln(\sin x)$

Let $u = \sin x \Rightarrow \frac{du}{dx} = \cos x$

$y = \ln u \Rightarrow \frac{dy}{du} = \frac{1}{u}$

Using the chain rule:

$$\frac{dy}{dx} = \frac{1}{u} \times \cos x$$

$$\frac{dy}{dx} = \frac{1}{\sin x} \times \cos x$$

$$\frac{dy}{dx} = \cot x$$

2. It is not necessary to use the chain rule "step by step" as in the previous example but instead think of it as "the derivative of the 'bit inside' multiplied by the derivative of the 'whole lot':

$y = \tan^3 x$

Rewrite as:

$y = (\tan x)^3$

$$\frac{dy}{dx} = \left(\sec^2 x\right) \times 3 \times (\tan x)^2$$

"Inside" × "Whole lot"

$$\frac{dy}{dx} = 3\sec^2 x \tan^2 x$$

This is much more efficient when working through a long problem.

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The Product Rule

Prove the product rule $\left[f(x).g(x)\right]' = f'(x).g(x) + f(x).g'(x)$

Let $p(x) = f(x).g(x)$

By first principles:

$$p'(x) = \lim_{h\to 0}\left[\frac{f(x+h).g(x+h) - f(x).g(x)}{h}\right]$$

$$p'(x) = \lim_{h\to 0}\left[\frac{f(x+h).g(x+h) + f(x+h).g(x) - f(x+h).g(x) - f(x).g(x)}{h}\right]$$

$$p'(x) = \lim_{h\to 0}\left[\frac{f(x+h).(g(x+h) - g(x)) + g(x)(f(x+h) - f(x))}{h}\right]$$

$$p'(x) = \lim_{h\to 0}\left[\frac{f(x+h).(g(x+h) - g(x))}{h}\right] + \lim_{h\to 0}\left[\frac{g(x).(f(x+h) - f(x))}{h}\right]$$

$$p'(x) = \lim_{h\to 0}\left[f(x+h)\right].\lim_{h\to 0}\left[\frac{g(x+h) - g(x)}{h}\right] + \lim_{h\to 0} g(x).\lim_{h\to 0}\left[\frac{f(x+h) - f(x)}{h}\right]$$

$$p'(x) = f(x).g'(x) + g(x).f'(x)$$

OR

If $y = uv$ $\quad \frac{dy}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$

Examples

1. $y = e^x \sin x$

$$\frac{dy}{dx} = e^x \sin x + e^x \cos x$$

$$\frac{dy}{dx} = e^x\left(\sin x + \cos x\right)$$

2. $y = \ln x^2 (3x-1)^8$

$$\frac{dy}{dx} = 2x.\frac{1}{x^2}(3x-1)^8 + \left(\ln x^2\right)\times 3\times 8\times(3x-1)^7$$

$$\frac{dy}{dx} = \frac{2}{x}(3x-1)^8 + 24\ln x^2 (3x-1)^7$$

In this second example it is necessary to use the chain rule and the product rule.
Using the chain rule by sight makes the process much more efficient.

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The Quotient Rule

Prove the quotient rule $\left[\frac{f(x)}{g(x)}\right]' = \frac{g(x)f'(x) - f(x)g'(x)}{(g(x))^2}$

Let $p(x) = \frac{f(x)}{g(x)}$

By first principles:

$$p'(x) = \lim_{h\to 0}\left[\frac{\frac{f(x+h)}{g(x+h)} - \frac{f(x)}{g(x)}}{h}\right]$$

$$p'(x) = \lim_{h\to 0}\left[\frac{g(x)f(x+h) - f(x)g(x+h)}{g(x)g(x+h)h}\right]$$

$$p'(x) = \lim_{h\to 0}\left[\frac{g(x)f(x+h) + f(x)g(x) - f(x)g(x) - f(x)g(x+h)}{g(x)g(x+h)h}\right]$$

$$p'(x) = \lim_{h\to 0}\left[\frac{g(x)\left(f(x+h) - f(x)\right) - f(x)\left(g(x+h) - g(x)\right)}{g(x)g(x+h)h}\right]$$

$$p'(x) = \lim_{h\to 0}\left[\frac{g(x)\left(f(x+h) - f(x)\right)}{g(x)g(x+h)h}\right] - \lim_{h\to 0}\left[\frac{f(x)\left(g(x+h) - g(x)\right)}{g(x)g(x+h)h}\right]$$

$$p'(x) = \frac{\lim_{h\to 0}\left(\frac{f(x+h) - f(x)}{h}\right)}{\lim_{h\to 0}\left(g(x+h)\right)} - \frac{\lim_{h\to 0}\left(f(x)\left(\frac{g(x+h) - g(x)}{h}\right)\right)}{\lim_{h\to 0}\left(g(x)g(x+h)\right)}$$

$$p'(x) = \frac{f'(x)}{g(x)} - \frac{f(x)g'(x)}{(g(x))^2}$$

$$p'(x) = \frac{g(x)f'(x) - f(x)g'(x)}{(g(x))^2}$$

Examples

1. $y = \frac{\tan 2x}{2x-1}$

$$\frac{dy}{dx} = \frac{2\sec^2 2x(2x-1) - 2\tan 2x}{(2x-1)^2}$$

2. $y = \cot x = \frac{\cos x}{\sin x}$

$$\frac{dy}{dx} = \frac{-\sin x.\sin x - \cos x.\cos x}{(\sin x)^2}$$

$$\frac{dy}{dx} = \frac{-\left(\sin^2 x + \cos^2 x\right)}{(\sin x)^2}$$

$$\frac{dy}{dx} = \frac{-1}{\sin^2 x} = -\operatorname{cosec}^2 x$$

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Ex 3.4 The Chain, Product and Quotient Rules

1. Differentiate $y = \left(1 - x^2\right)\ln\left(1 - x^2\right)$

2. Differentiate $y = \dfrac{\ln(\sin x)}{\sin x}$

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3.5 Implicit and Parametric Differentiation

Implicit Differentiation

A function expressed with y the subject is explicitly expressed in terms of y. If the y is "buried" in the function then it is expressed implicitly.

Examples of implicit expressions are:

$x = a^y$

$xy + x^2 - y = 0$

To differentiate an expression implicitly, use the chain rule:

Examples:

1. $x = \ln y$

   Differentiating with respect to x:

   $1 = \frac{dy}{dx} \times \frac{1}{y}$

   $\left(\frac{dy}{dx} \text{ is the derivative of the "bit inside" and } \frac{1}{y} \text{ the derivative of the "whole lot"} \right)$

   $\frac{dy}{dx} = y \quad \text{OR} \quad \frac{dy}{dx} = e^x$

2. $xy + y^2 + x^2 = 1$

   Use the product rule on the xy term and the chain rule on the y^2:

   $1.y + x.\frac{dy}{dx} + 2y\frac{dy}{dx} + 2x = 0$

   $\frac{dy}{dx}(x + 2y) = -(2x + y)$

   $\frac{dy}{dx} = \frac{-(2x + y)}{(x + 2y)}$

3. $y = a^x$

   $\ln y = x \ln a$

   $\frac{dy}{dx} \times \frac{1}{y} = \ln a$

   $\frac{dy}{dx} = y \ln a$

   $\frac{dy}{dx} = a^x \ln a$

Parametric Differentiation

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If function y = f(x) has parametric equations x = g(t) and y = h(t) then:

$$\frac{dy}{dx}=\frac{dy}{dt}\div\frac{dx}{dt} \quad \text{and} \quad \frac{d^2y}{dx^2}=\frac{d}{dt}\left(\frac{dy}{dx}\right)\div\frac{dx}{dt}$$

Examples:

1. Find the equation of the tangent to the ellipse defined by $x=2\cos\theta$, $y=3\sin\theta$ at $\theta=\frac{\pi}{4}$

$\frac{dy}{dt}=3\cos\theta$ $\qquad$ $\frac{dx}{dt}=-2\sin\theta$

$\frac{dy}{dx}=-\frac{3\cos\theta}{2\sin\theta}$

At $\theta=\frac{\pi}{4}$:

$\frac{dy}{dx}=-\frac{3\sqrt{2}/2}{2\sqrt{2}/2}$

$\frac{dy}{dx}=-\frac{3}{2}$

Equation of the tangent:

$y-3\sqrt{2}/2=-\frac{3}{2}\left(x-2\sqrt{2}/2\right)$

$2y-3\sqrt{2}=-3x+3\sqrt{2}$

$2y+3x-6\sqrt{2}=0$

2. Find $\frac{d^2y}{dx^2}$ for the curve defined by $x = \sin t$ and $y = \cos t$

$\frac{dy}{dt}=-\sin t$ $\qquad$ $\frac{dx}{dt}=\cos t$

$\frac{dy}{dx}=-\frac{\sin t}{\cos t}=-\tan t$

$\frac{d^2y}{dx^2}=-\sec^2 t\div\cos t$

$\frac{d^2y}{dx^2}=-\sec^3 t$

3. The tangent to the curve $x=2\sin t$, $y=2+\cos 2t$ at the point (2, 1) meets the x-axis at P and the y-axis at Q. Find the distance PQ in surd form.

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$\frac{dy}{dt} = -2\sin 2t$ $\frac{dx}{dt} = 2\cos t$

$\frac{dy}{dx} = \frac{-2\sin 2t}{2\cos t}$

At (2,1):

$2\sin t = 2$

$\sin t = 1$

$2 + \cos 2t = 1$

$\cos 2t = -1$

Hence $t = \frac{\pi}{2}$

$\frac{dy}{dx} = -\frac{\sin 2t}{\cos t}$

Substituting $t = \frac{\pi}{2}$ directly into this expression will result in $\frac{0}{0}$, but writing $\sin 2t = 2\sin t\cos t$:

$\frac{dy}{dx} = -\frac{2\sin t\cos t}{\cos t} = -2\sin t$

$t = \frac{\pi}{2}$

$\frac{dy}{dx} = -2$

Let the equation of the tangent be y = -2x +c

At (2,1):

$1 = -2\times 2 + c$

$c = 5$

Hence the equation of the tangent is $y = -2x + 5$

$x = 0 \Rightarrow y = 5$

$y = 0 \Rightarrow x = \frac{5}{2}$

$PQ = \sqrt{5^2 + \left(\frac{5}{2}\right)^2}$

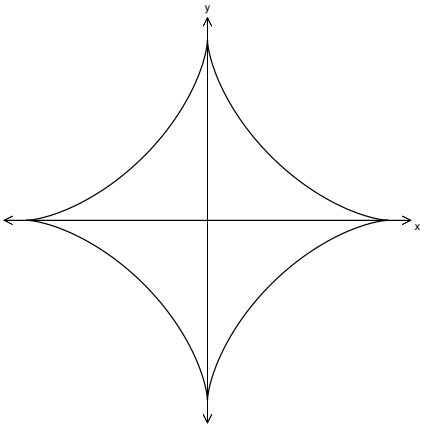
$PQ = \sqrt{5^2\left(1 + \frac{1}{4}\right)}$

$PQ = \frac{5}{2}\sqrt{5} = \frac{5\sqrt{5}}{2}$

Ex 3.5 Implicit and Parametric Differentiation

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1. The astroid has the parametric equations:



$x = 4\cos^3 t$ and $y = 4\sin^3 t$

Find the equation of the normal at

$t = \pi/4$

You must show all working.

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2. (a) If $x = f(t)$, $y = g(t)$ and $\frac{d^2y}{dx^2} = 0$,

(i) Prove that $\frac{dx}{dt}.\frac{d^2y}{dt^2} = \frac{dy}{dt}.\frac{d^2x}{dt^2}$

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(ii) Hence, or otherwise, show that the points of inflection of the curve defined parametrically by:

$$x = a\cos t + \frac{1}{2}b\cos 2t$$

$$y = a\sin t + \frac{1}{2}b\sin 2t$$

are given by $\cos t = -\frac{a^2 + 2b^2}{3ab}$, where a and b are constants.

You may assume that $\frac{d^3y}{dx^3} \neq 0$

(Scholarship 2005 Question Four (a))

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3. The nautilus is a marine creature that lives around coral reefs (Fig 5).

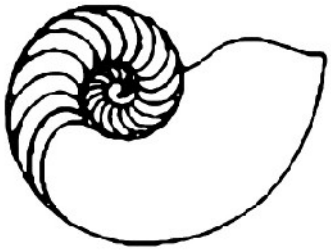


Fig. 5

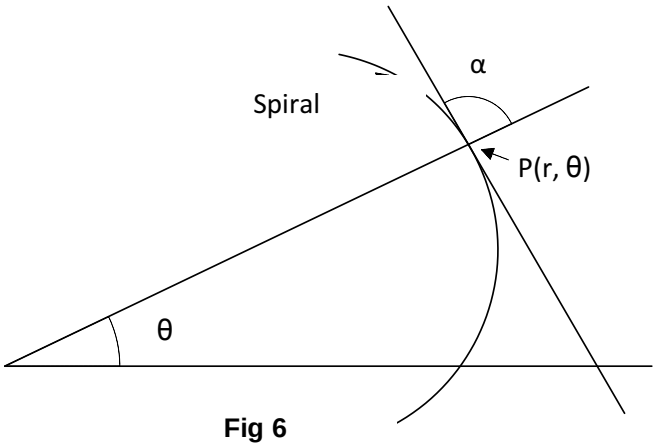


Fig 6

The mathematical model of a nautilus shell (Fig 6) is an equiangular spiral. Equiangular spirals have equations of the form $r = Ae^{k\theta}$, where k is a constant.

At every point P, the tangent to the curve makes the same angle, α, with the line OP from the point P to the origin (or pole) O.

The size of the angle α depends on the number k in this mathematical model where $r = Ae^{k\theta}$.

(a) Using the parametric equations for the Cartesian co-ordinates (x, y) of the point P in terms of θ, find $\frac{dy}{dx}$.

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(b) Hence, or otherwise, find the value of α in terms of k for this model.

(Scholarship 2004 Question Five)

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4. (a) If f(x), g(x) and h(x) are real functions of x, show that when $h(x)=\left[f(x)\right]^{g(x)}$

then $h'(x)=\left[f(x)\right]^{g(x)}\left[g'(x)\ln\left(f(x)\right)+g(x)\dfrac{f'(x)}{f(x)}\right]$

(b) Using the 'Super-Power rule' in part (a), or otherwise, find $\int\left(\ln x\right)^{x}p'(x)dx$ when $p(x)=x\ln\left(\ln x\right)$.

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(c) Find the condition for g(x) to be exactly divisible by x – 1, for

$g(x) = -(2x-a)^4 + bx + c$, where a, b and c are constants.

Hence, or otherwise, fully factorise g(x) and find ALL solutions of the equation

$-(2x-a)^4 + 8x - 7 = 0$.

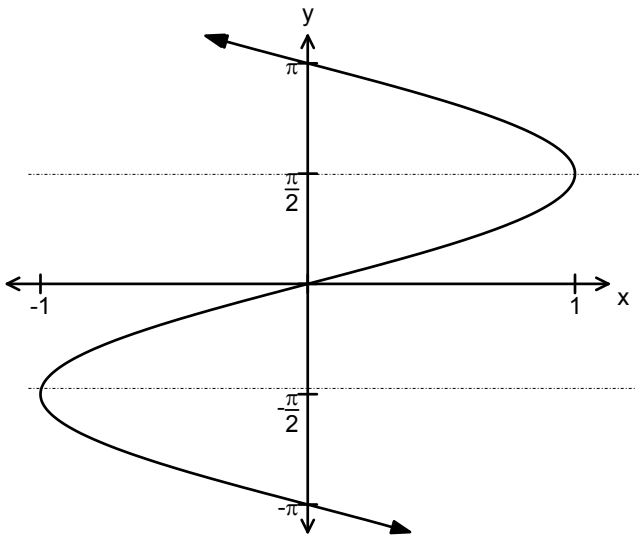
(Scholarship 2008 Question Four)

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3.6 Inverse Trigonometry Functions

The Inverse Trigonometric Functions

The inverse sine function $y = \sin^{-1}x$ is also written as arcsin(x), similarly for the other trigonometric functions.



The arcsin(x) is defined on the range $-\frac{\pi}{2} \geq y \leq \frac{\pi}{2}$ as outside those values the graph repeats and is no longer a function. To differentiate y = arcsin(x):

$$y = \sin^{-1} x$$
$$x = \sin y$$
$$\frac{dx}{dy} = \cos y$$
$$\frac{dx}{dy} = \sqrt{1 - \sin^2 y}$$
$$\frac{dx}{dy} = \sqrt{1 - x^2}$$
$$\frac{dy}{dx} = \frac{1}{\sqrt{1 - x^2}}$$

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The derivatives of the other inverse trig functions can similarly be found as:

Function	Range	Derivative
$\sin^{-1}x = \arcsin(x)$	$-\frac{\pi}{2} \geq y \leq \frac{\pi}{2}$	$\frac{1}{\sqrt{1-x^2}}$
$\cos^{-1}x = \arccos(x)$	$0 \geq y \leq \pi$	$\frac{-1}{\sqrt{1-x^2}}$
$\tan^{-1}x = \arctan(x)$	$-\frac{\pi}{2} < y < \frac{\pi}{2}$	$\frac{1}{1+x^2}$
$\sec^{-1}x = \operatorname{arcsec}(x)$	$0 \leq y < \pi/2$ or $\pi/2 < y \leq \pi$	$\frac{1}{x\sqrt{x^2-1}}$
$\operatorname{cosec}^{-1}x = \operatorname{arccosec}(x)$	$-\pi/2 \leq y < 0$ or $0 < y \leq \pi/2$	$\frac{-1}{x\sqrt{x^2-1}}$
$\cot^{-1}x = \operatorname{arccot}(x)$	$0 < y < \pi$	$\frac{-1}{1+x^2}$

These are particularly useful when integrating rational functions of the above type.

Examples:

1. $y = \cos^{-1} 5x$

$u = 5x \Rightarrow \frac{du}{dx} = 5$

$y = \cos^{-1} u$

$\frac{dy}{dx} = 5 \times \frac{-1}{\sqrt{1-u^2}}$

$\frac{dy}{dx} = \frac{-5}{\sqrt{1-25x^2}}$

2. $y = \sin^{-1}\left(1-x^2\right)$

$u = 1-x^2 \Rightarrow \frac{du}{dx} = -2x$

$y = \sin^{-1} u$

$\frac{dy}{du} = \frac{1}{\sqrt{1-u^2}}$

$\frac{dy}{dx} = \frac{-2x}{\sqrt{1-\left(1-2x^2+x^4\right)}}$

$\frac{dy}{dx} = \frac{-2x}{\sqrt{2x^2-x^4}}$

$\frac{dy}{dx} = \frac{-2x}{x\sqrt{2-x^2}}$

$\frac{dy}{dx} = \frac{-2}{\sqrt{2-x^2}}$

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Ex 3.6 Inverse Trigonometric Functions

1. Find the derivative of $y = x\arcsin x$.

2. Find the equation of the tangent to $y = \ln\left(x^2 + 4\right) - x\arctan\left(\frac{x}{2}\right)$ at x = 2.

Simplify the answer.

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3. Find the equation of the tangent to $y = \sin(\arccos x)$ at $x = \frac{1}{2}$.

Leave the answer in simplified surd form.

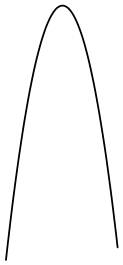
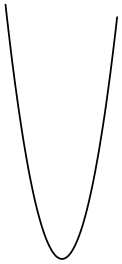
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3.7 Properties of curves

Stationary Points

At a stationary point $\frac{dy}{dx}=0$.

Stationary points are classified as maxima, minima and points of inflection:

Maximum	Minimum	Point of inflection
		
$\frac{d^2y}{dx^2}<0$	$\frac{d^2y}{dx^2}>0$	$\frac{d^2y}{dx^2}=0$

Points of Inflection and the Indeterminate case

A point of inflection is a point where the concavity of the curve changes.
This does **not** have to be a stationary point.

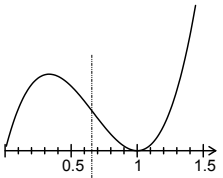
The equation of the cubic is $y=x^3-2x^2+x$

$\frac{dy}{dx}=3x^2-4x+1$

$\frac{d^2y}{dx^2}=6x-4$

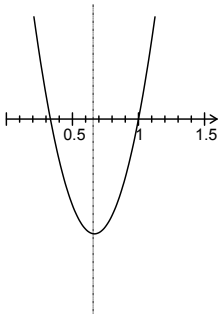
$6x-4=0$

$x=\frac{2}{3}$ is a point of inflection.



The graph is the derived function

$y'=3x^2-4x+1$



At $\frac{d^2y}{dx^2}=0$ (the minimum point of the derived function), the gradient of the original function is clearly not zero and the concavity changes.

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$\frac{d^2y}{dx^2}=0$ is a necessary but not sufficient condition for a point of inflection.

If $\frac{d^2y}{dx^2}=0$ **and** $\frac{dy}{dx}=0$, the stationary point may or may not be a point of inflection.

This is the indeterminate case.

Examples:

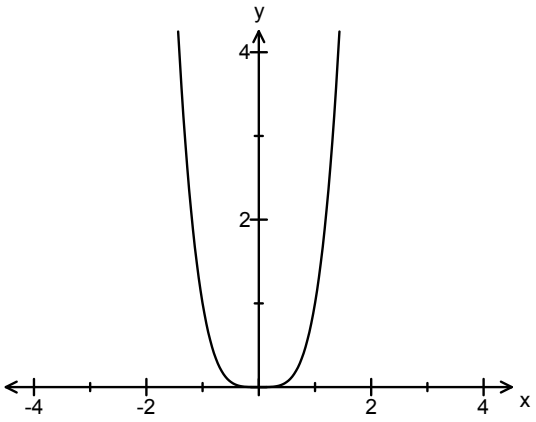
1.

$y=x^4$

$\frac{dy}{dx}=4x^3$ and $\frac{d^2y}{dx^2}=12x^2$

At $x=0$

$\frac{dy}{dx}=\frac{d^2y}{dx^2}=0$



But as the graph shows at x = 0 there is a minimum.

In the event that both first and second derivatives are zero it is necessary to test the first derivative either side of the stationary point:

$f'(^-1)=^-4$, $f'(1)=^+4$, hence there is a minimum at x = 0

2. Find all the stationary points and points of inflection of $f(x)=2x^3-3x^2-12x$ and determine their nature. **You must show all your working.**

$f'(x)=6x^2-6x-12$

$f'(x)=6\left(x^2-x-2\right)$

$f''(x)=12x-6$

$f'(x)=0$

$(x-2)(x+1)=0$

$x=2,-1$

$f''(2)=^+18$ ie: minimum at $(2, ^-28)$

$f''(^-1)=^-18$ ie: maximum $(^-1, ^+7)$

$f''(x)=0$

$12x-6=0$

$x=\frac{1}{2}$

There is a non stationary point of inflection at $\left(\frac{1}{2}, ^-6\frac{1}{2}\right)$

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Ex 3.7 Properties of Curves

1. Find all the stationary points and points of inflection of $y = \frac{x}{1+x^2}$ and determine their nature. **You must show all your working.**

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2. Find all the stationary points and points of inflection of $y = \sin x + \cos x,\ 0 \le x \le 2\pi$ and determine their nature. **You must show all your working.**

3. Find all the stationary points and points of inflection of $y = 2x + 10\operatorname{arccot} x$ where, $0 < \operatorname{arccot} x < \pi$ and determine their nature. **You must show all your working.**

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3.8 Related Rates of Change

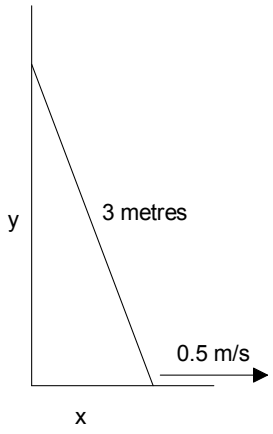
Related rates of change problems can simply be thought of as applications of the chain rule. Following are two different approaches to rate of change questions.

Examples:

Consider a simple example:

A 3-meter ladder is resting against a wall and its base slides away from the wall at 0.5 m/s.
How fast is the top of the ladder's sliding down the wall when the top is 2 meters from the ground?

$\frac{dx}{dt} = -0.5, \quad \text{find } \frac{dy}{dt}$



Link the two rates of change using the chain rule:

$\frac{dy}{dt} = \frac{dx}{dt} \times \frac{dy}{dx}$

Now, find a relationship between y and x (Pythagoras):

$x^2 + y^2 = 9$

$2x + 2y\frac{dy}{dx} = 0$

$\frac{dy}{dx} = -\frac{x}{y}$

By pythagoras when $y = 2, \; x = \sqrt{5}$

$\frac{dy}{dx} = -\frac{\sqrt{5}}{2}$

Using the chain rule:

$\frac{dy}{dx} = -0.5 \times -\frac{\sqrt{5}}{2} = \frac{\sqrt{5}}{4} \text{ m/s}$

Note:

$\frac{dy}{dt}$ was given a negative sign because y was decreasing and hence $\frac{dx}{dt}$ is positive because x is increasing.

The general process for this type of problem is:

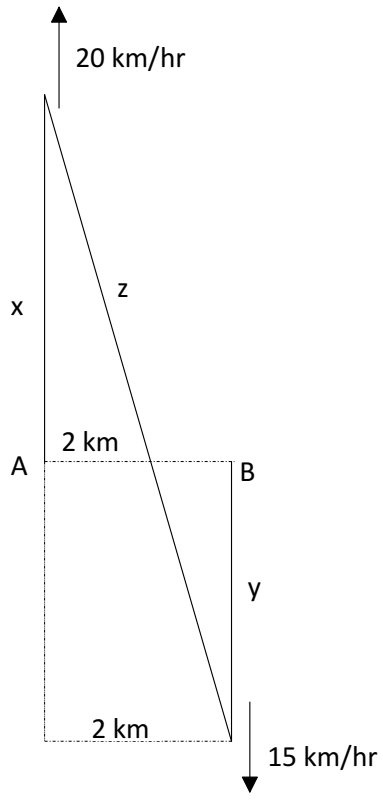
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1. Draw a diagram/define the variables.
2. Write down the given rate of change and an expression for that required.
3. Write down a chain rule expression linking the required rate of change to that given (this might require several rates of change).
4. Find a relationship between the variables of interest eg: use Pythagoras
5. Differentiate both sides of the relationship with respect to the independent variable (probably time), and use the chain rule expression to solve the problem.

The next example is slightly different and uses a different approach:

Two boats start 2km apart, with boat A due East of boat B. At 12 noon boat A heads due North at a speed of 20 km/hr and 10 minutes later boat B heads due South at a speed of 15 km/hr.
Find the rate at which the two boats are moving apart from each other at 12:30pm.

Find $\frac{dz}{dt}$ given that $\frac{dx}{dt} = 20$ and $\frac{dy}{dt} = 15$



Using Pythagoras:

$$z^2 = (x+y)^2 + 2^2 \qquad (1)$$

In this question, z is the dependent variable but there are independent variables, x & y.

Differentiating the expression above implicitly with respect to **t** gives:

$$2z\frac{dz}{dt} = 2\left(\frac{dx}{dt} + \frac{dy}{dt}\right)(x+y) \qquad (2)$$

Now after 30 minutes A has travelled 10 km and after 20 minutes B has travelled 5 km ie: x = 10 and y = 5:

Substituting in (1):

$$z^2 = (10+5)^2 + 2^2$$

$$z = \sqrt{229}$$

Substituting in (2):

$$2\sqrt{229}\frac{dz}{dt} = 2(20+15)(10+5)$$

$$\frac{dz}{dt} = 34.7 \text{ km/hr}$$

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In this example the expression for **z** was differentiated implicitly with respect to time, **t**.
It would have been possible to define a variable y representing the North-South distance between the boats, which is changing at a rate of $\frac{dy}{dt} = 35$ km/hr and then proceeding as in the previous example with $\frac{dz}{dt} = \frac{dy}{dt} \times \frac{dz}{dy}$ and using Pythagoras to establish the relationship between z & y.

Equally the first example could have been done using this second approach.

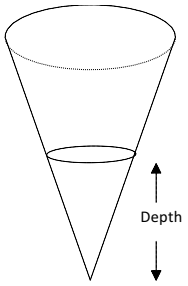
See Ex 3.9 Question 2c for a situation where the second approach works when the first does not.

Ex 3.8 Related Rates of Change

1. Water leaks out of a conical container at a rate of 6 cm^3 per sec. The height of the cone is 12 cm and its radius is 4cm.

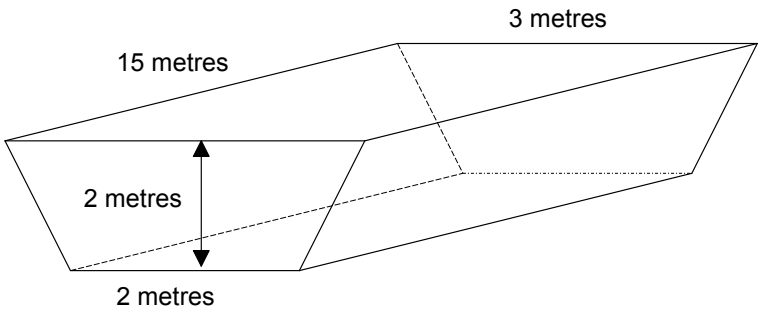
   Find the rate at which the depth of water changes when the depth of water is 3cm.

   Hint: the height of the cone at any point is always three times the radius



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2. A trough with trapezoidal ends is 15 metres long, 2 metres high, 3 metres wide at the top and 2 metres wide at the bottom. If water is being pumped in at 0.75 m^3/min, how fast is the depth increasing, when the water is 1.5 metres deep?



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3. One aeroplane is flying due North towards an airport at 200 km/hr and another is heading due East away from the airport at 300 km/hr.
Find the rate at which the aeroplanes are moving away from each other when the first plane is 50 km from the airport and the second 120 km away, to the nearest km/hr

4. Aeroplane A is flying directly over a town, heading due East at 300 km/hr.
At the same moment aeroplane B is 100 km away heading towards the town on a bearing of 330° and with a speed of 200 km/hr.

(a) Calculate the rate at which the distance between the planes is changing.

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(b) An hour later what is the rate at which the distance is changing?

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3.9 Optimisation Problems

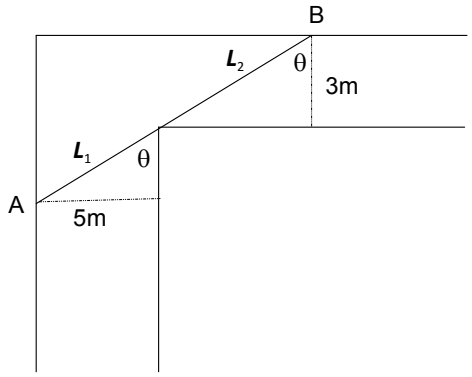
The approach to optimisation problems is the same as at Level Three, the context or skills required may be a little more complex.

1. Write down an expression for the variable to be optimised.
2. Use given information, if necessary, to reduce the problem to one independent variable.
3. Differentiate, set the derivative to zero and solve.
4. Verify maximum or minimum if required.

Examples:

1. A sheet of glass (which must be kept horizontal) is being carried down a corridor of width 5 metres. At the end of the corridor there is a right angled turn and the corridor narrows to 3 metres.
   What is the longest length of glass which can be manoeuvred around the turn in the corridor?

Let the length AB of the glass be L, the portion in the 5 metre corridor L_1 and in the other corridor L_2.



$L_1 = 5\operatorname{cosec}\theta$ and $L_2 = 3\sec\theta$

$L = 5\operatorname{cosec}\theta + 3\sec\theta$

$\dfrac{dL}{d\theta} = -5\operatorname{cosec}\theta\cot\theta + 3\sec\theta\tan\theta$

$\dfrac{dL}{d\theta} = 0$

$-5\operatorname{cosec}\theta\cot\theta + 3\sec\theta\tan\theta = 0$

$\dfrac{\sec\theta\tan\theta}{\operatorname{cosec}\theta\cot\theta} = \dfrac{5}{3}$

$\dfrac{\frac{1}{\cos\theta}\times\frac{\sin\theta}{\cos\theta}}{\frac{1}{\sin\theta}\times\frac{\cos\theta}{\sin\theta}} = \dfrac{5}{3}$

$\dfrac{\sin^3\theta}{\cos^3\theta} = \dfrac{5}{3}$

$\tan^3\theta = \dfrac{5}{3}$

$\theta = 49.9^\circ$

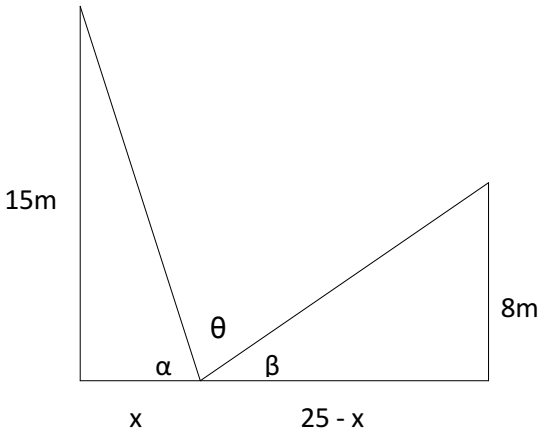
$L = \dfrac{5}{\sin 49.9} + \dfrac{3}{\cos 49.9}$

$L = 11.2\text{m}\ (1\text{ dp})$

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2. Two masts of height 8 metres and 15 metres are 25 metres apart on flat ground. A wire runs from the top of each mast and is fixed to the ground between the two masts.

How far from the taller mast should the wire be fixed to minimise the angle between the two of wires?



$$\tan\alpha = \frac{15}{x} \text{ and } \tan\beta = \frac{8}{25-x}$$

$$\alpha = \tan^{-1}\left(\frac{15}{x}\right) \text{ and } \beta = \tan^{-1}\left(\frac{8}{25-x}\right)$$

$$\theta = \pi - \tan^{-1}\left(\frac{15}{x}\right) - \tan^{-1}\left(\frac{8}{25-x}\right)$$

Note:

(1) the inverse tangents are in radians and hence it is necessary to use π

(2) when this expression is differentiated not only must we remember that the derivative of $\arctan x$ is $\frac{1}{1+x^2}$ but we must also use the chain rule.

$$\frac{d\theta}{dx} = -\frac{1}{1+\left(\frac{15}{x}\right)^2} \times -\frac{15}{x^2} - \frac{1}{1+\left(\frac{8}{25-x}\right)^2} \times \frac{8}{(25-x)^2}$$

$$\frac{d\theta}{dx} = \frac{15}{x^2\left[\frac{x^2+15^2}{x^2}\right]} - \frac{8}{(25-x)^2\left[\frac{(25-x)^2+8^2}{(25-x)^2}\right]}$$

$$\frac{d\theta}{dx} = \frac{15}{x^2+15^2} - \frac{8}{(25-x)^2+8^2}$$

$$\frac{d\theta}{dx} = \frac{15}{x^2+225} - \frac{8}{x^2-50x+689}$$

$$\frac{d\theta}{dx} = 0$$

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$$\frac{15}{x^2+225}=\frac{8}{x^2-50x+689}$$

$$15x^2-750x+10335=8x^2+1800$$

$$7x^2-750x+8535=0$$ Use the GC as it saves time.

$$x=94.2,\ 12.9$$

x must be less than 25, hence the wire is fixed 12.9 metres from the higher mast.

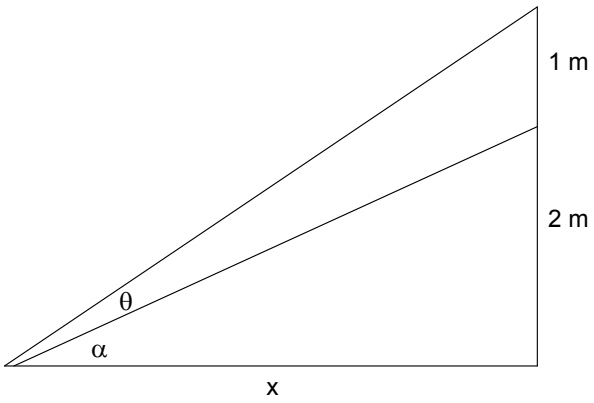
Ex 3.9 Optimisation Problems

1.

Ben has a wall mounted plasma TV set positioned 2 metres above eye level when seated.

The screen is 1 metre high.

How far from the screen should he sit to get the "best view"? (Maximum angle θ in the diagram).



You can assume that $\frac{d^2\theta}{dx^2}<0$

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2.

A square sheet of metal of side 2L centimetres has four pieces cut out symmetrically from the corners as shown in **Fig 1**.

The remaining piece of metal is made up of a square of side 2x centimetres and four isosceles triangles, as shown.

Assume L is a constant and L > 2x.

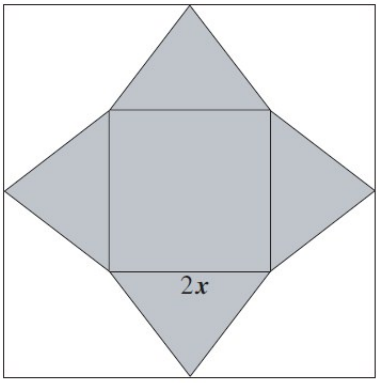


Fig. 1

(a) Find the value of x that gives a maximum volume of the square based pyramid that can be made from the shape in **Fig1**. You do not need to prove that it is a maximum.

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(b) A pyramid, cut from the square sheet as described above, has a fixed base of size 2k (ie: x = k).
This pyramid is turned upside down (see Fig 2), so that the square is horizontal, and a hole is drilled in the top, as shown.
Water is poured in through the hole so that the volume increases at a rate of $\frac{k^2}{3L}$ cm^3 per sec.

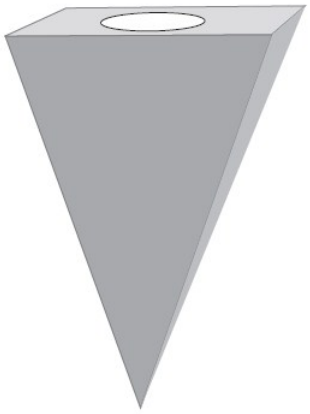


Fig. 2

For what values of h is the height, h centimetres, of the water in the container increasing at a rate $> \frac{L-2k}{h+l}$ cm per sec?

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(Scholarship 2004 Question One)

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3. (a) We are always looking for more efficient ways to store and stack things.
Cross sections tell us a lot about the stack and the space being filled. Figure 1 shows the cross section of a hexagonal stack.

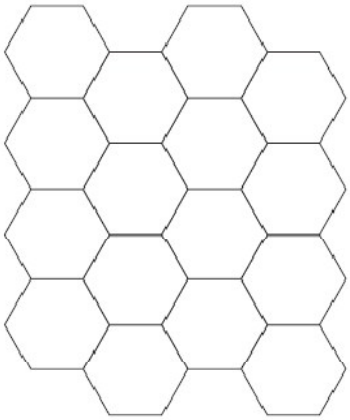


Figure 1

Show that the area of a regular hexagon with edge length s millimetres is $\frac{3\sqrt{3}}{2}s^2$ square millimetres.

Hence show that the total area of the hexagonal stack in Figure 1 is $24\sqrt{3}s^2$ square millimetres.

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(b) Although a single cell of bee's honeycomb has a hexagonal base, it is not a hexagonal prism.
The complete cell more commonly has the shape shown in Figure 2.

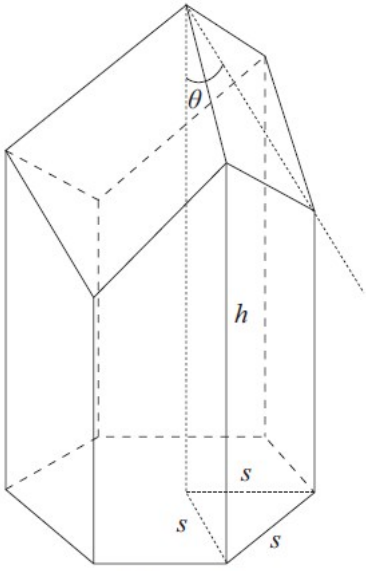


Figure 2

The surface area of the cell is given by:

$$A = 6hs + \frac{3}{2}s^2\left(\frac{-\cos\theta}{\sin\theta} + \frac{\sqrt{3}}{\sin\theta}\right)$$

where h, s, and θ are as shown in Figure 2.

Keeping h and s fixed, for what angle, θ, is the surface area a minimum?

You do not need to prove that it is a minimum.

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(c) Another cell, as described in part (b), has s not fixed, but increasing at a rate proportional to $\sin\theta$, where $0 < \theta < \frac{\pi}{2}$.

This rate is equal to $\sqrt{2}$ when $\theta = \frac{\pi}{4}$.

Keeping h fixed, at what rate is the surface area decreasing when θ is equal to the value found in part (b)? Give your answer in surd form in terms of h and s.

(Scholarship 2008 Question One)

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4. Integration

This chapter contains some revision of Level Three integration but also covers a great deal of work which either extends those ideas or covers ideas which go beyond the Level Three syllabus.

Topics included are:

(i) Trigonometric substitutions
(ii) Integration by parts
(iii) Harder trigonometric integration problems
(iv) The shell method for solids of revolution
(v) Arc length and surface of revolution
(vi) Solution of differential equations including substitution and reduction of second order equations to first order

4.1 Substitution

Standard Substitutions

Some substitution problems can be done by looking at them as utilising the chain rule in reverse:

Examples

1. $\int 2x\left(x^2-4\right)^6 dx$

In this very simple example the 2x at the front of the bracket is the derivative of the inside. Hence, reversing the chain rule:

$$\int 2x\left(x^2-4\right)^6 dx$$
$$= \frac{\left(x^2-4\right)^7}{7} + c$$

In the next example the numerical multiplier is adjusted, making 2x the derivative of x^2 :

2. $\int 6xe^{x^2} dx$
$= 3\int 2xe^{x^2} dx$
$= 3e^{x^2} + c$

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3. $\int \frac{4\cos 2x}{\sin 2x+1}dx$

$=2\int \frac{2\cos 2x}{\sin 2x+1}dx$

The derivative of $\sin 2x + 1$ is $2\cos 2x$.

Note the adjustment of the numerical multiplier.

Hence, reversing the chain rule:

$=2\ln|\sin 2x+1|+c$

If this property is obvious, it is much quicker to do the integration "on sight" as shown than use a formal substitution. However, in other cases, a formal substitution will be necessary.

4. $\int \frac{x}{1-x}dx$

Let $u=1-x$

$\frac{du}{dx}=-1$

$du=-dx$

$x=1-u$

$\int \frac{x}{1-x}dx$

$=-\int \frac{1-u}{u}du$

$=\int \frac{u-1}{u}du$

$=\int 1-\frac{1}{u}du$

$=u-\ln|u|+c$

$=1-x-\ln|1-x|+c$

5. $\int_{-1}^{8} \frac{x^2}{\sqrt{x+1}}dx$

Let $u=x+1$

$\frac{du}{dx}=1$

$du=dx$

Changing the limits of integration:

$x=8, u=9$ and $x=-1, u=0$

$\int_0^9 \frac{(u-1)^2}{\sqrt{u}}du$

$=\int_0^9 \frac{u^2-2u+1}{u^{1/2}}du$

$=\int_0^9 u^{3/2}-2u^{1/2}+u^{-1/2}du$

$=\left[\frac{2u^{5/2}}{5}-\frac{4u^{3/2}}{3}+2u^{1/2}\right]_0^9$

$=67\frac{1}{5}$

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Trigonometric Substitution

Integrals of the type $\int x\sqrt{1+x}dx$ can easily be integrated by using the substitution $u=\sqrt{x+1}$, but this type of substitution is often of no help if there is a quadratic under the square root sign.

Consider $\int\sqrt{9-x^2}dx$ which cannot be evaluated by such a substitution.

Trigonometric substitutions can sometimes be used to eliminate the root sign.
Three possible situations and substitutions are:

Integral	Substitution	Identity
$\int\sqrt{a^2-x^2}dx$	$x=a\sin\theta$	$\cos^2\theta=1-\sin^2\theta$
$\int\sqrt{a^2+x^2}dx$	$x=a\tan\theta$	$\sec^2\theta=1+\tan^2\theta$
$\int\sqrt{x^2-a^2}dx$	$x=a\sec\theta$	$\tan^2\theta=\sec^2\theta-1$

Examples:

1. $\int\frac{1}{\sqrt{1-x^2}}dx$

Let $x=\sin\theta$

$\frac{dx}{d\theta}=\cos\theta$

$dx=\cos\theta\,d\theta$

$\int\frac{1}{\sqrt{1-x^2}}dx$

$=\int\frac{1}{\sqrt{1-\sin^2\theta}}\cos\theta\,d\theta$

$=\int\frac{1}{\cos\theta}\cos\theta\,d\theta$

$=\theta+c$

$=\sin^{-1}x+c$

2. $\int_0^3\sqrt{9-x^2}$

$x=3\sin\theta$

$\frac{dx}{d\theta}=3\cos\theta$

$dx=3\cos\theta\,d\theta$

$x=3,\ \theta=\frac{\pi}{2}$ and $x=0,\ \theta=0$

$\int_0^{\frac{\pi}{2}}\sqrt{9-9\sin^2\theta}3\cos\theta d\theta$

$=9\int_0^{\frac{\pi}{2}}\sqrt{1-\sin^2\theta}\cos\theta d\theta$

$=9\int_0^{\frac{\pi}{2}}\cos^2\theta\,d\theta$

But $\cos^2\theta=\frac{1+\cos2\theta}{2}$

$=\frac{9}{2}\int_0^{\frac{\pi}{2}}1+\cos2\theta\,d\theta$

$=\frac{9}{2}\left[\theta+\frac{\sin2\theta}{2}\right]_0^{\frac{\pi}{2}}$

$=\frac{9}{2}\left(\frac{\pi}{2}+\frac{\sin\pi}{2}-0\right)=\frac{9\pi}{4}$

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3. Finally, a harder example:

$$\int \frac{dx}{\left(4+9x^2\right)^2}$$

Let $3x = 2\tan\theta$

$9x^2 = 4\tan^2\theta$ and $\frac{dx}{d\theta} = \frac{2}{3}\sec^2\theta$

$$\frac{2}{3}\int \frac{\sec^2\theta\, d\theta}{\left(4+4\tan^2\theta\right)^2}$$

$$= \frac{2}{3}\int \frac{\sec^2\theta\, d\theta}{16\sec^4\theta}$$

$$= \frac{1}{24}\int \frac{d\theta}{\sec^2\theta}$$

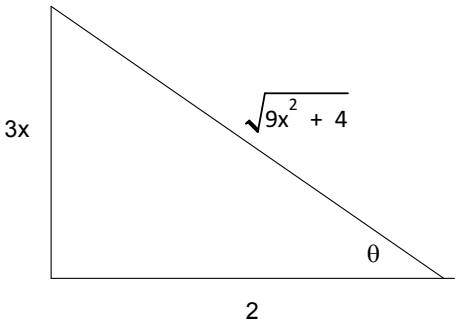
$$= \frac{1}{24}\int \cos^2\theta\, d\theta$$

$$= \frac{1}{48}\int 1+\cos 2\theta\, d\theta$$

$$= \frac{1}{48}\left(x+\frac{1}{2}\sin 2\theta\right)+c$$

$$= \frac{1}{48}\left(x+\sin\theta\cos\theta\right)+c$$

All that now remains is to get rid of the term in θ. To do this use Pythagoras in a right angled triangle:



From above:

$9x^2 = 4\tan^2\theta \Rightarrow \tan\theta = \frac{3x}{2}$

$\sin\theta = \frac{3x}{\sqrt{9x^2+4}}$ and $\cos\theta = \frac{2}{\sqrt{9x^2+4}}$

$\cos\theta\sin\theta = \frac{6x}{9x^2+4}$

Hence:

$$\int \frac{dx}{\left(4+9x^2\right)^2} = \frac{1}{48}\left(x+\frac{6x}{9x^2+4}\right)+c$$

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Ex 4.1 Substitutions

1. Evaluate the integrals:

(a) $\int x\sqrt{x+3}\,dx$

(b) $\int_5^8 \frac{x}{\sqrt{x-4}}\,dx$

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(c) $\int e^x \sqrt{e^x - 1}\, dx$

2. Use trigonometric substitutions to evaluate:

(a) $\int \frac{dx}{\sqrt{4 - x^2}}$

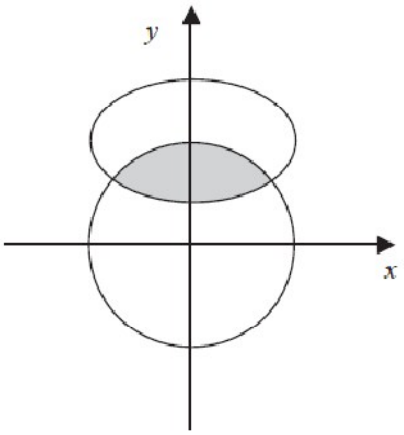
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(b) $\int \sqrt{9-4x^2}\, dx$

(c) $\int \frac{dx}{\left(4x^2-9\right)^{3/2}}$

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3. The diagram shows the circle $x^2 + y^2 = r^2$ and the ellipse $x^2 + 16(y - r)^2 = r^2$.



Find, in terms of a, the area between the ellipse and the circle (shaded).

You may use the substitution x = rsinu to find the integral $\int \sqrt{r^2 - x^2}\, dx$.

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(Scholarship 2005 Question Two (b))

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4.2 Integration by Parts

Derivation of the Formula

Starting with the differentiation product formula:

$\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}$, where u and v are functions of x.

As an integral this is written:

$uv = \int uv'dx + \int vu'dx$

$\int uv'dx = uv - \int vu'dx$

$\int udv = uv - \int vdu$

The integration by parts formula is:

$$\int uv'dx = uv - \int vu'dx \text{ OR } \int udv = uv - \int vdu$$

Examples

1. $\int 2x\cos 2xdx$

Let $u = 2x$ and $v' = \cos 2x \Rightarrow du = 2dx$ and $v = \frac{1}{2}\sin 2x$

$\int 2x\cos 2xdx = 2x.\frac{1}{2}.\sin 2x - \int \frac{1}{2}.\sin 2x.2dx$

$\int 2x\cos 2xdx = x\sin 2x + \frac{1}{2}\cos 2x + c$

As a general rule, v' should be the more complicated part that is easily integrated.

2. Consider $\int x^3 e^{x^2} dx$

xe^{x^2} is easily integrated. Let $u = x^2$ and $v' = xe^{x^2} \Rightarrow du = 2xdx$ and $v = \frac{1}{2}e^{x^2}$

$\int x^3 e^{x^2} dx = \frac{1}{2}x^2 e^{x^2} - \frac{1}{2}\int 2xe^{x^2} dx$

$\int x^3 e^{x^2} dx = \frac{1}{2}x^2 e^{x^2} - \frac{1}{2}e^{x^2} + c$

$\int x^3 e^{x^2} dx = \frac{1}{2}e^{x^2}\left(x^2 - 1\right) + c$

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3. $\int \ln x \, dx$

There is only one part, which cannot be easily integrated.

` So, let $v' = 1$ and $u = \ln x \Rightarrow du = \frac{dx}{x}$ and $v = x$

$$\int \ln x \, dx = x\ln x - \int x.\frac{1}{x} dx$$

$$\int \ln x \, dx = x\ln x - \int 1 dx$$

$$\int \ln x \, dx = x\ln x - x + c$$

$$\int \ln x \, dx = x(\ln x - 1) + c$$

4. $\int e^x \sin x \, dx$

$u = e^x$ and $v' = \sin x \Rightarrow du = e^x dx$ and $v = -\cos x$

$$\int e^x \sin x \, dx = -e^x \cos x + \int e^x \cos x \, dx$$

Repeating the process for $\int e^x \cos x \, dx$

$u = e^x$ and $v' = \cos x \Rightarrow du = e^x dx$ and $v = \sin x$

$$\int e^x \sin x \, dx = -e^x \cos x + e^x \sin x - \int e^x \sin x \, dx$$

Now we have $\int e^x \sin x \, dx$ on both sides

$$2\int e^x \sin x \, dx = -e^x \cos x + e^x \sin x$$

$$\int e^x \sin x \, dx = \frac{e^x}{2}(\sin x - \cos x) + c$$

5. Use integration by parts to show that $\int \sin^n x \, dx = \frac{\sin^{n-1} x \cos x}{n} + \frac{n-1}{n}\int \sin^{n-2} x \, dx$

and hence evaluate $\int \sin^4 x \, dx$.

$$\int \sin^n x \, dx = \int \sin x . \sin^{n-1} x \, dx$$

$v' = \sin x$ and $u = \sin^{n-1} x$ Þ $v = -\cos x$ and $u' = (n-1)\cos x . \sin^{n-2} x$

$$\int \sin^n x = -\cos x \sin^{n-1} x - \int -\cos x . (n-1) \cos x \sin^{n-2} x \, dx$$

$$\int \sin^n x = -\cos x \sin^{n-1} x + (n-1)\int \cos^2 x . \sin^{n-2} x \, dx$$

$$\int \sin^n x = -\cos x \sin^{n-1} x + (n-1)\int (1 - \sin^2 x) . \sin^{n-2} x \, dx$$

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$$\int \sin^n x = -\cos x \sin^{n-1} x + (n-1)\int \sin^{n-2} x\,dx - (n-1)\int \sin^2 x.\sin^{n-2} x\,dx$$

$$\int \sin^n x = -\cos x \sin^{n-1} x + (n-1)\int \sin^{n-2} x\,dx - n\int \sin^n x\,dx + \int \sin^n x\,dx$$

$$n\int \sin^n x = -\cos x \sin^{n-1} x + (n-1)\int \sin^{n-2} x\,dx$$

$$\int \sin^n x = -\frac{\cos x \sin^{n-1} x}{n} + \frac{(n-1)}{n}\int \sin^{n-2} x\,dx$$

Now, to evaluate $\int \sin^4 x\,dx$:

$$\int \sin^4 x\,dx = -\frac{\cos x \sin^3 x}{4} + \frac{3}{4}\int \sin^2 x\,dx$$

Using the expression again to evaluate $\int \sin^2 x\,dx$:

$$\int \sin^4 x\,dx = -\frac{\cos x \sin^3 x}{4} + \frac{3}{4}\left(-\frac{\cos x \sin x}{2} + \frac{1}{2}\int \sin^0 x\,dx\right)$$

$$\int \sin^4 x\,dx = -\frac{\cos x \sin^3 x}{4} + \frac{3}{4}\left(-\frac{\cos x \sin x}{2} + \frac{1}{2}x\right) + c$$

$$\int \sin^4 x\,dx = -\frac{\cos x \sin^3 x}{4} - \frac{3\cos x \sin x}{8} + \frac{3x}{8} + c$$

$$\int \sin^4 x\,dx = -\frac{\cos x \sin x\left(2\sin^2 x + 3\right)}{8} + \frac{3x}{8} + c$$

$$\int \sin^4 x\,dx = -\frac{\cos x \sin x\left(2\sin^2 x - 1 + 4\right)}{8} + \frac{3x}{8} + c$$

$$\int \sin^4 x\,dx = -\frac{4\cos x \sin x}{8} - \frac{\cos x \sin x\left(2\sin^2 x - 1\right)}{8} + \frac{3x}{8} + c$$

$$\int \sin^4 x\,dx = -\frac{\sin 2x}{4} - \frac{\sin 2x}{16}(-\cos 2x) + \frac{3x}{8} + c$$

$$\int \sin^4 x\,dx = -\frac{\sin 2x}{4} + \frac{\sin 4x}{32} + \frac{3x}{8} + c$$

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Ex 4.2 Integration by Parts

1. Evaluate the integrals:

(a) $\int x\cos x\,dx$

(b) $\int 2xe^{-x}\,dx$

(c) $\int \frac{\ln x}{x^2}\,dx$

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(d) $\int e^x \cos 2x \, dx$

2. Use integration by parts to show that $\int \ln^n x \, dx = x\ln^n x - n\int \ln^{n-1} x \, dx$ and hence evaluate $\int \ln^3 x \, dx$

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4.3 Further Trigonometric Integration Methods

The Level Three syllabus covers basic integration techniques using identities.
This section revises those techniques and then extends these techniques into more complex examples.

Examples

1. $\int \sin^2 2x\,dx$

Use the double angle formula $\cos 2x = 1 - 2\sin^2 x$.

In this case $\cos 4x = 1 - 2\sin^2 2x$

$$\frac{1}{2}\int 1 - \cos 4x\,dx$$

$$= \frac{1}{2}\left[x - \frac{1}{4}\sin 4x\right] + c$$

2. $\int \cos 4x \sin 2x\,dx$

Use $2\cos A \sin B = \sin(A+B) - \sin(A-B)$

$$\frac{1}{2}\int \sin(6x) - \sin(2x)\,dx$$

$$= \frac{1}{2}\left[-\frac{1}{6}\cos(6x) + \frac{1}{2}\cos(2x)\right] + c$$

3. $\int \sin^2 x \cos^2 x\,dx$

Use $\sin 2x = 2\sin x \cos x$

$$\int \frac{1}{4}\sin^2 2x\,dx$$

Using the result of Example 1:

$$= \frac{1}{8}\left[x - \frac{1}{4}\sin 4x\right] + c$$

The preceding examples used a straightforward application of a trig formula.

The following examples require some manipulation before using a trig identity and a substitution.

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4. $\int \sin^3 x\,dx$

Split off one sinx:

$= \int \sin^2 x.\sin x\,dx$

Rewrite $\sin^2 x$ as $1-\cos^2 x$:

$= \int \left(1-\cos^2 x\right)\sin x\,dx$

Let $u = \cos x$:

$\frac{du}{dx} = -\sin x$

$du = -\sin x\,dx$

$= -\int 1-u^2\,du$

$= \int u^2 - 1\,du$

$= \frac{u^3}{3} - u + c$

$= \frac{\cos^3 x}{3} - \cos x + c$

5. $\int \sin^4 x\cos^3 x\,dx$

The odd number of terms in cosx allows one to be split off leaving an even number, which can be replaced with sinx:

$= \int \sin^4 x\cos^2 x.\cos x\,dx$

$= \int \sin^4 x\left(1-\sin^2 x\right)\cos x\,dx$

Let $u = \sin x$

$\frac{du}{dx} = \cos x$

$du = \cos x\,dx$

$= \int u^4 - u^6\,du$

$= \frac{\sin^5 x}{5} - \frac{\sin^7 x}{7} + c$

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This technique will also work with secx and tanx, using the identity $\sec^2 x = \tan^2 x + 1$.

Now, whether secx or tanx is substituted depends on whether the power on the other function is odd or even ie: the the other must have an even power after the split of the derivative:

6. $\int \sec^3 x \tan^5 x \, dx$

$= \int \sec^2 x \tan^4 x \sec x \tan x \, dx$

$= \int \sec^2 x \left(\sec^2 x - 1\right)^2 \sec x \tan x \, dx$

Let $u = \sec x$

$\frac{du}{dx} = \sec x \tan x$

$du = \sec x \tan x \, dx$

$= \int u^2 \left(u^2 - 1\right)^2 du$

$= \int u^2 \left(u^4 - 2u^2 + 1\right) du$

$= \int u^6 - 2u^4 + u^2 \, du$

$= \frac{\sec^7 x}{7} - \frac{2\sec^5 x}{5} + \frac{\sec^3 x}{3} + c$

Note: tan is left with an even power after secx tanx is split off.

7. $\int \sec^4 x \tan^5 x \, dx$

$= \int \sec^2 x \tan^5 x \sec^2 x \, dx$

$= \int \left(1 + \tan^2 x\right) \tan^5 x \sec^2 x \, dx$

Let $u = \tan x$

$\frac{du}{dx} = \sec^2 x$

$du = \sec^2 x \, dx$

$= \int u^5 \left(1 + u^2\right) du$

$= \frac{\tan^6 x}{6} + \frac{\tan^8 x}{8} + c$

Note: sec is left with an even power.

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Finally, two examples using different techniques:

(i) The first example multiplies top and bottom by the same factor to make the integration easier.

(ii) The second uses integration by parts.

8. $\int \sec x \, dx$

Multiply by $\frac{\sec x + \tan x}{\sec x + \tan x}$:

$$= \int \sec x \times \frac{\sec x + \tan x}{\sec x + \tan x} dx$$
$$= \int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} dx$$
$$= \ln|\sec x + \tan x| = c$$

9. $\int \sec^3 x \, dx$

$$= \int \sec^2 x \sec x \, dx$$

Let $v' = \sec^2 x$ and $u = \sec x$

$v = \tan x$ and $u' = \sec x \tan x$

$$\int \sec^3 x \, dx = \sec x \tan x - \int \tan x \sec x \tan x \, dx$$
$$\int \sec^3 x \, dx = \sec x \tan x - \int \tan^2 x \sec x \, dx$$
$$\int \sec^3 x \, dx = \sec x \tan x - \int \left(\sec^2 x - 1\right) \sec x \, dx$$
$$\int \sec^3 x \, dx = \sec x \tan x - \int \left(\sec^3 x - \sec x\right) dx$$
$$\int \sec^3 x \, dx = \sec x \tan x - \int \sec^3 x dx + \int \sec x \, dx$$

$\sec^3 x$ has now appeared on the RHS as well, but moving it to the LHS:

$$2\int \sec^3 x \, dx = \sec x \tan x + \ln|\sec x + \tan x|$$
$$\int \sec^3 x \, dx = \frac{1}{2}\left(\sec x \tan x + \ln|\sec x + \tan x|\right)$$

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Ex 4.3 Trigonometric Integration Methods

1. Evaluate:

(a) $\int \sin^5 x \, dx$

(b) $\int \tan x \, dx$ by first writing $\tan x = \dfrac{\sin x}{\cos x}$

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(c) $\int \tan^3 x \, dx$

(d) $\int \sec^4 x \tan^2 x \, dx$

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(e) $\int \sin^5 x \cos^2 x \, dx$

2. (b) (i) Show that $\int_0^a f(x)dx = \int_0^a f(a-x)dx$

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(ii) Hence, or otherwise, calculate the value of the integral, showing clearly the steps in your working:

$$\int_0^{\frac{\pi}{2}} \frac{\sin^n x}{\sin^n x + \cos^n x} dx, \text{ for any integer } n.$$

(Scholarship 2004 Question Three (b))

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3. (b) Find $\frac{dy}{dx}$ when $y = x\sin nx + \frac{1}{n}\cos nx$, where n is a positive constant.

Hence, or otherwise, evaluate $I_n - I_{n-1}$, where

$$I_n = \int_0^{\pi}\left(\frac{1}{2}\pi - x\right)\sin\left(\left(n+\frac{1}{2}\right)x\right)\operatorname{cosec}\left(\frac{1}{2}x\right)dx \quad n \in \{1,2,3,.....\}$$

Hence find I_n in terms of n, given that $I_0 = 0$.

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4.4 Solids of Revolution: Disc and Shell Method

Disc Method

The standard Level Three technique is based on the summation of the volumes of a series of discs.

The volume of the solid of revolution formed by rotating f(x) about the x-axis from x = a to x = b is given by $V = \int_a^b \pi(f(x))^2\,dx$

If instead the rotation is about the y-axis the expression is $V = \int_c^d \pi(f(y))^2\,dy$

Examples:

1. Find the volume of the solid generated when $y = \tan x$ is rotated about the x-axis between $x = 0$ and $x = \frac{\pi}{4}$.

$$\text{Volume} = \pi\int_0^{\pi/4} \tan^2 x\,dx$$
$$\text{Volume} = \pi\int_0^{\pi/4} \sec^2 x + 1\,dx$$
$$\text{Volume} = \pi\left[\tan x + x\right]_0^{\pi/4}$$
$$\text{Volume} = \pi\left[1 + \frac{\pi}{4}\right] \text{ OR } \frac{\pi}{4}\left[\pi + 4\right]$$

2. Find the volume of the solid generated when the area bounded by $y = x^3$, the y-axis and y = 1 is rotated about the y-axis.

$$\text{Volume} = \pi\int_0^1 y^{2/3}\,dy$$
$$\text{Volume} = \pi\left[\frac{3y^{5/3}}{5}\right]_0^1$$
$$\text{Volume} = \frac{3\pi}{5}$$

3. Find the volume of the solid generated when the area bounded by $y = (x-1)^2$, the line x = 1 and y = 4 is rotated about the line x = 1.

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The formula for the volume applies to rotations about the x or y axes. In this case the line and function will have to be translated by one unit to the left so that the line x = 1 maps onto the y-axis. The function now becomes $y = x^2$ OR $x = \sqrt{y}$.

$$\text{Volume} = \pi\int_0^4 \left(\sqrt{y}\right)^2 dy$$

$$\text{Volume} = \pi\int_0^4 y\, dy$$

$$\text{Volume} = \pi\left[\frac{y^2}{2}\right]_0^4$$

$$\text{Volume} = 8\pi$$

Shell Method

Some volumes of solids of revolution cannot be calculated by the disc method or are easier calculated by another method.

The Shell Method is developed by rotating a vertical strip about the y-axis. The resultant shape is cylinder with a hole in it.

The volume of the solid of revolution formed by rotating about the y-axis from x = a to x = b is given by $V = \int_a^b 2\pi xy\, dx$

The volume of the solid of revolution formed by rotating about the x-axis from y = a to y = b is given by $V = \int_a^b 2\pi xy\, dy$

Examples

1. The area between the curve $y = x^3$, the x-axis and x = 1 is rotated about the y-axis. Find the volume of the resulting solid.

   The required volume consists of a series of discs (cylinders with a hole).
   If this is calculated by the disc method it must be calculated in two steps: (i) the volume between the curve and the y-axis and (ii) the volume of a cylinder minus the previously calculated volume:

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$$\text{Volume} = \text{Volume of the cylinder} - \text{Volume between the curve and the y axis}$$

$$\text{Volume} = \pi \times 1^2 \times 1 - \pi \int_0^1 (f(y))^2 \, dy$$

$$\text{Volume} = \pi - \pi \int_0^1 y^{2/3} dy$$

$$\text{Volume} = \pi - \pi \left[\frac{3}{5} y^{5/3} \right]_0^1$$

$$\text{Volume} = \pi - \pi \times \frac{3}{5} \times 1$$

$$\text{Volume} = \frac{2\pi}{5}$$

The shell method will do this very simply:

$$\text{Volume} = \int_0^1 2\pi x . x^3 \, dx$$

$$\text{Volume} = 2\pi \int_0^1 x^4 \, dx$$

$$\text{Volume} = 2\pi \left[\frac{x^5}{5} \right]_0^1$$

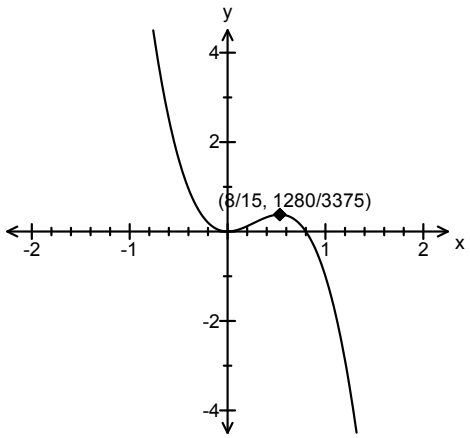
$$\text{Volume} = \frac{2\pi}{5}$$

In some situations it is either very difficult, or impossible, to use the disc method, whereas the shell method works perfectly well.

2. Find the volume of the solid of revolution generated by revolving the area bounded by the curve $y = 4x^2 - 5x^3$ and the x-axis about the y-axis.

To use the disc method it would be necessary to:
(i) rearrange $y = 4x^2 - 5x^3$ to make x the subject and
(ii) cope with the fact that for every y value between $\frac{1280}{3375}$ and zero there are three corresponding x values.

This is prohibitively difficult.



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Using the shell method:

$$\text{Volume} = \int_0^{4/5} 2\pi x\left(4x^2 - 5x^3\right)dx$$

$$\text{Volume} = 2\pi\int_0^{4/5} 4x^3 - 5x^4\,dx$$

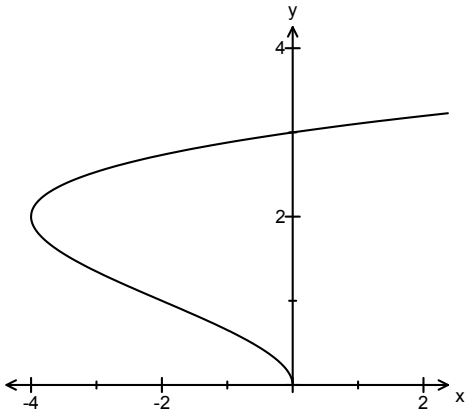
$$\text{Volume} = 2\pi\left[x^4 - x^5\right]_0^{4/5}$$

$$\text{Volume} = 2\pi\left(\left(\frac{4}{5}\right)^4 - \left(\frac{4}{5}\right)^5\right)$$

$$\text{Volume} = \frac{512\pi}{3125}$$

NB:

If, as in this graph, the area rotated is where x is negative then replace the x in $V = \int_a^b 2\pi xy\,dx$ with –x or the resultant volume will be negative.



Ex 4.4 Solids of Revolution

1. Use the disc method to evaluate the volume of the solid formed when the region bounded by the given curves is rotated about the x-axis:

   (a) $y = x^4$, the x-axis and $x = 1$

_____

_____

_____

_____

_____

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(b) $y = \sin x$, the x - axis, between $x = 0$ and $x = \pi$

2. Use the disc method to evaluate the volume of the solid formed when the area bounded by $y = \ln x$, the x-axis and $x = 2$ is rotated about $x = 1$

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3. Use the shell method to evaluate the volume of the solid generated when:

(a) The area bounded between $y = \ln x$, the x-axis and $x = e$ is rotated about the y-axis.

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(b) The area between $x = y^2 - 3y$ and the y-axis is rotated about the x-axis.

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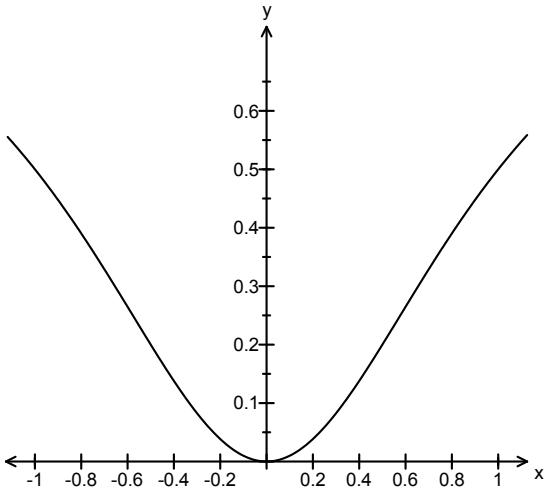
(c) The area between $y = \cos x$ and the x & y axes between $x = 0$ and $x = \frac{\pi}{2}$ is rotated about the line $x = \pi$

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4. (a) (i) The diagram below shows the function $y = \frac{x^2}{1+x^2}$ $-1 \le x \le 1$

The gradient at the point x = 1 is $\frac{1}{2}$

Hence show that there is a point with $\frac{1}{4} \le x \le \frac{1}{2}$, where the gradient is also $\frac{1}{2}$



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(ii) The shape of a wooden ornament is made by rotating the area between the graph of the function

$$y = \frac{(x-1)^2}{1+(x-1)^2} \quad 0 \le x \le 2$$

and the line $y = \frac{1}{2}$ through an angle 2π about the line $x = 1$.

Find the volume of this wooden ornament.

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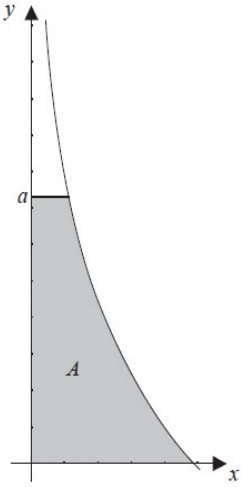
(Scholarship 2004 Question Three (a))

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5. (a) The shape of a rotor blade for a chemical mixing machine may be modelled by the function

$$y = k + \frac{1}{m}\ln\left(\frac{k}{x}\right), \quad \text{with } 0 < y \leq a$$

where m and k are positive constants.



(i) Find the value of a, in terms of m and p, when the blade's area, A is p% of the maximum possible area, $\frac{k}{m}e^{mk}$

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(ii) When the blade rotates in the chemical, it traces out a path equivalent to the rotation of the graph of the function $y = k + \frac{1}{m}\ln\left(\frac{k}{x}\right)$ through 2 π about the y-axis.

If V is the volume of revolution, then it is necessary for the stability of the blade that $\frac{V}{A} \leq e^{mk}$

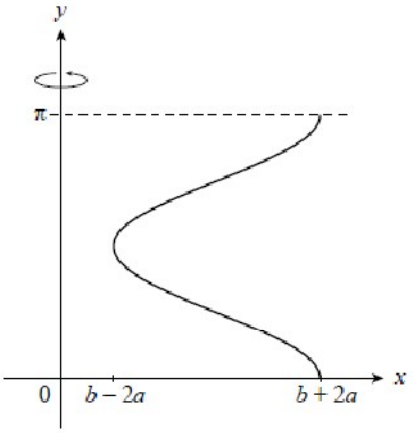
Show that, in this case the minimum value of a is $\frac{1}{m}\ln\left(\frac{\pi k}{2 - \pi k}\right)$

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(Scholarship 2005 Question Three (a))

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6. The diagram shows the graph of f(y) = x where $x = 2a\cos(2y) + b$, for a > 0, b > 0 and $0 \le y \le \pi$.



By rotating through 2π about the y-axis, we create a model of the shape of the **inner surface** of the vase.

Find an expression, in terms of a and b, for the volume contained within the inner surface of the vase.

(Scholarship 2006 Question One (c))

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4.5 Arc Length and Surface of Revolution

Arc Length

The arc length of any function f(x) is given by:

$$s=\int_m^n \sqrt{1+\left(f'(x)\right)^2}\,dx \quad \text{OR} \quad s=\int_a^b \sqrt{1+\left(f'(y)\right)^2}\,dy$$

If the function is in parametric form:

$$S=\int_{t_1}^{t_2} \sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2}\,dt$$

Example:

1. Find the arc length of $y=\dfrac{a\left(e^{x/a}+e^{-x/a}\right)}{2}$ from $x = a\ln a$ to $x = -a\ln a$

$$\frac{dy}{dx}=\frac{a\left(\frac{1}{a}e^{x/a}-\frac{1}{a}e^{-x/a}\right)}{2}$$

$$=\frac{\left(e^{x/a}-e^{-x/a}\right)}{2}$$

$$\left(\frac{dy}{dx}\right)^2=\frac{\left(e^{2x/a}-2e^{x/a}e^{-x/a}+e^{-2x/a}\right)}{4}$$

$$\left(\frac{dy}{dx}\right)^2=\frac{\left(e^{2x/a}-2+e^{-2x/a}\right)}{4}$$

$$S=\int_{-a\ln a}^{a\ln a}\sqrt{1+\frac{e^{2x/a}+e^{-2x/a}-2}{4}}\,dx$$

$$=\int_{-a\ln a}^{a\ln a}\sqrt{\frac{4+e^{2x/a}+e^{-2x/a}-2}{4}}\,dx$$

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$$= \frac{1}{2}\int_{-a\ln a}^{a\ln a} \sqrt{e^{2x/a} + e^{-2x/a} + 2}\, dx$$

$$= \frac{1}{2}\int_{-a\ln a}^{a\ln a} \sqrt{\left(e^{x/a} + e^{-x/a}\right)^2}\, dx$$

$$= \frac{1}{2}\int_{-a\ln a}^{a\ln a} \left(e^{x/a} + e^{-x/a}\right) dx$$

$$= \frac{1}{2}\left[ae^{x/a} - ae^{-x/a}\right]_{-a\ln a}^{a\ln a}$$

$$= \frac{a}{2}\left[\left(e^{a\ln a/a} - e^{-a\ln a/a}\right) - \left(e^{-a\ln a/a} - e^{a\ln a/a}\right)\right]$$

$$= \frac{a}{2}\left[\left(e^{\ln a} - e^{-\ln a}\right) - \left(e^{-\ln a} - e^{\ln a}\right)\right]$$

$$= \frac{a}{2}\left[\left(e^{\ln a} - e^{\ln(1/a)}\right) - \left(e^{\ln(1/a)} - e^{\ln a}\right)\right]$$

$$= \frac{a}{2}\left[2a - \frac{2}{a}\right]$$

$$= a^2 - 1$$

2. Find the arc length of y=ln(cosx) from x = 0 to x = $\frac{\pi}{4}$

$$\frac{dy}{dx} = \frac{1}{\cos x} \times -\sin x$$

$$\frac{dy}{dx} = -\tan x$$

$$s = \int_0^{\pi/4} \sqrt{1 + \tan^2 x}\, dx$$

$$s = \int_0^{\pi/4} \sqrt{\sec^2 x}\, dx$$

$$s = \int_0^{\pi/4} \sec x\, dx$$

Now using the result of Section 4.3 No 8:

$$s = \left[\ln\left|\sec x + \tan x\right|\right]_0^{\frac{\pi}{4}}$$

$$s = \ln\left|\sqrt{2} + 1\right|$$

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Surface of Revolution

If a function $y = f(x)$ has a continuous first derivative throughout the interval $a \le x \le b$, then the **area of the surface** generated by revolving the curve about the x-axis is given by:

$$S = \int_a^b 2\pi y\sqrt{1+\left(f'(x)\right)^2}\,dx$$

If the function is rotated about the y-axis then:

$$S = \int_a^b 2\pi x\sqrt{1+\left(f'(y)\right)^2}\,dy$$

If the function is in parametric form and rotated about the x-axis then:

$$S = \int_{t_1}^{t_2} 2\pi y\sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2}\,dt$$

Example

Find the surface area of the solid generated when $y = \sqrt{x}$ is rotated about the x-axis from x = 9 to x = 4. Leave the answer in simplified surd form.

$y = x^{1/2}$

$\frac{dy}{dx} = \frac{1}{2}x^{-1/2} = \frac{1}{2\sqrt{x}}$

$\left(\frac{dy}{dx}\right)^2 = \frac{1}{4x}$

$S = \int_4^9 2\pi x^{1/2}\sqrt{1+\frac{1}{4x}}\,dx$

Take $x^{1/2}$ inside the $\sqrt{\ }$ as x:

$S = \int_4^9 2\pi\sqrt{x+\frac{1}{4}}\,dx$

$S = \int_4^9 2\pi\sqrt{\frac{4x+1}{4}}\,dx$

$S = \pi\int_4^9 \sqrt{4x+1}\,dx$

$S = \pi\int_4^9 (4x+1)^{1/2}\,dx$

$S = \pi\left[\frac{1}{4}\times\frac{2}{3}(4x+1)^{3/2}\right]_4^9$

$S = \frac{\pi}{6}\left(37\sqrt{37}-17\sqrt{17}\right)$

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Ex 4.5 Arc Length and Surface of Revolution

1. Find the area of the resulting surface when the given function is rotated about the x-axis:

(a) $y^2 = 16x$, between $x = 0$ and $x = 4$

(b) $x = 3\cos^3\theta$, $y = 3\sin^3\theta$ from $\theta = 0$ to $\theta = \frac{\pi}{2}$

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2. Find the area of the resulting surface when the given function is rotated about the y-axis:

(a) $x = \sqrt{y}$ from $y = 0$ to $y = 2$

(b) $y = \ln x$ from $y = 0$ to $y = 1$

This is a very difficult question.

Hint: use two substitutions.

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3. Find the arc length of:

(a) $y = \sqrt{x^3}$ from $x = 0$ to $x = \frac{4}{3}$

(b) $x = e^t \sin t$, $y = e^t \cos t$ from $t = 0$ to $t = \pi$

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4.6 Differential equations

A differential equation is an equation involving one or more derivatives.
The solution of a differential equation is an equation.

The order of a differential equation is the power of the highest derivative:

$x\frac{dy}{dx}+xy=0$ is a first order differential equation

$x\frac{d^2y}{dx^2}+y=1$ is a second order differential equation

4.6.1 Separation of the variables

Separating the variables is a technique for solving differential equations. This involves rearranging the differential equation so that the dx is with the x terms and the dy with the y terms.

Examples:

1. Find the general solution of $\frac{dy}{dx}+xy=x$

$$\frac{dy}{dx}+xy=x$$
$$\frac{dy}{dx}=x(1-y)$$
$$\frac{dy}{1-y}=x\,dx$$
$$\int\frac{dy}{1-y}=\int x\,dx$$
$$-\ln|1-y|=\frac{x^2}{2}+c$$
$$\ln|1-y|=-\frac{x^2}{2}+c$$
$$|1-y|=e^{\left(-\frac{x^2}{2}+c\right)}$$

$1-y=\pm Ae^{-\frac{x^2}{2}}$, where $A=e^c$

$y=1+Ae^{-\frac{x^2}{2}}$

If A can take on + or - values then the ± can be dropped.

The general solution has unevaluated constant(s).

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2. Find the particular solution to $e^{x^2}\frac{dy}{dx}+xy=0$ if y = 1 when x = 0

$$\frac{e^{x^2}}{y}\frac{dy}{dx}=-x$$

$$\frac{dy}{y}=-\frac{x}{e^{x^2}}dx$$

$$\int\frac{dy}{y}=\int -xe^{-x^2}\,dx$$

$$\ln|y|=\frac{1}{2}e^{-x^2}+c$$

At $x=0$ $y=1$:

$$0=\frac{1}{2}+c$$

$$\ln|y|=\frac{1}{2}e^{-x^2}-\frac{1}{2}$$

3. In a chemical reaction a compound A is formed from a compound B.
The masses of compounds A and B at any time are x and y respectively and the sum of x and y at any time t is a constant amount p.
If the rate at which x is increasing at any time t is proportional to the product of the mass of A and B present at that time, form a differential equation to represent this situation.

At the start of the reaction compound A forms 25% of the total mass present and at t = 1 it forms 30% of the total mass.
Find the time at which compound A forms 75% of the total mass.

The differential equation:

$$\frac{dx}{dt}=kx(p-x)$$

Separating the variables:

$$\int\frac{dx}{x(p-x)}=\int k\,dt$$

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To integrate the LHS it is necessary to use partial fractions to express the rational expression as two fractions:

$$\frac{1}{x(p-x)}=\frac{A}{x}+\frac{B}{p-x}$$

$$\text{RHS}=\frac{A(p-x)+Bx}{x(p-x)}$$

$$\text{RHS}=\frac{(B-A)x+Ap}{x(p-x)}$$

Equating the numerators on the LHS & RHS:

$$Ap=1\Rightarrow A=\frac{1}{p}$$

$$B-A=0\Rightarrow A=B=\frac{1}{p}$$

Hence, $\frac{1}{x(p-x)}=\frac{1}{px}+\frac{1}{p(p-x)}$ and so:

$$\int\frac{1}{px}+\frac{1}{p(p-x)}dx=\int k dt$$

$$\frac{1}{p}\left(\ln|x|-\ln|p-x|\right)=kt+c$$

$$\ln\left|\frac{x}{p-x}\right|=k't+c'$$

$$\frac{x}{p-x}=e^{k't+c'}$$

$$\frac{x}{p-x}=Ae^{k't}$$

At $t=0,\ x=0.25p$

$$\frac{0.25p}{p-0.25p}=A$$

$$A=\frac{1}{3}$$

$$\frac{x}{p-x}=\frac{1}{3}e^{k't}$$

At $t=1,\ x=0.3p$

$$\frac{0.3p}{p-0.3p}=\frac{1}{3}e^{k'}$$

$$\frac{3}{7}=\frac{1}{3}e^{k'}$$

$$k'=\ln\left(\frac{9}{7}\right)$$

$$k'=0.2513\ (4dp)$$

$$\frac{x}{p-x}=\frac{1}{3}e^{0.2513t}$$

$$x=\frac{p}{3}e^{0.2513t}-\frac{x}{3}e^{0.2513t}$$

$$\left(3+e^{0.2513t}\right)x=pe^{0.2513t}$$

$$x=\frac{pe^{0.2513t}}{3+e^{0.2513t}}$$

Substituting $x = 0.75p$:

$$0.75p=\frac{pe^{0.2513t}}{3+e^{0.2513t}}$$

$$0.75=\frac{e^{0.2513t}}{3+e^{0.2513t}}$$

$$9=e^{0.2513t}$$

$$t=8.743$$

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Exercise 4.6.1 Differential Equations

1. Solve these differential equations by separating the variables:

(a) $(x-1)\frac{dy}{dx} = y+1$, where $(0,1)$ is a point on the curve

(b) $\tan x \frac{dy}{dx} = y^2 \sec^2 x$

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(c) $y^2e^{y^3}\frac{dy}{dx} = e^{2x}$, where (0, 0) is a point on the curve.

2. (a) $\frac{A}{x} + \frac{B}{P-x} = \frac{1}{x(P-x)}$ where x is a variable and P is a constant.

Find A and B in terms of P.

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(b) When a rumour about a teacher is started in a school of size P students, it spreads at a rate (in students per day) that is proportional to the product of the number of students who know the rumour, N, and those who do not.

Find an expression for the number of students, N, who know the rumour after t days.

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(c) For a particular rumour about a teacher, 0.5% of the students know the rumour initially. The principal will need to act to stop the rumour once more than half the school's students know it.

When $\frac{1}{5}$ of the students know the rumour, the number who know the rumour is increasing at a rate of 0.08P students per day.

How long will it be before the principal must act?

(Scholarship 2008 Question Three)

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4.6.2 Homogeneous Differential Equations

Solution of Level Three Differential Equations generally involves separation of the variables.
It is not always possible to separate the variables.
If an equation is homogeneous the substitution y = vx can be made and the variables can then be separated.

A differential equation f(x, y)= 0 is homogeneous if $f(zx,\ zy) = z^n f(x,\ y)$

For example, replacing x & y with zx & zy in the differential equation $xy\frac{dy}{dx}+x^2+y^2=0$ results in

$$z^2xy\frac{dy}{dx}+z^2x^2+z^2y^2=0 \text{ OR } z^2\left(xy\frac{dy}{dx}+x^2+y^2=0\right)$$

Examples

1. Solve $xy\frac{dy}{dx}+x^2+y^2=0$, $x>0$

Dividing by x^2:

$$\frac{y}{x}\frac{dy}{dx}+1+\frac{y^2}{x^2}=0$$

Substitute $v=\frac{y}{x}$.

Rearranging, y = vx, and using the product rule:

$$\frac{dy}{dx}=v+\frac{dv}{dx}x$$

Now substituting for $\frac{dy}{dx}$ and $\frac{y}{x}$ in the DE will allow the variables x & v to be separated:

$$v\left(v+\frac{dv}{dx}x\right)=-1-v^2$$

$$v+\frac{dv}{dx}x=-\frac{1+v^2}{v}$$

$$\frac{dv}{dx}x=-\frac{1+v^2}{v}-v$$

$$\frac{dv}{dx}x=-\left(\frac{2v^2+1}{v}\right)$$

$$\int\frac{v\,dv}{2v^2+1}=-\int\frac{dx}{x}$$

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$$\frac{1}{4}\left(\ln\left|2v^2+1\right|\right) = -\ln|kx|$$

$\ln\left(2v^2+1\right) = -4\ln kx$ - the $|\ |$ signs can be removed as x and $2v^2+1>0$

$$\ln\left(2v^2+1\right) = \ln\left(\frac{A}{x^4}\right)$$

$$2v^2+1 = \frac{A}{x^4}$$

$$v = \pm\sqrt{\frac{A-x^4}{2x^4}}$$

Substituting back:

$$\frac{y}{x} = \pm\sqrt{\frac{A-x^4}{2x^4}}$$

$$y = \pm\sqrt{\frac{A-x^4}{2x^2}}$$

2. $x\frac{dy}{dx} - x - y = 0$

$$\frac{dy}{dx} = \frac{x+y}{x}$$

$$\frac{dy}{dx} = 1+\frac{y}{x}$$

Let $y = vx$

$$\frac{dy}{dx} = v + x\frac{dv}{dx}$$

$$v + x\frac{dv}{dx} = 1+v$$

$$x\frac{dv}{dx} = 1$$

$$\int dv = \int \frac{dx}{x}$$

$$v = \ln|x| + c$$

$$\frac{y}{x} = \ln|x| + c$$

$$y = x\ln|x| + cx \ OR \ y = x\ln|kx|$$

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Exercise 4.6.2 Homogeneous differential equations

1. Solve these differential equations using the substitution $z = \frac{y}{x}$:

(a) $x^3 \frac{dy}{dx} = x^2 y + y^3$

(b) $x^2 \frac{dy}{dx} = x^2 - y^2 + xy$, where $(1, 0)$ is a point on the curve

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(c) $x\frac{dy}{dx} - x\sec\left(\frac{y}{x}\right) - y = 0$, where (1, 0) is a point on the curve

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2. The family of curves y = f(x) has a gradient represented by $y\frac{dy}{dx} = 2y + x$.

The curves of this family are perpendicular to a second family y = g(x) at every point of the co-ordinate plane.

Find an implicit equation, in terms of y and x, for this second family of curves.

You may find the substitution $z = \frac{x}{y}$ useful.

You may also use the relationship $\frac{z+2}{z(z+1)} = \frac{2}{z} - \frac{1}{z+1}$.

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(Scholarship 2006 Question Four (b))

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4.6.3 Linear differential equations and integrating factors

A linear differential equation is of the form $\frac{dy}{dx}+P(x).y=Q(x)$, where P(x) and Q(x) are functions of x.

These can often be integrated by multiplying by an integrating factor of $\mu=e^{\int P(x)dx}$

Examples:

1. $\frac{dy}{dx}+2xy=x$

$\mu=e^{\int 2xdx}=e^{x^2}$

Multipling by μ:

$e^{x^2}\frac{dy}{dx}+2xe^{x^2}y=xe^{x^2}$

This approach relies on the fact that the LHS can be reduced to $\frac{d\left(ye^{x^2}\right)}{dx}$ by reversing the product rule.

If the RHS can be integrated the differential equation is then easily solved:

$\frac{d\left(ye^{x^2}\right)}{dx}=xe^{x^2}$

$ye^{x^2}=\int xe^{x^2}dx$

$ye^{x^2}=\frac{1}{2}e^{x^2}+c$

$y=\frac{1}{2}+ce^{-x^2}$

2. $\frac{dy}{dx}+\frac{y}{x}=\cos x$, when $x=\pi$, $y=0$

$\mu=e^{\int\frac{1}{x}dx}=e^{\ln x}=x$

$x\frac{dy}{dx}+x\frac{y}{x}=x\cos x$

$x\frac{dy}{dx}+y=x\cos x$

$\frac{d(xy)}{dx}=x\cos x$

$xy=\int x\cos xdx$

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Integrating the RHS by parts:

$v' = \cos x$ and $u = x$

$v = \sin x$ and $u' = 1$

$xy = x\sin x - \int \sin x\,dx$

$xy = x\sin x + \cos x + c$

$y = \sin x + \frac{\cos x}{x} + \frac{c}{x}$

$x = \pi,\ y = 0$

$0 = 0 - \frac{1}{\pi} + \frac{c}{\pi}$

$c = 1$

$y = \sin x + \frac{\cos x}{x} + \frac{1}{x}$

Exercise 4.6.3 Linear DE's and Integrating Factors

1. Solve $\frac{dy}{dx} + 3x^2y = x^2$

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2. Solve $x\frac{dy}{dx} + y = x$, when x = -1, y = $-2\frac{1}{2}$

3. Solve $\frac{dy}{dx} + y\tan x = \sec x$, when x = 0, y = 1

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4. Solve $\frac{dy}{dx} + 2xy = x^3$, when $x = 0$, $y = \frac{1}{2}$

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4.6.4 Reducing the order of second order Differential Equations

Some second order differential equations can be reduced to first order differential equations. Two situations are looked at here.

1. The dependent variable is not present in the differential equation.

Example

$$\frac{d^2y}{dx^2}+\frac{dy}{dx}=x$$

Let $w=\frac{dy}{dx}$:

$$\frac{dw}{dx}+w=x$$

$$\frac{dw}{dx}=x-w$$

We now have a first order differential equation where the variables cannot be separated. Using the technique of the previous section, let v = x – w

$$\frac{dv}{dx}=1-\frac{dw}{dx}$$

$$\frac{dw}{dx}=1-\frac{dv}{dx}$$

Substituting into the changed DE:

$$1-\frac{dv}{dx}=v$$

$$\frac{dv}{dx}=1-v$$

$$\int\frac{dv}{1-v}=\int dx$$

$$-\ln|1-v|=x+c$$

$$\ln|1-v|=-x+c$$

$$1-v=\pm Ae^{-x}$$

$$v=1\pm Ae^{-x}$$

$$x-w=1\pm Ae^{-x}$$

$$w=x-1\pm Ae^{-x}$$

$$\frac{dy}{dx}=x-1\pm Ae^{-x}$$

$$y=\frac{x^2}{2}-x\pm Ae^{-x}+c$$

We have played “foot loose and fancy free” with the constants, but as nothing has been evaluated it does not matter.

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2. The independent variable is not present.

The substitution $w = \frac{dy}{dx}$ is made as above but instead of $\frac{d^2y}{dx^2} = \frac{dw}{dx}$ it is replaced with $w\frac{dw}{dy}$.

Proof:

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right)$$

$$\frac{d^2y}{dx^2} = \frac{dw}{dx} = \frac{dw}{dy} \times \frac{dy}{dx} = \frac{dw}{dy} \times w$$

Example

Solve $y^3\frac{d^2y}{dx^2} - \left(\frac{dy}{dx}\right)^3 = 0$

$$y^3 w\frac{dw}{dy} - w^3 = 0$$

$$w = 0 \text{ OR } y^3\frac{dw}{dy} - w^2 = 0$$

$$w = 0 \Rightarrow y = c$$

$$y^3\frac{dw}{dy} - w^2 = 0$$

$$y^3\frac{dw}{dy} = w^2$$

$$\int w^{-2}\,dw = \int y^{-3}\,dy$$

$$-\frac{1}{w} = -\frac{1}{2y^2} + c$$

$$\frac{dx}{dy} = \frac{1}{2y^2} + c$$

$$x = -\frac{1}{2y} + cx + d$$

$$\frac{1}{2y} = (c-1)x + d$$

$$2y = \frac{1}{(c-1)x + d}$$

$$y = \frac{1}{2(c-1)x + 2d}$$

$$y = \frac{1}{ax + b}$$

In fact, for this to be true in the DE, $a = -2$ or 0.

But $a = 0$ yields $y = c$, which is already a solution.

So, $y = \frac{1}{b - 2x}$

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Ex 4.6.4 Reducing the Order of Differential Equations

1. Solve these second order differential equations by reducing their order:

(a) $\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^2=0$, when $x=0$ $y=0$ and $\frac{dy}{dx}=1$

(b) $\frac{d^2y}{dx^2}+y\frac{dy}{dx}=0$, when $x=0$, $y=0$ *and* $\frac{dy}{dx}=2$

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(c) $\frac{d^2y}{dx^2} - \frac{dy}{dx} = 1$, when $x = 0$, $y = 1$ and $\frac{dy}{dx} = 0$

2 Given the function $y = \sin(\ln x)$, $x > 0$, find the value of k when

$$x^2 \frac{d^2y}{dx^2} + x\frac{dy}{dx} + y = k.$$

Hence, or otherwise, find a solution of the differential equation

$$\frac{d\left(x^2 \frac{dy}{dx}\right)}{dx} = x\frac{dy}{dx} - y + 5$$

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(Scholarship 2006 Question Three (a))

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3. For a function y = f(x) find the value of the gradient $\frac{dy}{dx}$ at the point where y = 2, given that:

$$\frac{dx}{dy}.\frac{d^2y}{dx^2}=k\frac{dy}{dx}$$

where k is a constant and the gradient $\frac{dy}{dx}$, is 1 at the point (0, 1).

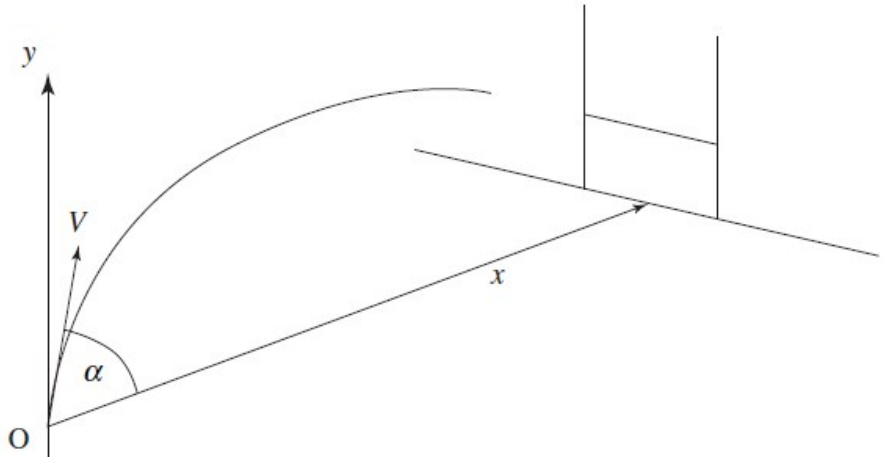
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(Scholarship 2005 Question Four (b))

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Ex 4.6.5 Mixed Integration Questions

1.



(a) During a penalty conversion attempt a rugby ball may be considered to be a particle that moves as a projectile. Hence you may ignore the effects of air resistance etc.
Initially, before the ball is kicked, the particle is considered to be at the origin, O, relative to the horizontal x-axis and a perpendicular y-axis (diagram above). Assume that the particle moves only in the plane of the x-y axes.

The acceleration of the particle after the ball has been kicked, measured relative to these two axes is given by:

$$\frac{d^2x}{dt^2} = 0 \quad \text{and} \quad \frac{d^2y}{dt^2} = -g \text{ , where g is the acceleration due to gravity.}$$

The particle leaves the ground with a speed of V m per sec at an angle of α to the x-axis.

By using integration, or otherwise, find the equation of the path of the particle.

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(b) On a particular conversion attempt the co-ordinates of the centre of the goalpost crossbar are (kh, h)

(i) Show that there are two possible paths by which the particle may hit the centre of the crossbar if $V^2 > gh\left(1+\sqrt{1+k^2}\right)$

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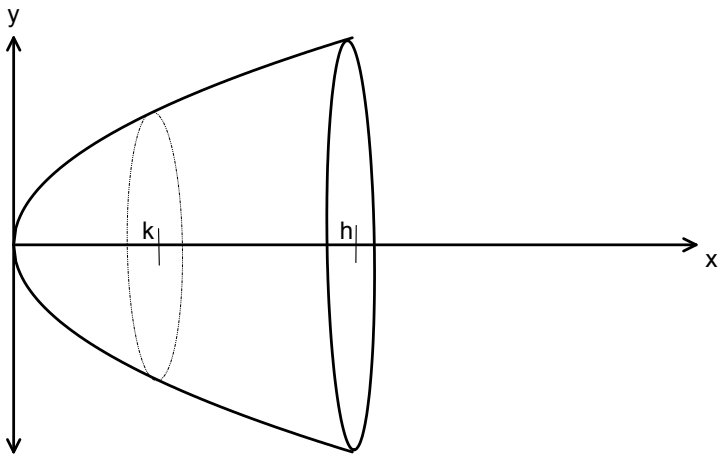
(ii) In this case, show that for these two possible angles α_1 and α_2,

$$\alpha_1 + \alpha_2 = \tan^{-1}(-k)$$

(Scholarship 2004 Question Six)

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2. An icemaker produces ice in the shape of paraboloids that may be modelled by rotating the graph of $y^2 = 4ax$ through 360° about the x-axis:



(a) Find, in terms of a and h, the volume of an ice paraboloid of length h.

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(b) The surface area of a solid of revolution such as that above is given by:

$$\text{Surface of revolution} = \int_a^b 2\pi y \sqrt{1+\left(\frac{dy}{dx}\right)^2}\, dx$$

Find, in terms of h, the area of the surface of revolution of an ice paraboloid of length h.

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(c) Each paraboloid piece of ice produced melts such that the rate of change of its volume is proportional to the area of the' surface of revolution' of the ice.

If the length h is changing at a rate of 0.26 mm/sec when h = 8a, find the rate at which the volume of a piece of ice is changing when h = 3a, in terms of a and π.

(Scholarship 2007 Question One)

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5. Conic Sections

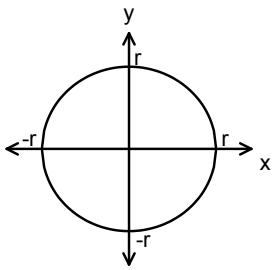
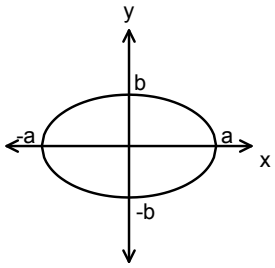
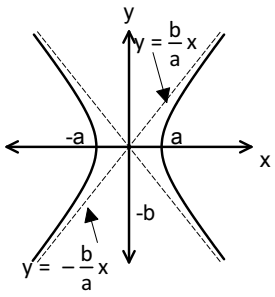
This chapter starts with a brief revision of Level Three Conic Sections and then looks at ideas which are useful at scholarship level:

(i) an alternative means of working out the equations of tangents

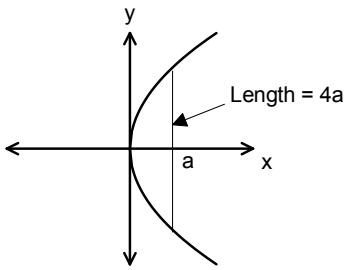
(ii) simplifying conic sections problems by translation

5.1 The Cartesian and Parametric forms of Conic Sections

Summary

Shape		Cartesian	Parametric
Circle		$x^2 + y^2 = r^2$	$x = r\cos\theta,\ y = r\sin\theta$
Ellipse		$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$	$x = a\cos\theta,\ y = b\sin\theta$
Hyperbola		$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$	$x = a\sec\theta,\ y = b\tan\theta$

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Parabola		$y^2 = 4ax$	$x = at^2,\ y = 2at$

All the basic shapes in the summary have centre on the origin (vertex in the case of the parabola).

If the centre moves to (f, g), the x and y in the equations are simply replaced with (x – f) and (y – g)

Example:

The ellipse with a = 2, b = 5, centre (1, -2) has equation $\frac{(x-1)^2}{4}+\frac{(y+2)^2}{25}=1$

To get the equation into the basic form it may be necessary to complete the square.

Example:

$4x^2-8x-y^2-6x-9=0$

$4\left(x^2-2x\right)-\left(y^2+6x\right)=9$

$4(x-1)^2-(y+3)^2=9+4-9$

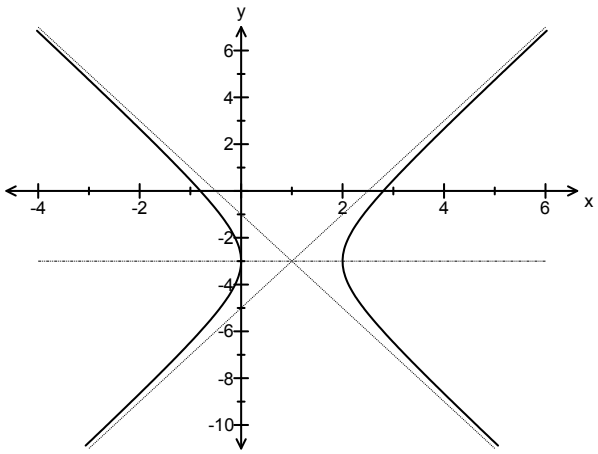
$4(x-1)^2-(y+3)^2=4$

$(x-1)^2-\frac{(y+3)^2}{4}=1$

So, a = 1 and b = 2

Centre at (1, -3)

Vertices (2, -3) and (0, -3)



Assymptotes:

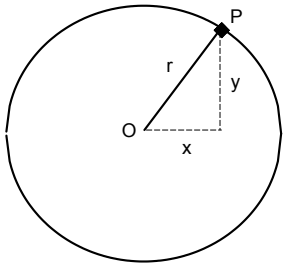
y + 3 = 2(x – 1) OR y = 2x – 5 and y + 3 = -2(x – 1) OR y= -2x - 1

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5.2 Focus, Eccentricity and Directrix

Scholarship students should have an understanding of the derivation of the equations and the properties of focus, eccentricity and directrix.

1. A circle is simply the locus of points P from a fixed point O such that OP = r, where O is the centre of the circle and r it's radius.

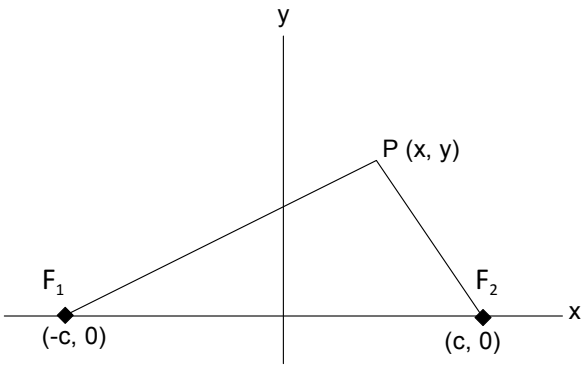


By Pythagoras:

$x^2 + y^2 = r^2$

2. An ellipse is the locus of point P such that the sum of the distances from two fixed points F_1 and F_2 (the foci) is a constant

ie: $PF_1 + PF_2$ = constant



Let $PF_1 + PF_2 = 2a$, where a is a constant.

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$$\sqrt{(x+c)^2+y^2}+\sqrt{(x-c)^2+y^2}=2a$$

$$\sqrt{(x+c)^2+y^2}=2a-\sqrt{(x-c)^2+y^2}$$

$$(x+c)^2+y^2=4a^2-4a\sqrt{(x-c)^2+y^2}+(x-c)^2+y^2$$

$$x^2+2cx+c^2+y^2=4a^2-4a\sqrt{(x-c)^2+y^2}+x^2-2cx+c^2+y^2$$

$$4a\sqrt{(x-c)^2+y^2}=4a^2-4cx$$

$$a\sqrt{(x-c)^2+y^2}=a^2-cx$$

$$a^2\left(x^2-2cx+c^2+y^2\right)=a^4-2a^2cx+c^2x^2$$

$$a^2x^2-2a^2cx+a^2c^2+a^2y^2=a^4-2a^2cx+c^2x^2$$

$$\left(a^2-c^2\right)x^2+a^2y^2=a^4-a^2c^2$$

$$\left(a^2-c^2\right)x^2+a^2y^2=a^2\left(a^2-c^2\right)$$

$$\frac{x^2}{a^2}+\frac{y^2}{a^2-c^2}=1$$

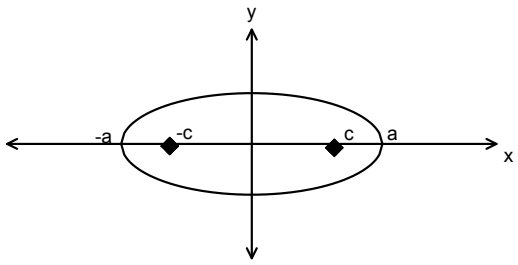
Now let $a^2-c^2=b^2$

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$$

This is the standard ellipse with centre (0, 0) semi – major axis of length a.

Now the eccentricity of the ellipse can be defined as the ratio $e=\frac{c}{a}$.

Sketching the standard ellipse:



It can be seen that a > c and therefore, in the case of an ellipse, 0 < e < 1

Using the definition of eccentricity viz: $e=\frac{c}{a}$ we can derive an expression for the eccentricity of an ellipse, in terms of **a** and **b**:

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$$e = \frac{c}{a}$$

$$e^2 = \frac{c^2}{a^2}$$

But $b^2 = a^2 - c^2$

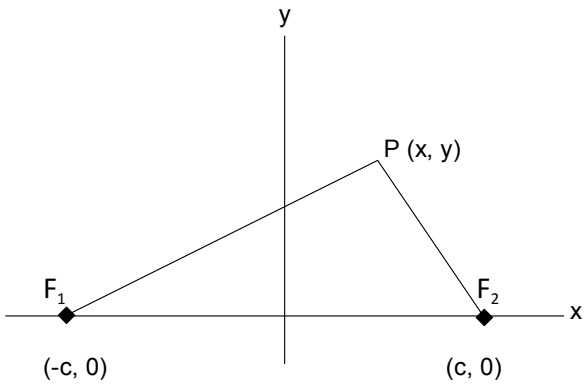
$$e^2 = \frac{a^2 - b^2}{a^2}$$

$$e^2 = 1 - \frac{b^2}{a^2}$$

$$e = \sqrt{1 - \frac{b^2}{a^2}}$$

3. A hyperbola is the locus of point P such that the difference between the distances from two fixed points F_1 and F_2 (the foci) is a constant

ie: $PF_1 - PF_2$ = constant



$$\sqrt{(x+c)^2 + y^2} - \sqrt{(x-c)^2 + y^2} = 2a$$

$$\sqrt{(x+c)^2 + y^2} = 2a + \sqrt{(x-c)^2 + y^2}$$

$$(x+c)^2 + y^2 = 4a^2 + 4a\sqrt{(x-c)^2 + y^2} + (x-c)^2 + y^2$$

$$x^2 + 2xc + c^2 + y^2 = 4a^2 + 4a\sqrt{(x-c)^2 + y^2} + x^2 - 2xc + c^2 + y^2$$

$$4a\sqrt{(x-c)^2 + y^2} = 4cx - 4a^2$$

$$a\sqrt{(x-c)^2 + y^2} = cx - a^2$$

$$a^2\left(x^2 - 2cx + c^2 + y^2\right) = a^4 - 2a^2cx + c^2x^2$$

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$$a^2x^2 - 2a^2cx + a^2c^2 + a^2y^2 = a^4 - 2a^2cx + c^2x^2$$

$$\left(c^2 - a^2\right)x^2 - a^2y^2 = a^2c^2 - a^4$$

$$\left(c^2 - a^2\right)x^2 + a^2y^2 = a^2\left(c^2 - a^2\right)$$

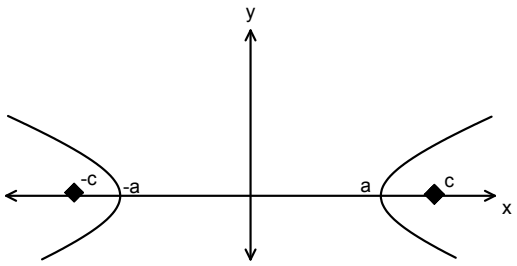
$$\frac{x^2}{a^2} - \frac{y^2}{c^2 - a^2} = 1$$

Now let $c^2 - a^2 = b^2$

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

This is the standard equation of a hyperbola with centre (0, 0), vertices at (a, 0) (-a, 0). Its asymptotes are $y = \pm\frac{b}{a}x$

Sketching the hyperbola:



It can be seen that c > a and hence e > 1

Using the definition of eccentricity viz: $e = \frac{c}{a}$ we can derive an expression for the eccentricity of a hyperbola, in terms of **a** and **b**:

$$e = \frac{c}{a}$$

$$e^2 = \frac{c^2}{a^2}$$

But $b^2 = c^2 - a^2$

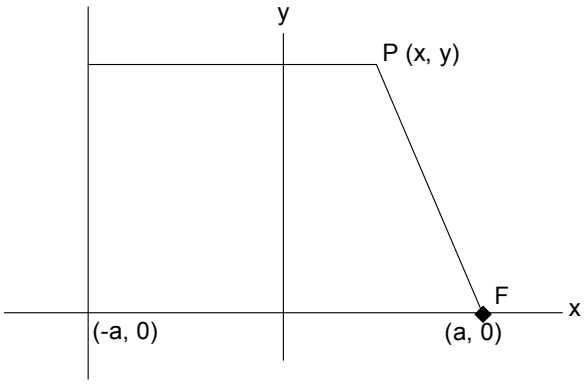
$$e^2 = \frac{a^2 + b^2}{a^2}$$

$$e^2 = 1 + \frac{b^2}{a^2}$$

$$e = \sqrt{1 + \frac{b^2}{a^2}}$$

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4. A parabola is the locus of point P such that the distance between a fixed line (the directrix) and a fixed point (the focus) is the same.



$$x + a = \sqrt{(x-a)^2 + y^2}$$

$$x^2 + 2ax + a^2 = x^2 - 2ax + a^2 + y^2$$

$$y^2 = 4ax$$

In this case a = c ie: e = 1

Summary of Eccentricity

Curve	Interval	e
Circle	e = 0	0
Ellipse	0 < e < 1	$\sqrt{1-\frac{b^2}{a^2}}$
Parabola	e = 1	1
Hyperbola	e > 1	$\sqrt{1+\frac{b^2}{a^2}}$

The eccentricity can be interpreted as the fraction of the distance along the semi-major axis at which the focus lies.

$$e = \frac{c}{a}$$

where c is the distance from the centre of the conic section to the focus.

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5.3 Tangents and Normals

Working out the equation of the tangent to a conic section is the same as for any curve:

1. Differentiate to find the gradient

2. Use the gradient and a point on the curve to work out the equation of the curve

Because of the form in which the Cartesian equation of a conic is usually expressed the first step will normally involve implicit differentiation. It might also, in some problems, be preferable to differentiate parametrically instead.

To work out the equation of the normal use the fact that if the gradient of the tangent is m_1 and the gradient of the normal is m_2, then:

$$\boxed{m_1 \times m_2 = -1 \quad \text{OR} \quad m_2 = -\frac{1}{m_1}}$$

Examples:

1. Find the equation of the tangent to the ellipse $\frac{(x-3)^2}{4} + y^2 = 1$ at $\left(2, \frac{\sqrt{3}}{2}\right)$

$$\frac{2(x-3)}{4} + 2y\frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{x-3}{4y}$$

At $\left(2, \frac{\sqrt{3}}{2}\right)$:

$$\frac{dy}{dx} = -\frac{(2-3)}{4 \times \sqrt{3}/2} = \frac{\sqrt{3}}{6}$$

Equation of tangent:

$$y - \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{6}(x-2)$$

$$6y - 3\sqrt{3} = \sqrt{3}x - 2\sqrt{3}$$

$$6y - \sqrt{3}x - \sqrt{3} = 0$$

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2. Find the equation of the normal to the parabola defined by $x = 4t^2 + 1$ and $y = 8t - 3$ when $t = 1$

$\frac{dx}{dt} = 8t$ and $\frac{dy}{dt} = 8$

$\frac{dy}{dx} = \frac{8}{8t} = \frac{1}{t}$

At $t = 1$:

$\frac{dy}{dx} = 1,\ x = 5,\ y = 5$

$y - 5 = 1(x - 5)$

$y = x$

A related skill which can also be useful when dealing with tangents is to remember that if a line is to be tangent to a curve then the discriminant of the quadratic formed by substituting the equation of the line into that of the conic section must be zero

5.4 An alternative approach to tangents

The Ellipse and Hyperbola

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$\frac{2x}{a^2} + \frac{2y}{b^2}\frac{dy}{dx} = 0$

$\frac{dy}{dx} = -\frac{b^2x}{a^2y}$

Equation of the tangent:

$y - y_1 = -\frac{b^2x_1}{a^2y_1}(x - x_1)$

$yy_1 - y_1^2 = -\frac{b^2}{a^2}\left(xx_1 - x_1^2\right)$

$\frac{yy_1}{b^2} - \frac{y_1^2}{b^2} = -\frac{xx_1}{a^2} + \frac{x_1^2}{a^2}$

$\frac{yy_1}{b^2} + \frac{xx_1}{a^2} = \frac{x_1^2}{a^2} + \frac{y_1^2}{b^2}$

$\frac{yy_1}{b^2} + \frac{xx_1}{a^2} = 1$ is the equation of the tangent

ie: the y^2 is replaced with yy_1

Similarly the equation of the tangent to the hyperbola is $\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$

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The Parabola

$y^2 = 4ax$

$2y\frac{dy}{dx} = 4a$

$\frac{dy}{dx} = \frac{2a}{y}$

Equation of the tangent:

$y - y_1 = \frac{2a}{y_1}(x - x_1)$

$yy_1 - y_1^2 = 2ax - 2ax_1$

$yy_1 - 4ax_1 = 2ax - 2ax_1$

$yy_1 = 2ax + 2ax_1$

$yy_1 = 2a(x + x_1)$

$yy_1 = 4a\left(\frac{x + x_1}{2}\right)$

In general, replace x^2 with xx_1 and x with $\frac{x + x_1}{2}$

Examples:

1. Find the equations of the tangents to the curves at the given points:

a) $\frac{x^2}{4} + \frac{y^2}{9} = 1$ at $\left(1, \frac{3\sqrt{3}}{2}\right)$

$\frac{xx_1}{4} + \frac{yy_1}{9} = 1$

$\frac{x}{4} + \frac{3\sqrt{3}y}{2 \times 9} = 1$

$3x + 2\sqrt{3}y = 12$

b) $x^2 + 2x + y^2 = 8$ at $\left(1, \sqrt{5}\right)$

$xx_1 + 2\left(\frac{x + x_1}{2}\right) + yy_1 = 8$

$x + x + 1 + \sqrt{5}y = 8$

$2x + \sqrt{5}y = 7$

2. The tangent to the hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$ cuts the y axis at -4.

Find the coordinates of the point where the tangent touches the hyperbola.

The equation of the tangent is:

$\frac{xx_1}{9} - \frac{yy_1}{4} = 1$

At $x = 0,\ y = -4$

$y_1 = 1$

Substituting in the equation of the hyperbola gives $x_1 = \frac{3\sqrt{5}}{2}$

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5.5 Translation of Conic Sections

Some conic sections problems are best solved by translating the conic section to provide a simpler problem. Sometimes it is best to translate the conic section so that the centre becomes the origin but in other cases a different transformation may better simplify the problem.

An important fact to remember is that if a curve is translated then the intercepts of tangents and normals will change but **not** the gradient

Examples:

1. Find the equation of the tangent to the ellipse $\frac{(x-5)^2}{16}+\frac{(y+4)^2}{12}=1$ at the point (7, -1).

   Translate the ellipse so its centre is on the origin.

   The transformed ellipse has equation $\frac{x^2}{16}+\frac{y^2}{12}=1$ and the point is now (2, 3)

   Using the approach of the previous section, the equation of the tangent to this transformed curve is now:

   $$\frac{2x}{16}+\frac{3y}{12}=1$$
   $$\frac{x}{8}+\frac{y}{4}=1$$
   $$x+2y=8$$

   Note: the line has gradient -0.5

   To get the required tangent to the original ellipse:

   Either, translate the line 5 to the right and 4 down:

   $$x-5+2(y+4)=8$$
   $$x+2y=5$$

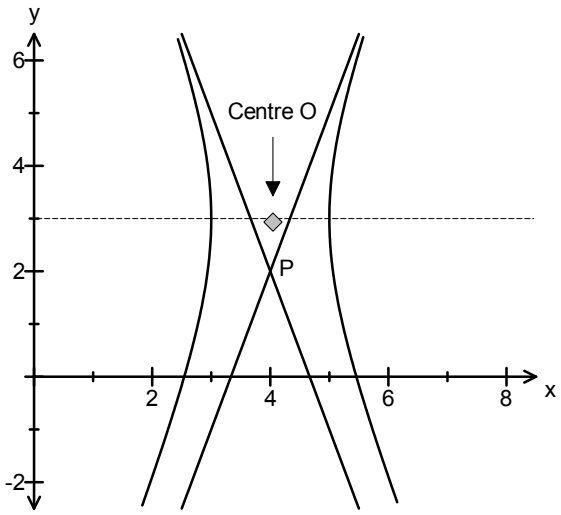
   Or, using the gradient of -0.5 and point (7, -1):

   $$y+1=-0.5(x-7)$$
   $$x+2y=5$$

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2. The hyperbola $(x-4)^2 - \frac{(y-3)^2}{8} = 1$ has tangents passing through point P(4, 2).

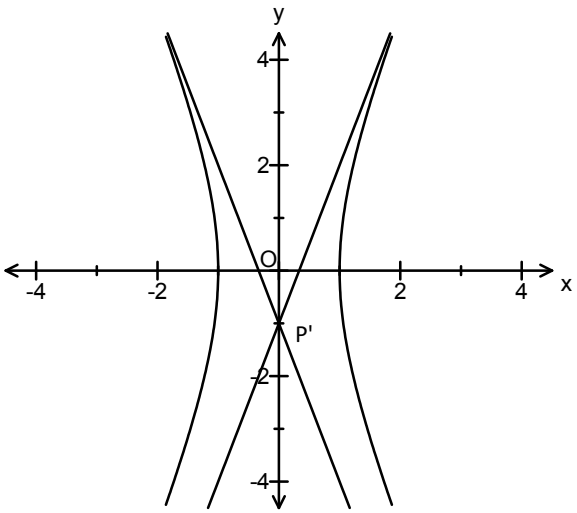
Find the equations of the tangents.



Translating the graph so that the centre is on the origin will also move point P onto the y axis, thus giving the y intercepts of the two tangents.

It, then only remains to find the gradients and then translate the graph back.

The equation of the translated graph is now $x^2 - \frac{y^2}{8} = 1$ and the point $P'(0, -1)$



Let the equation of the tangent to the transformed graph be y = mx – 1 and substitute into the equation of the graph:

$$x^2 - \frac{(mx-1)^2}{8} = 1$$

$$8x^2 - m^2x^2 + 2mx - 1 = 8$$

$$\left(8 - m^2\right)x^2 + 2mx - 9 = 0$$

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For the line to be a tangent the discriminant must be zero:

$$4m^2 - 4\left(8 - m^2\right) \times 9 = 0$$

$$4m^2 + 288 - 36m^2 = 0$$

$$288 - 32m^2 = 0$$

$$m^2 = 9$$

$$m = \pm 3$$

Hence the equations to the original curve are:

$$y - 2 = 3(x - 4)$$

$$y = 3x - 10$$

and

$$y - 2 = -3(x - 4)$$

$$y = -3x + 14$$

Note:

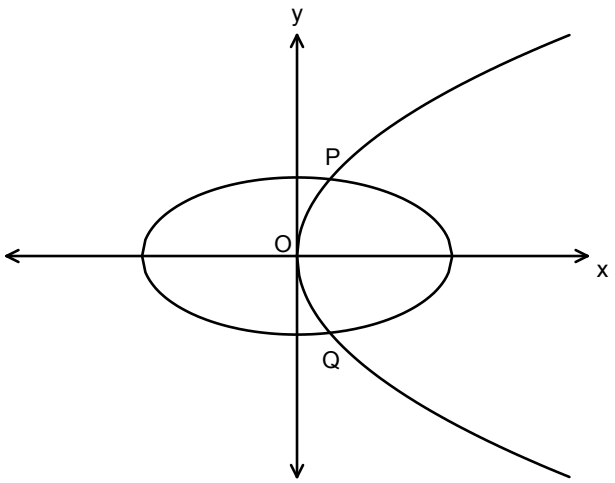
In this problem transforming the graph to give centre the origin worked but the crucial fact was that this transformation made the external point the y intercept for the tangents.

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Ex 5 Conic Sections

1.

The parabola given by $y^2 = 4ax$ and the ellipse given by $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, where $a > 0$ and $b > 0$, meet at points P and Q , as shown:



(a) The two curves intersect in such a way that the tangent at the parabola at P is perpendicular to the tangent of the ellipse at P.

(i) Show that $b^2 = 2a^2$

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(ii) Hence, find in terms of a, the distance of the point P from the origin O.

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(b) The tangent to the parabola at P meets the x-axis at M.

The tangent to the ellipse at P meets the x-axis at N.

Show that the length of $MN = 2\sqrt{2}a$

(Scholarship 2004 Question Two)

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2.

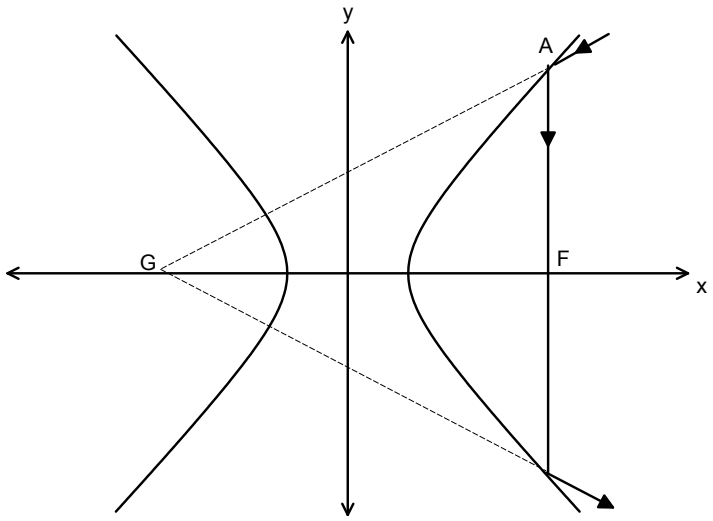
(a) Show that the equation $4x^2 - y^2 - 16hx + 2hy + 15h^2 - 4a^2 = 0$ where h and a are positive constants, represents a hyperbola.

If the tangent to this hyperbola at the point (p, q) is parallel to the straight line $y = \left(e^2 - 1\right)x$, where e is the eccentricity of the hyperbola, show that p – q = h.

(The eccentricity of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is given by $e^2 = 1 + \frac{b^2}{a^2}$)

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(b) A beam of light aimed at the focus G = (-ae, 0) of a hyperbolic reflector whose equation is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, will be reflected through the focus (ae, 0). The diagram shows the case where the light passes through the point a vertically above F.



For the hyperbola given in part (a) above, $4x^2 - y^2 - 16hx + 2hy + 15h^2 - 4a^2 = 0$, a similar beam of light aimed at its focus G_1 , is reflected along the corresponding line A_1F_1 , perpendicular to the x axis.

Find the equation of the line A_1G_1 that the light was initially travelling along.

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(c) Given that the equation of the tangent at the point (x_1, y_1) on the hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \text{ is } \frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$$

find the equation of this tangent in terms of a, b and its gradient m.

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(d) For the hyperbola $4x^2 - y^2 - 16hx + 2hy + 15h^2 - 4a^2 = 0$, a tangent with positive gradient passes through the point $\left(\frac{2a}{3} + 2h, h\right)$.

Show that the size of the angle, α, between the tangent and the asymptote with the positive gradient may be expressed as $\alpha = \tan^{-1}\left(\frac{2(41 - 15\sqrt{5})}{139}\right)$

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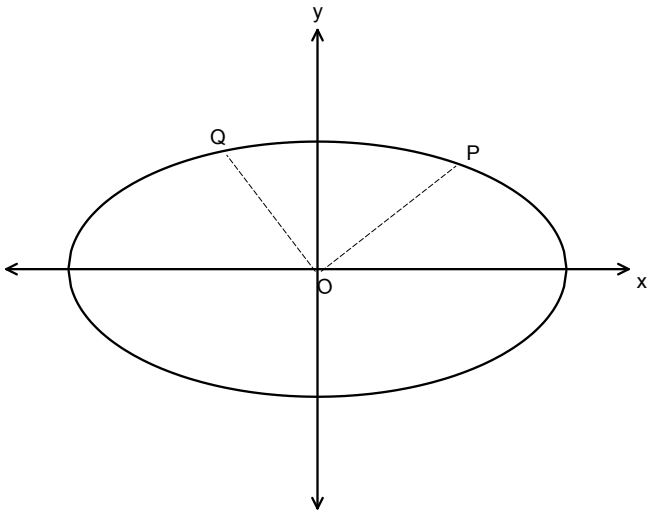
(Scholarship 2005 Question Six)

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3.

Two satellites, P and Q, are travelling anticlockwise along the same elliptical orbit around a planet centre Q, such that the lines OP and OQ are always perpendicular.

The elliptical orbit may be represented by $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, a > b, as shown in the diagram below, and the satellite P is represented by the point $(a\cos\theta, b\sin\theta)$.



(a) The satellite P shines a microwave beam in the directions perpendicular to its direction of motion.

Show that, when $\theta=\frac{\pi}{4}$, the beam cuts the vertical plane, represented by the y-axis, at a vertical distance $\frac{\sqrt{2}a^2}{2b}$ below the level of the satellite.

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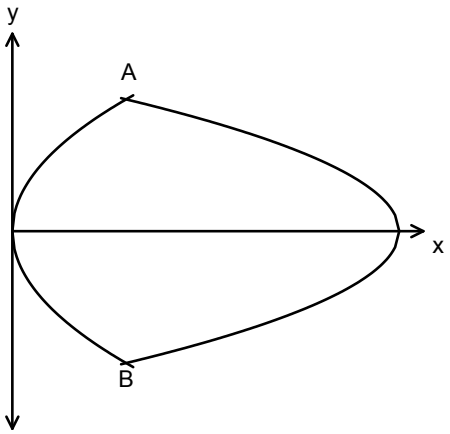
(b) Find the coordinates of Q in terms of θ

(Scholarship 2006 Question Five)

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4. A surfing company logo has the basic shape shown immediately below.
The two curves have parametric equations of the form:

Curve 1: $x = at^2 + b$, $y = 2at$ and Curve 2: $x = 3b - as^2$, $y = \sqrt{2}as$ respectively, where a, b are constants.



(a) Find, in terms of a and b, the co-ordinates (x, y) of the two points of intersection A and B, of the curves.

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(b) The chord AB divides the area enclosed by the curves in two parts.
Find the ratio of the two areas.

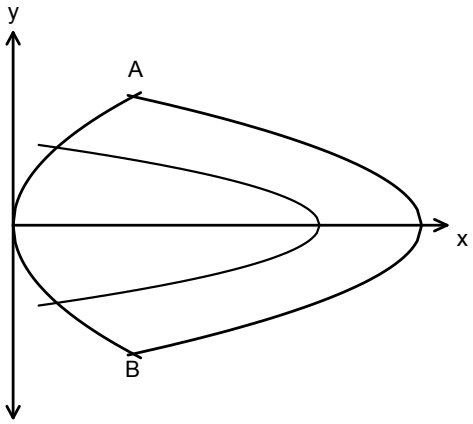
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(c) The logo is enhanced by the addition of Curve 3 as shown below.
This curve, which is the locus of the point M, is obtained by:

- taking the normal to Curve 2 at the point $P = \left(3b - as^2, \sqrt{2}as\right)$
- finding the point Q where the normal meets the x-axis
- taking the midpoint M of the chord PQ
- drawing the curve through all the points M for different values of s

Show that the Cartesian equation of Curve 3 has the form $y^2 = ka\left(x + ma + nb\right)$, where k, m and n are constants, and find the values of k, m and n.



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(Scholarship 2007 Question Five)

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5. A curve is defined by its parametric equations

$x = a\sec\theta - 1$, $y = 2\tan\theta + 2$, where $-\pi < \theta < \pi$, and $\theta - \pm\frac{\pi}{2}$

(a) Find the Cartesian equation of the curve

(b) The normal at the point P, where $\theta = \frac{\pi}{4}$, meets the curve again at the point Q, in the region where $x < 0$.

Find the y-ordinate of Q in the form $\frac{a}{b}$, where a and b are integers.

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(c) The tangent at the point R is parallel to the tangent at P.

Find the y-ordinate of the point S where the normal at R meets the curve again, in the region where $x > 0$.

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(Scholarship 2008 Question Five)

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Answers and Worked Solutions

Chapter 1 Algebra

Ex 1.1 Logarithms and Exponentials (page 3)

1. Let $\log_2 x = y$

Then $\log_4 y = 1$

$y=4$

Hence $\log_2 x = 4$

$x = 2^4$

$x = 16$

2. The minimum value of 3^{x^2-4x} occurs when $x^2 - 4x$ is a minimum

$x^2 - 4x = (x-2)^2 - 4$

Since $(x-2)^2 \geq 0$ the minimum value is -4

Hence the minimum value of 3^{x^2-4x} is $3^{-4} = 1/81$

3. $3^x - 3^{x-1} = 162$

$3.3^{x-1} - 3^{x-1} = 162$

$2.3^{x-1} = 162$

$3^{x-1} = 81$

$3^{x-1} = 3^4$

$x - 1 = 4$

$x = 5$

4. $\sqrt{3^x} - \dfrac{18}{\sqrt{3^x}} = 3$

Let $\sqrt{3^x} = a$

$a - \dfrac{18}{a} = 3$

$a^2 - 3a - 18 = 0$

$(a-6)(a+3) = 0$

$a = 6, -3$

But a cannot be negative.

Hence:

$\sqrt{3^x} = 6$

$3^x = 36$

$x = \dfrac{\log 36}{\log 3}$

$x = 3.26$ (2dp)

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5. $a^2+b^2=14ab$

$(a+b)^2-2ab=14ab$

$(a+b)^2=16ab$

$\log(a+b)^2=\log(16ab)$

$2\log(a+b)=\log16+\log a+\log b$

$2\log(a+b)-\log16=\log a+\log b$

$2\log(a+b)-\log4^2=\log a+\log b$

$2\log(a+b)-2\log4=\log a+\log b$

$\log(a+b)-\log4=\frac{1}{2}(\log a+\log b)$

$\log\left(\frac{a+b}{4}\right)=\frac{1}{2}(\log a+\log b)$

6. $\log_x x+\log_x\left(4x^2+x-4\right)=4$

$\log_x x\left(4x^2+x-4\right)=4$

$x\left(4x^2+x-4\right)=x^4$

$x^4-4x^3-x^2+4x=0$

$x\left(x^3-4x^2-x+4\right)=0$

$x(x+1)(x-1)(x-4)=0$

$x=0,-1,1,4$

But is undefined for $x=0,-1,1$

Hence $x=4$

7. $y=\log_4 16x$

$16x=4^y$

$x=\frac{4^y}{4^2}$

$x=4^{y-2}$ (1)

$y=\log_{16}x$

$x=16^y$

$x=4^{2y}$ (2)

Equating (1) & (2):

$4^{2y}=4^{y-2}$

$2y=y-2$

$y=-2$

$x=\frac{1}{256}$

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8. $k3^x + 3^{-x} = 4$

If k = 0 then the equation reduces to $3^{-x} = 4$ which has exactly one solution

If $k \neq 0$ then;

$$k3^x + \frac{1}{3^x} = 4$$

$$k3^{2x} - 4.3^x - 1 = 0$$

This is a quadratic in 3^x

Hence,

$$3^x = \frac{4 \pm \sqrt{16 - 4k}}{2k}$$

$$3^x = \frac{4 \pm 2\sqrt{4 - k}}{2k}$$

$$3^x = \frac{2 \pm \sqrt{4 - k}}{k}$$

Now,

(1) If 4 – k = 0 there is exactly one solution

(2) If 0 < k < 4 then there are two solutions

(3) If $3^x = \frac{2 + \sqrt{4 - k}}{k}$ and k < 0 then $3^x < 0$ (+ve numerator and –ve denominator) and hence there is no solution.

If $3^x = \frac{2 - \sqrt{4 - k}}{k}$ and k < 0 then $3^x > 0$ (-ve numerator and –ve denominator) and hence there is one solution.

Therefore the equation has only one solution if $k \leq 0$ or k = 4

Ex 1.2 Surd Equations (page 7)

1. x = 5, 2
2. x = 7
3. x = 5
4. x = 23
5. x = 0

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Ex 1.3 Sum and Product of the Roots (page 13)

1. $\alpha + \beta = -\frac{b}{a}$ and $\alpha\beta = \frac{c}{a}$.

Set a= 1 and hence $\alpha + \beta = -b$ and $\alpha\beta = c$:

(a) $\alpha + \beta = 5$ and $\alpha\beta = 3$

$x^2 - 5x + 3 = 0$

(b) $\alpha + \beta = -\frac{4}{5}$ and $\alpha\beta = \frac{1}{10}$

$x^2 + \frac{4}{5}x + \frac{1}{10} = 0$

$10x^2 + 8x + 1 = 0$

(c) $\alpha + \beta = 5a$ and $\alpha\beta = 3a^2$

$x^2 - 5ax + 3a^2 = 0$

2. (a) Let the two roots be α and $\frac{1}{\alpha}$

$(3p - 1)x^2 + (2p + 5)x + p = 0$

So, $\alpha \times \frac{1}{\alpha} = \frac{p}{3p-1}$

$\frac{p}{3p-1} = 1$

$3p - 1 = p$

$p = \frac{1}{2}$

Hence the equation $\frac{1}{2}x^2 + 6x + \frac{1}{2} = 0$ OR $x^2 + 12x + 1 = 0$ is

Now, to verify that the roots are reciprocals use the quadratic formula:

$x = \frac{-12 \pm \sqrt{144 - 4}}{2}$

$x = \frac{-12 \pm 2\sqrt{35}}{2}$

$x = -6 + \sqrt{35},\ -6 - \sqrt{35}$

Finally to show that they are reciprocals:

$\frac{1}{6+\sqrt{35}} \times \frac{6-\sqrt{35}}{6-\sqrt{35}}$

$= 6 - \sqrt{35}$

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(b) $\alpha + \beta = 2\alpha\beta$

$$-\frac{b}{a} = \frac{2c}{a}$$

$$b = -2c$$

So, $-p = -2p - 2$

$$p = -2$$

Substituting in the equation:

$$-4x^2 + 2x - 1 = 0$$

$$4x^2 - 2x + 1 = 0$$

3. $(\alpha + \beta)^2 = \alpha^2 + 2\alpha\beta + \beta^2$

$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$ and $(\alpha\beta)^2 = \alpha^2\beta^2$

$$3x^2 - 11x + 6 = 0$$

$$\alpha\beta = 2$$

$$\alpha + \beta = \frac{11}{3}$$

$$(\alpha + \beta)^2 - 2\alpha\beta = \frac{121}{9} - 4 = \frac{85}{9}$$

$$\alpha^2\beta^2 = 4$$

Hence the equation with roots α^2, β^2 is:

$$x^2 - \frac{85}{9}x + 4 = 0$$

$$9x^2 - 85x + 36 = 0$$

4. $$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta}$$

$$\frac{1}{\alpha} \times \frac{1}{\beta} = \frac{1}{\alpha\beta}$$

$$x^2 - 5x - 9 = 0$$

$\alpha + \beta = 5$, $\alpha\beta = -9$

So, $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{5}{-9}$

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$$\frac{1}{\alpha} \times \frac{1}{\beta} = \frac{1}{-9}$$

So the equation is:

$$x^2 + \frac{5}{9}x - \frac{1}{9} = 0$$

$$9x^2 + 5x - 1 = 0$$

5. To find the equation with roots $2 + \sqrt{3}$, $2 - \sqrt{3}$ then

EITHER:

$\alpha + \beta = 2 + \sqrt{3} + 2 - \sqrt{3} = 4$

$\alpha\beta = (2 + \sqrt{3})(2 - \sqrt{3}) = 4 - 3 = 1$

So the equation is:

$x^2 - 4x + 1 = 0$

OR

$(x - (2 + \sqrt{3}))(x - (2 - \sqrt{3})) = 0$

$((x - 2) - \sqrt{3})((x - 2) + \sqrt{3}) = 0$

$(x - 2)^2 - 3 = 0$

$x^2 - 4x + 4 - 3 = 0$

$x^2 - 4x + 1 = 0$

Ex 1.4 Rational Expressions (page 17)

1. $\dfrac{19x + 16}{(2x + 3)(x - 1)}$

2. $\dfrac{(x - 1)}{(x + 1)(x + 2)}$, $x \neq -3$

3. $\dfrac{1}{(x + 1)(x + 2)}$, $x \neq -3$

4. $\dfrac{x - 1}{(x + 1)(x + 2)}$, $x \neq -2/3$

5. $\dfrac{1}{(x - 1)(x + 3)}$, $x \neq -4$

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Ex 1.5 Partial Fractions (page 23)

1. $\frac{1}{x+2}+\frac{1}{x+1}$

2. $\frac{3}{x+3}+\frac{2}{x-1}$

3. $\frac{8}{x+2}+\frac{4}{x-1}$

4. $\frac{5}{x}-\frac{10}{2x+1}$

5. $\frac{2}{x-1}-\frac{2}{x-3}+\frac{3}{(x-3)^2}$

6. $\frac{3}{x-2}+\frac{3}{x^2+2}$

Ex 1.6 Infinite Series (page 31)

1. (a) $1 - 4x + 16x^2 - 64x^3 + 256x^4 \ldots, -\frac{1}{4} < x < \frac{1}{4}$

   (b) $3-9x^2+27x^4\ldots., -\sqrt{3} < x < \sqrt{3}$

2. (a) $\cos x = 1-\frac{x^2}{2!}+\frac{x^4}{4!}-\ldots.$

   (b) $e^x = 1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\frac{x^4}{4!}+\ldots$

3. (a) $\frac{n(3n+5)}{4(n+1)(n+2)}$ (b) $\frac{11n^3-18n^2-11n+6}{18n(n-1)(n+1)}$

4. $\frac{2k-1}{k(k+1)(k+2)}=\frac{A}{k}+\frac{B}{k+1}+\frac{C}{k+2}$

   $\frac{2k-1}{k(k+1)(k+2)}=\frac{A(k+1)(k+2)+Bk(k+2)+Ck(k+1)}{k(k+1)(k+2)}$

   Let k = 0 and comparing numerators:

   2A = -1

   Let k = -1 and comparing numerators:

   -B = -3

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Let k = -2 and comparing numerators:

-2C = -5

Hence $A=-\frac{1}{2}, B=3, C=\frac{5}{2}$ and therefore $f(n)=\sum_{k=1}^{k=n}\left(-\frac{1}{2k}+\frac{3}{k+1}-\frac{5}{2(k+2)}\right)$

This can equally well be done in the reverse direction by adding the fractions on the right hand side.

To proceed with the second part of the question it is necessary to expand each of the three series far enough to see what is happening. We must also include the general terms on the end. This is a common technique for such questions.

$$f(n)=\sum_{k=1}^{k=n}\left(-\frac{1}{2k}+\frac{3}{k+1}-\frac{5}{2(k+2)}\right)$$

$$f(n)=\sum_{k=1}^{k=n}-\frac{1}{2k}+\sum_{k=1}^{k=n}\frac{3}{k+1}-\sum_{k=1}^{k=n}\frac{5}{2(k+2)}$$

$$f(n)=-\frac{1}{2}\sum_{k=1}^{k=n}\frac{1}{k}+3\sum_{k=1}^{k=n}\frac{1}{k+1}-\frac{5}{2}\sum_{k=1}^{k=n}\frac{1}{(k+2)}$$

$$f(n)=-\frac{1}{2}\left(1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\ldots\frac{1}{n}\right)+3\left(\frac{1}{2}+\frac{1}{3}+\ldots+\frac{1}{n}+\frac{1}{n+1}\right)-\frac{5}{2}\left(\frac{1}{3}+\frac{1}{4}+\ldots.+\frac{1}{n}+\frac{1}{n+1}+\frac{1}{n+2}\right)$$

$$f(n)=\left(-\frac{1}{2}+3-\frac{5}{2}\right)\left(\frac{1}{3}+\frac{1}{4}+\ldots\frac{1}{n}\right)-\frac{1}{2}\left(1+\frac{1}{2}\right)+3\left(\frac{1}{2}+\frac{1}{n+1}\right)-\frac{5}{2}\left(\frac{1}{n+1}+\frac{1}{n+2}\right)$$

$$_f(n)=0-\frac{3}{4}+\frac{3}{2}+\frac{3}{n+1}-\frac{5}{2}\left(\frac{1}{n+1}+\frac{1}{n+2}\right)$$

$$f(n)=\frac{3}{4}+\frac{3}{n+1}-\frac{5}{2}\left(\frac{1}{n+1}+\frac{1}{n+2}\right)$$

$$\lim_{n\to\infty} f(n)=\lim_{n\to\infty}\left(\frac{3}{4}+\frac{3}{n+1}-\frac{5}{2}\left(\frac{1}{n+1}+\frac{1}{n+2}\right)\right)$$

$$\lim_{n\to\infty} f(n)=\frac{3}{4}+\lim_{n\to\infty}\left(\frac{3}{n+1}-\frac{5}{2}\left(\frac{1}{n+1}+\frac{1}{n+2}\right)\right)$$

$$\lim_{n\to\infty} f(n)=\frac{3}{4}+0$$

$$\lim_{n\to\infty} f(n)=\frac{3}{4}$$

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Ex 1.7 General Complex Numbers (page 39)

1. $z^5 - 1 = (z - 1)(z^4 + z^3 + z^2 + z + 1)$

So the remaining four (complex) solutions of $z^5 = 1$ will provide the factors of $z^4 + z^3 + z^2 + z + 1$

$z^5 = 1$

$z^5 = \text{cis}(0 + 2\pi n) = \text{cis}\, 2\pi n$

Let $z = r\text{cis}\theta$

So, $z^5 = r^5\text{cis}5\theta$

Comparing gives $r = 1$ and $\theta = \dfrac{2\pi n}{5}$

$$z = \text{cis}\left(\frac{2\pi n}{5}\right)$$

The complex roots are:

$$\text{cis}\left(\frac{2\pi}{5}\right), \text{cis}\left(\frac{4\pi}{5}\right),$$

$$\text{cis}\left(\frac{6\pi}{5}\right) = \text{cis}\left(-\frac{4\pi}{5}\right), \text{cis}\left(\frac{8\pi}{5}\right) = \text{cis}\left(-\frac{2\pi}{5}\right)$$

Grouping in their complex conjugate pairs to obtain real coefficients when multiplied:

$$z^4 + z^3 + z^2 + z + 1$$

$$= \left(z - \text{cis}\left(\frac{2\pi}{5}\right)\right)\left(z - \text{cis}\left(-\frac{2\pi}{5}\right)\right)\left(z - \text{cis}\left(\frac{4\pi}{5}\right)\right)\left(z - \text{cis}\left(-\frac{4\pi}{5}\right)\right)$$

$$= \left(z^2 - \left(\text{cis}\left(\frac{2\pi}{5}\right) + \text{cis}\left(-\frac{2\pi}{5}\right)\right)z + \text{cis}\left(\frac{2\pi}{5}\right)\text{cis}\left(-\frac{2\pi}{5}\right)\right) \times$$

$$\left(z^2 - \left(\text{cis}\left(\frac{4\pi}{5}\right) + \text{cis}\left(-\frac{4\pi}{5}\right)\right)z + \text{cis}\left(\frac{4\pi}{5}\right)\text{cis}\left(-\frac{4\pi}{5}\right)\right)$$

Now, useful and easily proved, are the facts that:

$\text{cis}\vartheta + \text{cis}(-\vartheta) = 2\cos\vartheta$ and $\text{cis}\vartheta.\text{cis}(-\vartheta) = 1$

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Hence:

$$z^4+z^3+z^2+z+1$$
$$=\left(z^2-2\cos\left(\frac{2\pi}{5}\right)z+1\right)\left(z^2-2\cos\left(\frac{4\pi}{5}\right)z+1\right)$$

Finally, comparing the coefficients of z^2 on the LHS & RHS of the above:

$$4\cos\left(\frac{2\pi}{5}\right)\cos\left(\frac{4\pi}{5}\right)+2=1$$
$$\cos\left(\frac{2\pi}{5}\right)\cos\left(\frac{4\pi}{5}\right)=-\frac{1}{4}$$

2. $z^5-1=0$

If α is a root then:

$$\alpha^5-1=0$$
$$(\alpha-1)\left(\alpha^4+\alpha^3+\alpha^2+\alpha+1\right)=0$$

But α is a complex root and hence:

$$\alpha^4+\alpha^3+\alpha^2+\alpha+1=0$$
$$\alpha^4+\alpha^3+\alpha^2+\alpha=-1$$

From the above:

$$(\alpha+\alpha^4)+(\alpha^3+\alpha^2)=-1$$

ie: the sum of the roots is -1

The product of the roots:

$$(\alpha+\alpha^4)(\alpha^3+\alpha^2)=\alpha^3+\alpha^4+\alpha^6+\alpha^7$$
$$(\alpha+\alpha^4)(\alpha^3+\alpha^2)=\alpha^3+\alpha^4+\alpha^5(\alpha+\alpha^2)$$
But $\alpha^5=1$
$$(\alpha+\alpha^4)(\alpha^3+\alpha^2)=\alpha^3+\alpha^4+\alpha+\alpha^2$$
$$(\alpha+\alpha^4)(\alpha^3+\alpha^2)=-1$$

The sum and product of the roots expressions are regularly useful for scholarship and appear again here.

So, the equation must be the quadratic $z^2+z-1=0$

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3.

$$\cos\left(\frac{\pi}{4}\right)=2\cos^2\left(\frac{\pi}{8}\right)-1$$

$$\cos^2\left(\frac{\pi}{8}\right)=\frac{\cos\left(\frac{\pi}{4}\right)+1}{2}$$

$$\cos\left(\frac{\pi}{8}\right)=\sqrt{\frac{\cos\left(\frac{\pi}{4}\right)+1}{2}}$$

$$\cos\left(\frac{\pi}{8}\right)=\sqrt{\frac{\frac{\sqrt{2}}{2}+1}{2}}$$

$$\cos\left(\frac{\pi}{8}\right)=\sqrt{\frac{\sqrt{2}+2}{2}}$$

$$\cos\left(\frac{\pi}{8}\right)=\frac{\sqrt{\sqrt{2}+2}}{2}$$

$$z^2=\frac{1+7i}{-3+4i}$$

$$z^2=\frac{1+7i}{-3+4i}\times\frac{-3-4i}{-3-4i}$$

$$z^2=\frac{25-25i}{25}$$

$$z^2=1-i$$

In polar form:

$$z^2=\sqrt{2}cis\left(-\frac{\pi}{4}+2\pi n\right)$$

Let $z = \text{rcis}\theta$

$z^2 = r^2\text{cis}2\theta$

Comparing :

$$r=\sqrt[4]{2}\ and\ \theta=-\frac{\pi}{8}+\pi n$$

$$z=\sqrt[4]{2}cis\left(-\frac{\pi}{8}+\pi n\right)$$

$$z=\sqrt[4]{2}cis\left(-\frac{\pi}{8}\right)\ or\ \sqrt[4]{2}cis\left(\frac{7\pi}{8}\right)$$

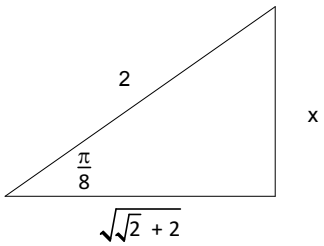
We have an expression for $\cos\left(\frac{\pi}{8}\right)$ from above but also need an expression for $\sin\left(\frac{\pi}{8}\right)$

The use of special triangles and, in this case, using pythagoras to find trig expressions is an essential skill for scholarship candidates.

$$x^2 = 4-\left(\sqrt{\sqrt{2}+2}\right)^2$$

$$x^2 = 4-\sqrt{2}-2$$

$$x = \sqrt{2-\sqrt{2}}$$



$$\sin\left(\frac{\pi}{8}\right)=\frac{\sqrt{2-\sqrt{2}}}{2}$$

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Returning to the roots:

$$z = \sqrt[4]{2}\,\mathrm{cis}\left(-\frac{\pi}{8}\right) \text{ or } \sqrt[4]{2}\,\mathrm{cis}\left(\frac{7\pi}{8}\right)$$

$z = \sqrt[4]{2}\,\mathrm{cis}\left(-\frac{\pi}{8}\right)$:

$$z = \sqrt[4]{2}\left(\cos\left(-\frac{\pi}{8}\right) + i\sin\left(-\frac{\pi}{8}\right)\right)$$

$$z = \sqrt[4]{2}\left(\cos\left(\frac{\pi}{8}\right) - i\sin\left(\frac{\pi}{8}\right)\right)$$

$$z = \sqrt[4]{2}\left(\frac{\sqrt{2+\sqrt{2}}}{2} - i\frac{\sqrt{2-\sqrt{2}}}{2}\right)$$

$$z = \sqrt{\sqrt{2}}\left(\frac{\sqrt{2+\sqrt{2}}}{2} - i\frac{\sqrt{2-\sqrt{2}}}{2}\right)$$

$$z = \frac{\sqrt{2\sqrt{2}+2}}{2} - i\frac{\sqrt{2\sqrt{2}-2}}{2}$$

To calculate the other root we can either repeat the procedure above or note that $\frac{7\pi}{8}$ ***is in the second quadrant whereas*** $-\frac{\pi}{8}$ ***is in the fourth quadrant where sine and cosine have the opposite signs to what they have in the second quadrant.***

Hence:

$$z = -\frac{\sqrt{2\sqrt{2}+2}}{2} + i\frac{\sqrt{2\sqrt{2}-2}}{2}$$

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4. ***This question looks very complicated but don't get overwhelmed by the square root and multiple function signs and just take the questions little bit at a time.***
Part (a) simply requires a perfect square and difference of two squares expansion.

(a) $$z^2 = \frac{1}{2}\left(a+\sqrt{\left(a^2+b^2\right)}\right)+2i\sqrt{\frac{1}{2}\left(a+\sqrt{\left(a^2+b^2\right)}\right)}\sqrt{\frac{1}{2}\left(-a+\sqrt{\left(a^2+b^2\right)}\right)}+i^2\frac{1}{2}\left(-a+\sqrt{\left(a^2+b^2\right)}\right)$$

$$z^2 = \frac{1}{2}\left(a+\sqrt{\left(a^2+b^2\right)}\right)+2i\sqrt{\frac{1}{2}\left(a+\sqrt{\left(a^2+b^2\right)}\right)}\sqrt{\frac{1}{2}\left(-a+\sqrt{\left(a^2+b^2\right)}\right)}-\frac{1}{2}\left(-a+\sqrt{\left(a^2+b^2\right)}\right)$$

$$z^2 = a+2i\sqrt{\frac{1}{2}\left(a+\sqrt{\left(a^2+b^2\right)}\right)\frac{1}{2}\left(-a+\sqrt{\left(a^2+b^2\right)}\right)}$$

$$z^2 = a+2i\frac{1}{2}\sqrt{a^2+b^2-a^2}$$

$$z^2 = a+ib$$

(b) ***Evaluate $f^n(0)$ until the modulus > 2 or repeats a pattern with the modulus always < 2.***

(i)

$$c = 1+\frac{i}{2}$$

$$f(0) = 1+\frac{i}{2}$$

$$\left|1+\frac{i}{2}\right| = \sqrt{\left(1+\tfrac{1}{4}\right)} < 2$$

$$f^2(0) = \left(1+\frac{i}{2}\right)^2+1+\frac{i}{2}$$

$$= 1+i-\frac{1}{4}+1+\frac{i}{2}$$

$$= 2+\frac{3}{2}i$$

$$\left|f^2(0)\right| = \sqrt{\left(4+\tfrac{9}{4}\right)} > 2$$

So, $c = 1+\frac{i}{2}$ is not part of the Mandelbrot set.

$$c = i$$

$$f(0) = i$$

$$\left|f(0)\right| = 1 < 2$$

$$f^2(0) = -1+i$$

$$\left|f^2(0)\right| = \sqrt{2} < 2$$

$$f^3(0) = (-1+i)^2+i$$

$$= 1-2i-1+i$$

$$= -i$$

$$\left|f^3(0)\right| = 1 < 2$$

$$f^4(0) = -1+i = f^2(0)$$

The pattern will now repeat with $\left|f^n(0)\right| < 2$

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(b) ***The first part can be done using the result of part (a) or evaluating $c^2 + c$.***

(ii) ***Evaluating the modulus is simply a matter of having faith in your algebraic skills. Practice at algebraic manipulation helps a lot here.***

$$c=\sqrt{\left(\frac{1}{2}\left(a+\sqrt{a^2+3a^2}\right)\right)}+i\sqrt{\left(\frac{1}{2}\left(-a+\sqrt{a^2+3a^2}\right)\right)}$$

$$c=\sqrt{\left(\frac{3a}{2}\right)}+i\sqrt{\left(\frac{a}{2}\right)}$$

$$f^2(0)=c^2+c$$

But from (a):

$$c^2=a+ib$$

$$c^2=a+i\sqrt{3}a$$

$$f^2(0)=a+\sqrt{\left(\frac{3a}{2}\right)}+i\left(\sqrt{3}a+\sqrt{\left(\frac{a}{2}\right)}\right)$$

$$a=\frac{1}{8}$$

$$f^2(0)=\frac{1}{8}+\sqrt{\frac{3}{16}}+i\left(\frac{\sqrt{3}}{8}+\sqrt{\left(\frac{1}{16}\right)}\right)$$

$$f^2(0)=\frac{1}{8}+\frac{\sqrt{3}}{4}+i\left(\frac{\sqrt{3}}{8}+\frac{1}{4}\right)$$

$$\left|f^2(0)\right|=\sqrt{\left(\frac{1}{8}+\frac{\sqrt{3}}{4}\right)^2+\left(\frac{\sqrt{3}}{8}+\frac{1}{4}\right)^2}$$

$$\left|f^2(0)\right|=\sqrt{\left(\frac{1}{64}+2\times\frac{1}{8}\times\frac{\sqrt{3}}{4}+\frac{3}{16}+\frac{3}{64}+2\times\frac{\sqrt{3}}{8}\times\frac{1}{4}+\frac{1}{16}\right)}$$

$$\left|f^2(0)\right|=\sqrt{\left(\frac{5}{16}+\frac{\sqrt{3}}{8}\right)}$$

$$\left|f^2(0)\right|=\sqrt{\left(\frac{5}{16}+\frac{2\sqrt{3}}{16}\right)}$$

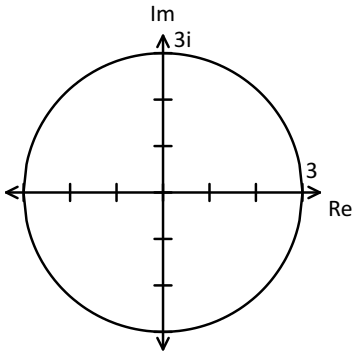
$$\left|f^2(0)\right|=\frac{\sqrt{\left(5+2\sqrt{3}\right)}}{4}$$

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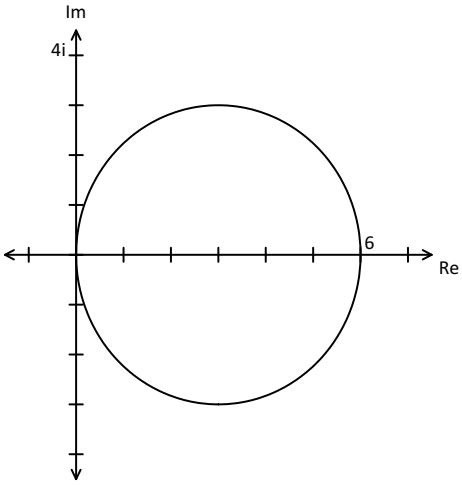
Ex 1.8 The Geometry of Complex numbers (page 49)

1.

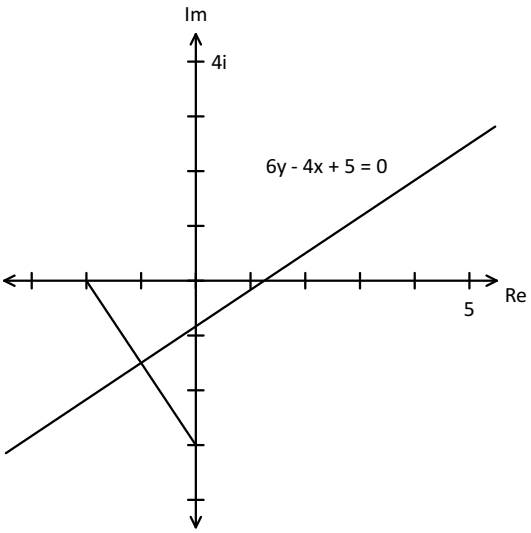
(a)



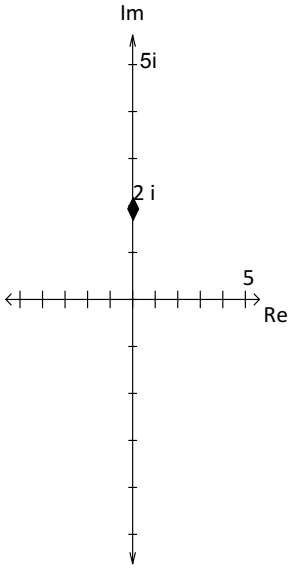
(b)



(c)



(d)



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2.

$$|z| = 2|z-6|$$

$$|z|^2 = 4|z-6|^2$$

Let $z = x + iy$

$$x^2 + y^2 = 4[(x-6)^2 + y^2]$$

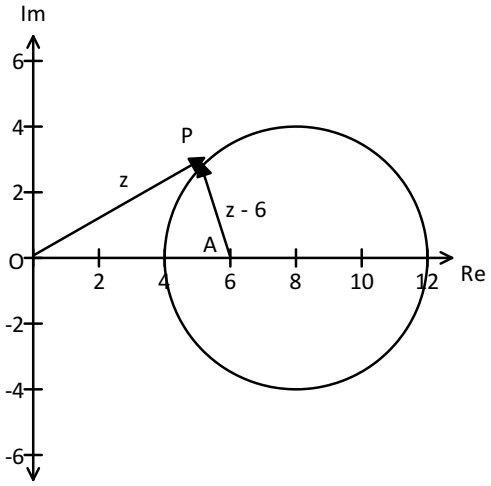
$$x^2 + y^2 = 4[x^2 - 12x + 36 + y^2]$$

$$x^2 + y^2 = 4x^2 - 48x + 144 + 4y^2$$

$$3x^2 + 3y^2 - 48x + 144 = 0$$

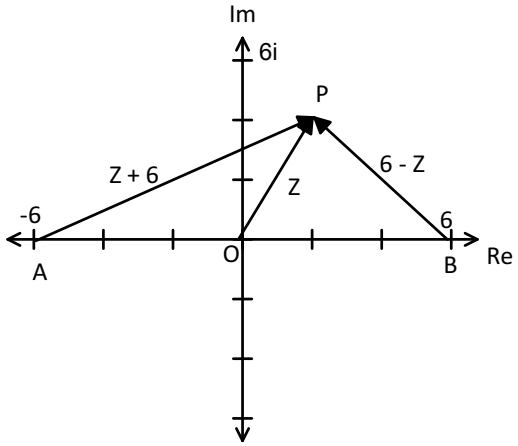
$$x^2 + y^2 - 16x + 48 = 0$$

$$(x-8)^2 + y^2 = 16$$



3. By definition, if the difference of the distances of P from two fixed points A and B is a constant then, the locus of P must be a hyperbola.

Now setting up the axes as shown, and using the obvious fact that $(6 + z) - (6 - z) = 2z$:



$$|z| = \left|\frac{1}{2}((6+z)-(6-z))\right|$$

$$|z| = \frac{1}{2}|(6+z)-(6-z)|$$

But $|a-b| \geq |a| - |b|$, so

$$|z| \geq \frac{1}{2}(|6+z| - |6-z|)$$

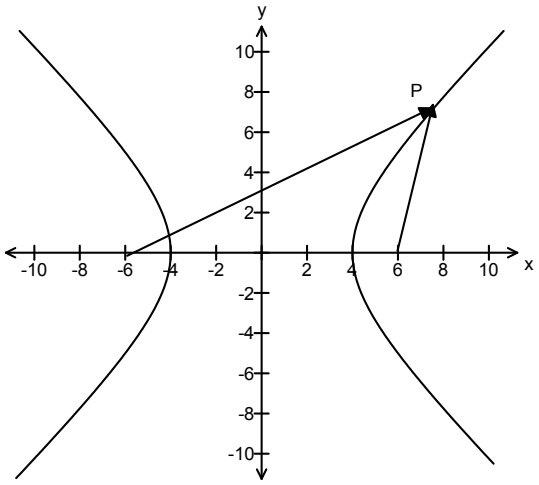
$$|z| \geq \frac{1}{2} \times 8$$

$$|z| \geq 4$$

The minimum value of $|z| = 4$ will occur on the x axis at the vertices of the hyperbola

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The points A & B are the foci of the hyperbola.



So we have a = 4, c = 6 and using the relationship $b^2 = c^2 - a^2$ then $b^2 = 20$

Hence the equation of the hyperbola is:

$$\frac{x^2}{16} - \frac{y^2}{20} = 1$$

An alternative solution, replacing z = x + iy:

$$|6+z| - |6-z| = 8$$

$$|6+x+iy| - |6-x-iy| = 8$$

$$\sqrt{(6+x)^2+y^2} - \sqrt{(6-x)^2+y^2} = 8$$

$$\sqrt{(6+x)^2+y^2} = \sqrt{(6-x)^2+y^2} + 8$$

$$(6+x)^2 + y^2 = (6-x)^2 + y^2 + 64 + 16\sqrt{(6-x)^2+y^2}$$

$$36 + 12x + x^2 + y^2 = 36 - 12x + x^2 + y^2 + 64 + 16\sqrt{(6-x)^2+y^2}$$

$$24x - 64 = 16\sqrt{(6-x)^2+y^2}$$

$$3x - 8 = 2\sqrt{(6-x)^2+y^2}$$

$$9x^2 - 48x + 64 = 4\left(36 - 12x + x^2 + y^2\right)$$

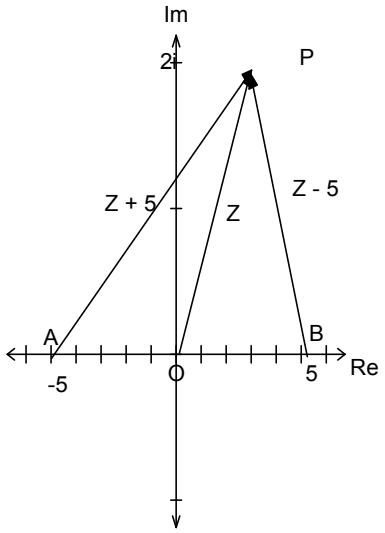
$$9x^2 - 48x + 64 = 144 - 48x + 4x^2 + 4y^2$$

$$5x^2 - 4y^2 = 80$$

$$\frac{x^2}{16} - \frac{y^2}{20} = 1$$

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4. By definition, if the sum of the distances of P from two fixed points A and B is a constant then the locus of P must be an ellipse.
Now setting up the axes as shown,



$$|z| = \left|\frac{1}{2}((z+5)+(z-5))\right|$$

$$|z| = \frac{1}{2}|((z+5)+(z-5))|$$

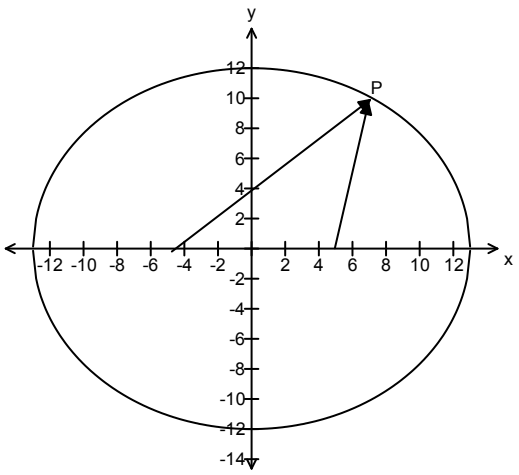
But $|a+b| \le |a|+|b|$, so

$$|z| \le \frac{1}{2}(|z+5|+|z-5|)$$

$$|z| \le \frac{1}{2} \times 26$$

$$|z| \le 13$$

The maximum value of $|z| = 13$ will occur on the x axis. Hence a = 13.



The points A & B are the foci of the ellipse.

So we have a = 13, c = 5 and using the relationship $b^2 = a^2 - c^2$ then $b^2 = 144$

Hence the equation of the ellipse is:

$$\frac{x^2}{169} + \frac{y^2}{144} = 1$$

5. (i) $|v| = \sqrt{2}$

$$|vz| = |v||z| = \sqrt{2}\sqrt{x^2+y^2}$$

$$|z-v| = |vz|$$

$$|z-v|^2 = |vz|^2$$

$$|(x-1)+i(y-1)|^2 = 2(x^2+y^2)$$

$$(x-1)^2+(y-1)^2 = 2x^2+2y^2$$

$$x^2-2x+1+y^2-2y+1 = 2x^2+2y^2$$

$$x^2+2x+y^2+2y = 2$$

$$(x+1)^2+(y+1)^2 = 4$$

Which is a circle radius 2 and centre (-1, -1).

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This can be done from the beginning substituting v = 1 + I and z = x + iy but it is usually better to work through a few steps before making the substitution.

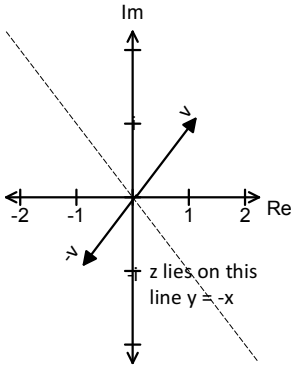
Note:

(1) the use of $|vz| = |v||z|$ ***and (2) a common mistake evaluating the modulus is to leave the i in the calculation***

(ii) $|z - v| = |z + v|$ means that the distance from z to v is the same as the distance from z to –v

ie: z lies on the perpendicular bisector of the line between (1, 1) and (-1, -1).

This is the line y = -x.



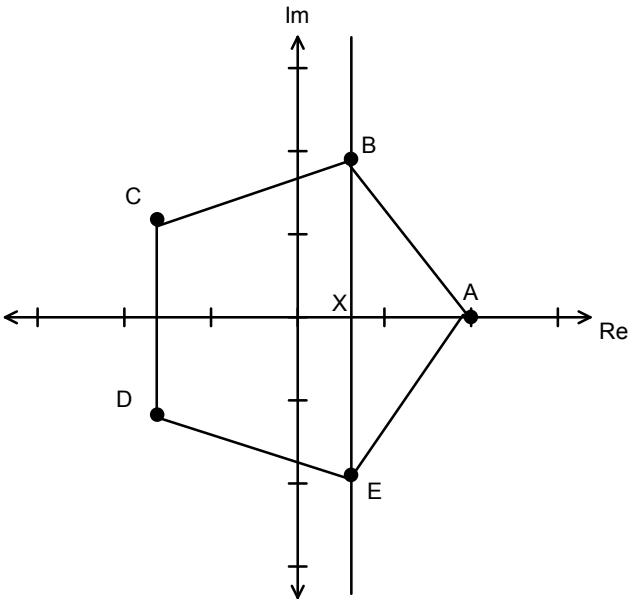
Familiarity with the definition of the perpendicular bisector makes the first part of (ii) very easy, otherwise the approach is to substitute v = 1 + I and z = x + iy and proceed from there.

Now, substitute y = -x into $(x-1)^2 + (y-1)^2 = 4$

$x^2 - 2x + 1 + x^2 + 2x + 1 = 4 \Rightarrow 2x^2 = 2 \Rightarrow x = \pm 1$

So the intersection is at (1, -1) and (-1, 1)

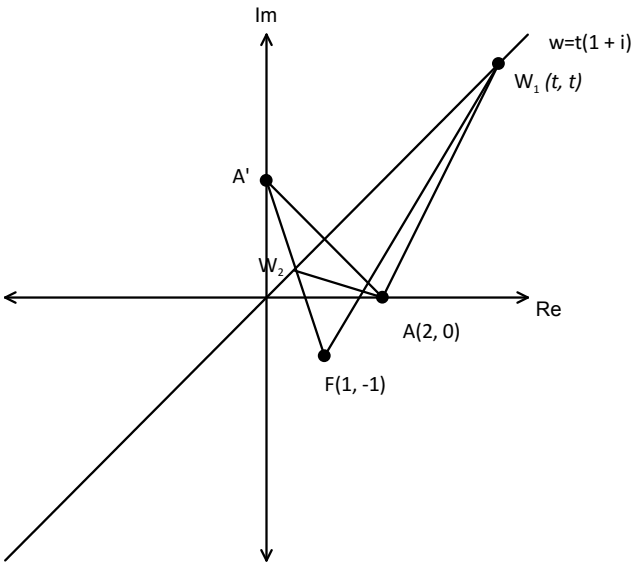
6. (i)



By symmetry BX = CX. BX = b, hence BE = 2b. But BE = CE, so CE = 2b.

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(ii)



Now, if w = t(1 + i) then W lies on the line y = x

There are several ways to do this question.
We will look at two methods using (i) calculus and (ii) geometry

Using calculus:

Minimise $AW_1 + FW_1$
Using Pythagoras or the distance formula:

$$AW_1 = \sqrt{(2-t)^2 + t^2} = \sqrt{4 - 4t + 2t^2}$$

$$FW_1 = \sqrt{(t-1)^2 + (t+1)^2} = \sqrt{2t^2 + 2} = \sqrt{2}\sqrt{t^2 + 1}$$

$$AW_1 + FW_1 = \sqrt{4 - 4t + 2t^2} + \sqrt{2}\sqrt{t^2 + 1}$$

$$\frac{d(AW_1 + FW)}{dt} = (4t - 4) \times \frac{1}{2} \times \left(4 - 4t + 2t^2\right)^{-\frac{1}{2}} + \sqrt{2} \times 2t \times \frac{1}{2} \times \left(t^2 + 1\right)^{-\frac{1}{2}}$$

$$\frac{d(AW_1 + FW)}{dt} = \frac{2t - 2}{\sqrt{4 - 4t + 2t^2}} + \frac{\sqrt{2} \times t}{\sqrt{t^2 + 1}}$$

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$$\frac{d(AW_1+FW)}{dt}=\frac{(2t-2)\sqrt{t^2+1}+\sqrt{2}\times t\times\sqrt{4-4t+2t^2}}{\sqrt{4-4t+2t^2}\sqrt{t^2+1}}$$

$$\frac{d(AW_1+FW)}{dt}=0$$

$$(2t-2)\sqrt{t^2+1}+\sqrt{2}\times t\times\sqrt{4-4t+2t^2}=0$$

$$2(t-1)\sqrt{t^2+1}=-\left(\sqrt{2}\times t\times\sqrt{4-4t+2t^2}\right)$$

$$4\left(t^2-2t+1\right)\left(t^2+1\right)=2t^2\left(4-4t+2t^2\right)$$

$$4\left(t^2-2t+1\right)\left(t^2+1\right)=4t^2\left(2-2t+t^2\right)$$

$$\left(t^2-2t+1\right)\left(t^2+1\right)=t^2\left(2-2t+t^2\right)$$

$$t^4-2t^3+t^2+t^2-2t+1=2t^2-2t^3+t^4$$

$$-2t+1=0$$

$$t=\frac{1}{2}$$

$$AW_1+FW_1=\sqrt{4-4\times\frac{1}{2}+2\left(\frac{1}{2}\right)^2}+\sqrt{2}\sqrt{\left(\frac{1}{2}\right)^2+1}$$

$$AW_1+FW_1=\sqrt{2\frac{1}{2}}+\sqrt{2}\sqrt{1\frac{1}{4}}$$

$$AW_1+FW_1=\sqrt{2\frac{1}{2}}+\sqrt{2\frac{1}{2}}$$

$$AW_1+FW_1=2\sqrt{2\frac{1}{2}}$$

$$AW_1+FW_1=\sqrt{10}$$

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Using Geometry:

Returning to the diagram.

A' is the image of A. A' is (0, 2)

By symmetry $AW_1 = A'W_1$

The minimum value of AW + FW will occur when A', W and F all lie on a straight line

ie:$A'W_2F$

Using Pythagoras or the distance formula:

$$A'F = \sqrt{(0-1)^2 + (2-{}^{-}1)^2}$$
$$A'F = \sqrt{10}$$

Clearly this method is much quicker and suggests that taking a minute to think "is there an easier method" can be very fruitful in terms of time saved.

7. (a) By symmetry, the three solutions to $z^3 = 1$ are 1, $\text{cis}\left(\frac{2\pi}{3}\right)$, $\text{cis}\left(\frac{4\pi}{3}\right)$.

If $\text{cis}\left(\frac{2\pi}{3}\right) = w$ then, by de Moivre, $\text{cis}\left(\frac{4\pi}{3}\right) = w^2$, so the roots of $z^3 = 1$ are 1, w, w^2.

$$(z+1)^3 = 8$$
$$\frac{(z+1)^3}{8} = 1$$
$$\left(\frac{z+1}{2}\right)^3 = 1$$

So, $\frac{z+1}{2} = 1, w, w^2$

$z = 1, 2w-1, 2w^2-1$

The sum of the roots

$$= 1 + 2w - 1 + 2w^2 - 1$$
$$= 2w^2 + 2w - 1$$
$$= 2(w^2 + w + 1) - 3$$

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But $w^3 = 1$

$w^3 - 1 = 0$

$(w-1)(w^2 + w + 1) = 0$

w, w^2 must be the complex roots of $w^2 + w + 1 = 0$

ie: $w^2 + w + 1 = 0$

Hence, the sum of the roots $= -3$

Alternatively solve $(z+1)^3 = 8cis(2\pi n)$ to get 1, $-2+\sqrt{3}i$, $-2-\sqrt{3}i$ and add these roots. This is a longer method and you should try to understand the above solution.

(b)

$(z+1)^3 = 8(z-1)^3$

$\left(\frac{z+1}{2(z-1)}\right)^3 = 1$

$\frac{z+1}{2(z-1)} = 1, w, w^2$

$\frac{z+1}{2(z-1)} = 1$

$z + 1 = 2z - 2$

$z = 3$

$\frac{z+1}{2(z-1)} = w$

$z + 1 = 2zw - 2w$

$(2w-1)z = 2w + 1$

$z = \frac{2w+1}{2w-1}$

$w = cis\left(\frac{2\pi}{3}\right) = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$

$z = \frac{-1+\sqrt{3}i+1}{-1+\sqrt{3}i-1}$

$z = \frac{\sqrt{3}i}{-2+\sqrt{3}i} \times \frac{-2-\sqrt{3}i}{-2-\sqrt{3}i}$

$z = \frac{-2\sqrt{3}i+3}{7}$

The remaining root is the complex conjugate of the above ie: $z = \frac{3+2\sqrt{3}i}{7}$

So the roots are 3, $\frac{3}{7} + \frac{2\sqrt{3}i}{7}$ and $\frac{3}{7} - \frac{2\sqrt{3}i}{7}$

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(c) Continuing using the approach from (a) and (b):

The solutions for z are 3, $\frac{3+2\sqrt{3}i}{7}$ and $\frac{3-2\sqrt{3}i}{7}$

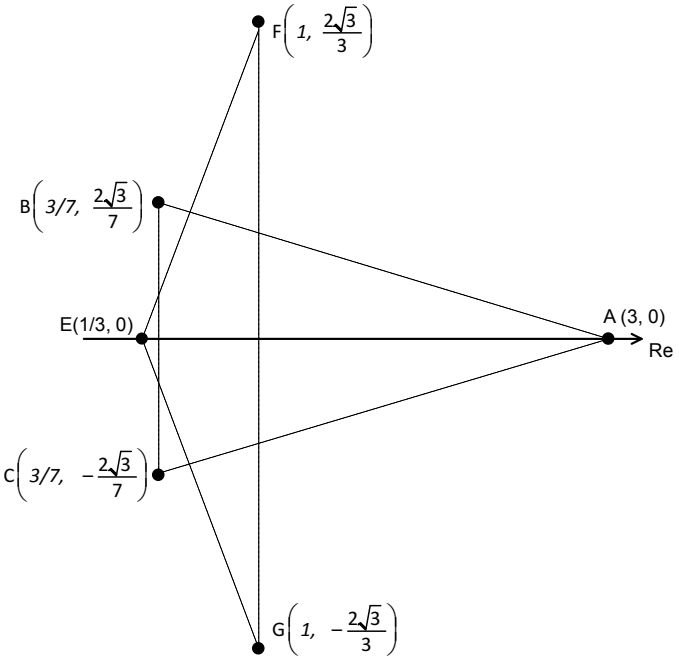
Hence $\frac{1}{z}=\frac{1}{3}, \frac{7}{3+2\sqrt{3}i}, \frac{7}{3-2\sqrt{3}i}$

$$\frac{7}{3+2\sqrt{3}i}=\frac{7}{3+2\sqrt{3}i}\times\frac{3-2\sqrt{3}i}{3-2\sqrt{3}i}$$
$$=\frac{21-14\sqrt{3}i}{21}$$
$$=\frac{3-2\sqrt{3}i}{3}$$

The other complex root is $\frac{3+2\sqrt{3}i}{3}$

So the roots are $\frac{1}{3}, 1+\frac{2\sqrt{3}i}{3}, 1-\frac{2\sqrt{3}i}{3}$

Plotting the solutions from (b) and (c):



$$\frac{\Delta ABC}{\Delta EFG}=\frac{\left(\frac{1}{2}\times\frac{4\sqrt{3}}{7}\times\left(3-\frac{3}{7}\right)\right)}{\left(\frac{1}{2}\times\frac{4\sqrt{3}}{3}\times\left(1-\frac{1}{3}\right)\right)}$$

$$\frac{\Delta ABC}{\Delta EFG}=\frac{\left(\frac{2\sqrt{3}}{7}\times\frac{18}{7}\right)}{\left(\frac{2\sqrt{3}}{3}\times\frac{2}{3}\right)}$$

$$\frac{\Delta ABC}{\Delta EFG}=\frac{9\times18}{2\times49}$$

$$\frac{\Delta ABC}{\Delta EFG}=\frac{81}{49}$$

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Chapter 2 Trigonometry

Ex 2.1 Graphs and Modelling (page 66)

1. (a) $T = 18 - 11\cos\left(\frac{\pi t}{12}\right)$ or equivalent

(b) T = 10.2° (1dp)

(c) 9 am or 9pm

2. (a) $h'(x) = -2\sin x \qquad f'(x) = 6\cos 2x$

$6\cos 2x = -2\sin x$

$3\left(1 - 2\sin^2 x\right) = -\sin x$

$6\sin^2 x - \sin x - 3 = 0$

$\sin x = 0.7963$ or $\sin x = -0.6286$ (GC)

$x = 0.9196, 2.222, 3.8214, 5.6033$

(b) $p(x) = -a(x-b)^2 - c$ is an inverted parabola. If it cuts f(x) and g(x) at one point only and the intersection with f(x) is at x = 2.575 then the maximum value of h(x) must touch the minimum value of g(x).

Hence $b = \pi$ and c = 2. So, $p(x) = -a(x-\pi)^2 - 2$

At the intersection of f(x) and p(x), $f'(x) = p'(x)$ ie:

$-2a(x-\pi) = 6\cos 2x$

But $x = 2.575$

$a = \frac{-6\cos(5.15)}{2.575 - \pi} = 2.244$

(c) f(x) and h(x)first touch when their gradients are equal ie: x = 0.9196 (from part (a))

When h(0.9196) = f(0.9196) then:

$k + 2\cos 0.9196 = 3\sin(2 \times 0.9196)$

$k = 1.68$

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Before this the graphs cut more than twice and immediately after this they touch only twice.

As k increases above 1.68 the graphs intersect only twice, until they touch at x = 3.8214 after which they no longer intersect:

When h(3.8214) = f(3.8214) then:

$$k + 2\cos 3.8214 = 3\sin\left(2 \times 3.8214\right)$$
$$k = 4.4887$$

So, for k > 0 they cut twice for $1.68 < k < 4.4887$

Ex2.2 Identities (page 73)

1.
$$\text{LHS} = \frac{2\cos\left(\frac{A+B}{2}\right)\sin\left(\frac{A-B}{2}\right)}{2\sin\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)}$$
$$= \cot\left(\frac{A+B}{2}\right)\tan\left(\frac{A-B}{2}\right) = \text{RHS}$$

2.
$$\text{LHS} = \frac{2\sin\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)}{2\cos\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)}$$
$$\text{LHS} = \tan\left(\frac{A+B}{2}\right)$$
$$A + B + C = 180^{o}$$
$$\frac{A+B}{2} = 90^{o} - \frac{C}{2}$$
$$\text{LHS} = \tan\left(90^{o} - \frac{C}{2}\right)$$
$$= \frac{\sin\left(90^{o} - \frac{C}{2}\right)}{\cos\left(90^{o} - \frac{C}{2}\right)}$$
$$= \frac{\cos\left(\frac{C}{2}\right)}{\sin\left(\frac{C}{2}\right)}$$
$$= \cot\left(\frac{C}{2}\right) = \text{RHS}$$

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Ex 2.3 Equations (page 81)

1. $\cos(2A+B) = \cos 2A\cos B - \sin 2A\sin B$

$\cos(2A+B) = (2\cos^2 A - 1)\cos B - 2\sin A\cos A\sin B$

$A = B = \theta$

$\cos(3\theta) = 2\cos^3\theta - \cos\theta - 2\sin^2\theta\cos\theta$

$\cos(3\theta) = 2\cos^3\theta - \cos\theta - 2\cos\theta(1 - \cos^2\theta)$

$\cos(3\theta) = 2\cos^3\theta - \cos\theta - 2\cos\theta + 2\cos^3\theta$

$\cos(3\theta) = 4\cos^3\theta - 3\cos\theta$

$\frac{1}{4}\cos(3\theta) = \cos^3\theta - \frac{3}{4}\cos\theta$ (1)

Replacing x with $\frac{2}{3}\cos\theta$ in the equation:

$27x^3 - 9x = 1$

$27\left(\frac{2}{3}\cos\theta\right)^3 - 9\times\frac{2}{3}\cos\theta = 1$

$8\cos^3\theta - 6\cos\theta = 1$

$\cos^3\theta - \frac{3}{4}\cos\theta = \frac{1}{8}$

Using the result (1) above:

$\frac{1}{4}\cos 3\theta = \frac{1}{8}$

$\cos 3\theta = \frac{1}{2}$

$3\theta = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}$

$\theta = \frac{\pi}{9}, \frac{5\pi}{9}, \frac{7\pi}{9}$

There are only three solutions as the equation is a cubic.

Hence:

$x = \frac{2}{3}\cos\left(\frac{\pi}{9}\right), \frac{2}{3}\cos\left(\frac{5\pi}{9}\right), \frac{2}{3}\cos\left(\frac{7\pi}{9}\right)$

If these are the roots of the equation $27x^3 - 9x - 1 = 0$ then their product is $\frac{1}{27}$:

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$$\frac{2}{3}\cos\left(\frac{\pi}{9}\right)\times\frac{2}{3}\cos\left(\frac{5\pi}{9}\right)\times\frac{2}{3}\cos\left(\frac{7\pi}{9}\right)=\frac{1}{27}$$

$$\cos\left(\frac{\pi}{9}\right)\cos\left(\frac{5\pi}{9}\right)\cos\left(\frac{7\pi}{9}\right)=\left(\frac{3}{2}\right)^3\times\frac{1}{27}$$

Hence, using $\cos\left(\frac{3\pi}{9}\right)=\frac{1}{2}$:

$$\cos\left(\frac{\pi}{9}\right)\cos\left(\frac{3\pi}{9}\right)\cos\left(\frac{5\pi}{9}\right)\cos\left(\frac{7\pi}{9}\right)=\left(\frac{3}{2}\right)^3\times\frac{1}{2}\times\frac{1}{27}$$

$$\cos\left(\frac{\pi}{9}\right)\cos\left(\frac{3\pi}{9}\right)\cos\left(\frac{5\pi}{9}\right)\cos\left(\frac{7\pi}{9}\right)=\frac{1}{16}$$

2.

(a) $10\pi\cos\left(\frac{\pi x}{2}\right)+10\pi\cos\left(\frac{2\pi x}{3}\right)=0$

$$\cos\left(\frac{\pi x}{2}\right)+\cos\left(\frac{2\pi x}{3}\right)=0$$

$$2\cos\left(\frac{7\pi x}{12}\right)\cos\left(-\frac{\pi x}{12}\right)=0$$

$$\cos\left(\frac{7\pi x}{12}\right)=0 \text{ OR } \cos\left(\frac{\pi x}{12}\right)=0$$

$$\frac{7\pi x}{12}=\frac{\pi}{2},\frac{3\pi}{2},\frac{5\pi}{2},\frac{7\pi}{2}\ldots. \text{ OR } \frac{\pi x}{12}=\frac{\pi}{2},\frac{3\pi}{2},\frac{5\pi}{2},\frac{7\pi}{2}\ldots.$$

$$x=\frac{6}{7},\frac{18}{7},\frac{30}{7},6,\ldots \text{ OR } \quad x=6,18,\ldots.$$

$$x=\frac{6}{7},\frac{18}{7},\frac{30}{7},6$$

(b) (i) Let the angular velocity of the arm be $\frac{d\theta}{dt}$ and the wheel $\frac{d\alpha}{dt}$

$$\frac{d\theta}{dt}=\frac{2\pi}{4}=\frac{\pi}{2} \qquad \frac{d\alpha}{dt}=\frac{2\pi}{3}$$

$$\theta=\frac{\pi t}{2} \qquad \alpha=\frac{2\pi t}{3}$$

Equation:

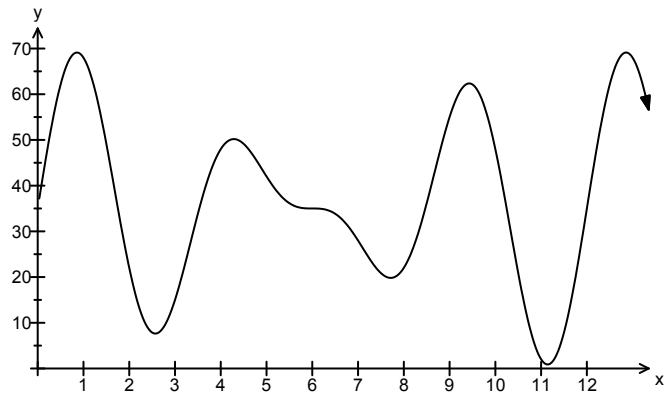
$$h(t)=35+20\sin\left(\frac{\pi t}{2}\right)+15\sin\left(\frac{2\pi t}{3}\right)$$

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(ii) $h'(t) = 10\pi\cos\left(\frac{\pi t}{2}\right) + 10\pi\cos\left(\frac{2\pi t}{3}\right)$

This is the same equation as (a) and hence $t = \frac{6}{7}, \frac{18}{7}, \frac{30}{7}$, 6 and continuing to 4π

Investigating the graph of h(t) up to 4π:



There are a series of local maxima but the first (and later repeated one) appear to be the greatest:

$$h''(t) = -5\pi^2 \sin\left(\frac{\pi t}{2}\right) - 5\pi^2 \sin\left(\frac{2\pi t}{3}\right)$$

$$h''\left(\frac{6}{7}\right) = -5\pi^2 \sin\left(\frac{6\pi}{14}\right) - 5\pi^2 \sin\left(\frac{12\pi t}{21}\right)$$

Both expressions in the brackets are less than $\frac{\pi}{2}$, hence the sine is positive and

$h''\left(\frac{6}{7}\right) < 0$

Hence a maximum of

$$h\left(\frac{6}{7}\right) = 35 + 20\sin\left(\frac{6\pi t}{14}\right) + 15\sin\left(\frac{12\pi t}{21}\right) = 69.1\ldots$$

Testing all other values of t will confirm that this is the global maximum.

Hence the maximum height is 69.1 m (1 dp)

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Ex 2.4 Triangles (page 87)

1.

$$\frac{c}{SinC}=\frac{b}{sinB}=k$$

$$c=ksinC \quad \text{and} \quad b=ksinB$$

$$\frac{b-c}{b+c}=\frac{ksinB-ksinC}{ksinB+ksinC}$$

$$=\frac{sinB-sinC}{sinB+sinC}$$

$$=\frac{2cos\left(\frac{B+C}{2}\right)sin\left(\frac{B-C}{2}\right)}{2sin\left(\frac{B+C}{2}\right)cos\left(\frac{B-C}{2}\right)}$$

$$=tan\left(\frac{B-C}{2}\right)\frac{cos\left(\frac{B+C}{2}\right)}{sin\left(\frac{B+C}{2}\right)}$$

But $A+B+C=180$

$$\frac{B+C}{2}=90^o-\frac{A}{2}$$

$$\frac{b-c}{b+c}=tan\left(\frac{B-C}{2}\right)\frac{cos\left(90^o-\frac{A}{2}\right)}{sin\left(90^o-\frac{A}{2}\right)}$$

$$=tan\left(\frac{B-C}{2}\right)\frac{sin\left(\frac{A}{2}\right)}{cos\left(\frac{A}{2}\right)}$$

$$=tan\left(\frac{B-C}{2}\right)tan\left(\frac{A}{2}\right)$$

2.

(a) $<AQC=<ABC\left(<'s \text{ in the same segment}\right)$ ie: $\alpha=B$

ΔQAC:

$<QAC=90^o$ (< in semicircle)

$$sin\alpha=\frac{b}{2R}$$

$$\text{But } \alpha=B \Rightarrow sinB=\frac{b}{2R} \text{ or } \frac{b}{sinB}=2R$$

$\therefore$ in ΔABC:

$$\frac{a}{sinA}=\frac{b}{sinB}=\frac{c}{sinC}=2R$$

(b) (i) $a=\theta_2, b=\theta_4, c=\theta_1, d=\theta_3$ $\left(\text{all } <'s \text{ in the same segment}\right)$

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(ii) ΔABC:

$$\frac{S_1}{\sin\theta_1} = \frac{S_2}{\sin\theta_2} = 2R$$

ΔADC:

$$\frac{S_3}{\sin\theta_3} = \frac{S_4}{\sin\theta_4} = 2R$$

ΔDBC:

$$\frac{S_3}{\sin\theta_2} = \frac{S_2}{\sin\theta_2} = 2R$$

Hence:

$$\frac{S_1}{\sin\theta_1} = \frac{S_2}{\sin\theta_2} = \frac{S_3}{\sin\theta_3} = \frac{S_4}{\sin\theta_4} = 2R$$

Hence:

$S_1 = 2R\sin\theta_1$, $S_2 = 2R\sin\theta_2$, $S_3 = 2R\sin\theta_3$, $S_4 = 2R\sin\theta_4$

$$\therefore S_1 S_3 + S_2 S_4 = 4R^2\sin\theta_1\sin\theta_3 + 4R^2\sin\theta_2\sin\theta_4$$

$$S_1 S_3 + S_2 S_4 = 4R^2\left(\sin\theta_1\sin\theta_3 + \sin\theta_2\sin\theta_4\right)$$

ΔABC:

$$\frac{AC}{\sin(\theta_3+\theta_4)} = 2R$$

ΔABD:

$$\frac{DB}{\sin(\theta_3+\theta_2)} = 2R$$

$$AC \times BD = 4R^2\sin(\theta_3+\theta_4)\sin(\theta_3+\theta_2)$$

Applying Ptolemy's Theorem:

$$4R^2\left(\sin\theta_1\sin\theta_3 + \sin\theta_2\sin\theta_4\right) = 4R^2\sin(\theta_3+\theta_4)\sin(\theta_3+\theta_2)$$

$$\sin\theta_1\sin\theta_3 + \sin\theta_2\sin\theta_4 = \sin(\theta_3+\theta_4)\sin(\theta_3+\theta_2)$$

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3. (a) $\Delta BB'C$:

$$\frac{BB'}{\sin C}=\frac{BC}{\sin B'}$$

$$\frac{BB'}{\sin C}=\frac{BC}{\sin\left(180-\left(C+\frac{B}{2}\right)\right)}$$

$$\frac{BB'}{\sin C}=\frac{BC}{\sin\left(C+\frac{B}{2}\right)}$$

Similarly:

$$\frac{CC'}{\sin B}=\frac{BC}{\sin\left(B+\frac{C}{2}\right)}$$

Now, $C+\frac{B}{2}=C+\frac{180-A-C}{2}=\frac{180-A+C}{2}=90-\frac{A-C}{2}$

Hence:

$$\frac{BB'}{\sin C}=\frac{BC}{\sin\left(90-\frac{A-C}{2}\right)}=\frac{BC}{\cos\left(\frac{A-C}{2}\right)}$$

Similarly:

$$\frac{CC'}{\sin B}=\frac{BC}{\cos\left(\frac{A-B}{2}\right)}$$

$CC'=BB'$ so,

$$\frac{BC\sin C}{\cos\left(\frac{A-C}{2}\right)}=\frac{BC\sin B}{\cos\left(\frac{A-B}{2}\right)}$$

$$\sin C\cos\left(\frac{A-B}{2}\right)=\sin B\cos\left(\frac{A-C}{2}\right)$$

$$2\sin\left(\frac{C}{2}\right)\cos\left(\frac{C}{2}\right)\cos\left(\frac{A-B}{2}\right)=2\sin\left(\frac{B}{2}\right)\cos\left(\frac{B}{2}\right)\cos\left(\frac{A-C}{2}\right)$$

It now remains to replace the sine terms:

$$\frac{C}{2}=\frac{180-A-B}{2}=90-\frac{A+B}{2}\quad\text{and}\quad\frac{B}{2}=\frac{180-A-C}{2}=90-\frac{A+C}{2}$$

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Hence:

$$2\sin\left(90-\frac{A+B}{2}\right)\cos\left(\frac{C}{2}\right)\cos\left(\frac{A-B}{2}\right)=2\sin\left(90-\frac{A+C}{2}\right)\cos\left(\frac{B}{2}\right)\cos\left(\frac{A-C}{2}\right)$$

$$\cos\left(\frac{C}{2}\right)\cos\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)=\cos\left(\frac{B}{2}\right)\cos\left(\frac{A+C}{2}\right)\cos\left(\frac{A-C}{2}\right)$$

(b) $\cos(x+y)\cos(x-y)=\frac{1}{2}(\cos 2x+\cos 2y)$

$$=\frac{1}{2}\left(2\cos^2 x-1+2\cos^2 y-1\right)$$

$$=\frac{1}{2}\left(2\cos^2 x+2\cos^2 y-2\right)$$

$$=\cos^2 x+\cos^2 y-1$$

Using this expression in the result from (i):

$$\cos\left(\frac{C}{2}\right)\left(\cos^2\left(\frac{A}{2}\right)+\cos^2\left(\frac{B}{2}\right)-1\right)=\cos\left(\frac{B}{2}\right)\left(\cos^2\left(\frac{A}{2}\right)+\cos^2\left(\frac{C}{2}\right)-1\right)$$

$$\cos\left(\frac{C}{2}\right)\left(\cos^2\left(\frac{B}{2}\right)-\sin^2\left(\frac{A}{2}\right)\right)=\cos\left(\frac{B}{2}\right)\left(\cos^2\left(\frac{C}{2}\right)-\sin^2\left(\frac{A}{2}\right)\right)$$

$$\cos\left(\frac{C}{2}\right)\cos^2\left(\frac{B}{2}\right)-\cos\left(\frac{C}{2}\right)\sin^2\left(\frac{A}{2}\right)=\cos\left(\frac{B}{2}\right)\cos^2\left(\frac{C}{2}\right)-\cos\left(\frac{B}{2}\right)\sin^2\left(\frac{A}{2}\right)$$

$$\sin^2\left(\frac{A}{2}\right)\left(\cos\left(\frac{B}{2}\right)-\cos\left(\frac{C}{2}\right)\right)+\cos\left(\frac{C}{2}\right)\cos^2\left(\frac{B}{2}\right)-\cos\left(\frac{B}{2}\right)\cos^2\left(\frac{C}{2}\right)=0$$

$$\sin^2\left(\frac{A}{2}\right)\left(\cos\left(\frac{B}{2}\right)-\cos\left(\frac{C}{2}\right)\right)+\cos\left(\frac{C}{2}\right)\cos\left(\frac{B}{2}\right)\left(\cos\left(\frac{B}{2}\right)-\cos\left(\frac{C}{2}\right)\right)=0$$

$$\left(\cos\left(\frac{B}{2}\right)-\cos\left(\frac{C}{2}\right)\right)\left(\sin^2\left(\frac{A}{2}\right)+\cos\left(\frac{C}{2}\right)\cos\left(\frac{B}{2}\right)\right)=0$$

But the second bracket is always positive and hence:

$$\cos\left(\frac{B}{2}\right)=\cos\left(\frac{C}{2}\right)$$

$B=C$

ie: the triangle is isosceles

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Chapter 3 Differentiation

Ex 3.2 Limits (page 98)

1. $\frac{3}{2}$ 2. 1 3. $\frac{5}{4}$

Ex 3.3 First Principles (page 101)

1.

$$f'(x)=\lim_{h\to 0}\left(\frac{\operatorname{cosec}(x+h)-\operatorname{cosec}x}{h}\right)$$

$$f'(x)=\lim_{h\to 0}\left(\frac{\frac{1}{\sin(x+h)}-\frac{1}{\sin x}}{h}\right)$$

$$f'(x)=\lim_{h\to 0}\left(\frac{\frac{\sin x-\sin(x+h)}{\sin(x+h)\sin x}}{h}\right)$$

$$f'(x)=\lim_{h\to 0}\left(\frac{\sin x-\sin(x+h)}{h\sin(x+h)\sin x}\right)$$

$$f'(x)=\lim_{h\to 0}\left(\frac{2\cos\left(\frac{2x+h}{2}\right)\sin\left(-\frac{h}{2}\right)}{h\sin(x+h)\sin x}\right)$$

$$f'(x)=\lim_{h\to 0}\left(\frac{\cos\left(\frac{2x+h}{2}\right)\sin\left(-\frac{h}{2}\right)}{\frac{h}{2}\sin(x+h)\sin x}\right)$$

$$f'(x)=-\lim_{h\to 0}\left(\frac{\sin\left(\frac{h}{2}\right)}{\frac{h}{2}}\right)\lim_{h\to 0}\left(\frac{\cos\left(\frac{2x+h}{2}\right)}{\sin(x+h)\sin x}\right)$$

$$f'(x)=-1\times\lim_{h\to 0}\left(\frac{\cos\left(\frac{2x+h}{2}\right)}{\sin(x+h)\sin x}\right)$$

$$f'(x)=-1\times\left(\frac{\cos x}{\sin x\sin x}\right)$$

$$f'(x)=-\frac{\cos x}{\sin x}\times\frac{1}{\sin x}$$

$$f'(x)=-\operatorname{cosec}x\cot x$$

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2.
$$f'(x) = \lim_{h\to 0}\left(\frac{\frac{1}{\sqrt{(x+h)}} - \frac{1}{\sqrt{x}}}{h}\right)$$

$$f'(x) = \lim_{h\to 0}\left(\frac{\sqrt{x} - \sqrt{(x+h)}}{h\sqrt{(x+h)}\sqrt{x}}\right)$$

$$f'(x) = \lim_{h\to 0}\left(\frac{\sqrt{x} - \sqrt{(x+h)}}{h\sqrt{(x+h)}\sqrt{x}} \times \frac{\sqrt{x} + \sqrt{(x+h)}}{\sqrt{x} + \sqrt{(x+h)}}\right)$$

$$f'(x) = \lim_{h\to 0}\left(\frac{-h}{h\sqrt{(x+h)}\sqrt{x}} \times \frac{1}{\sqrt{x} + \sqrt{(x+h)}}\right)$$

$$f'(x) = \lim_{h\to 0}\left(\frac{-1}{\sqrt{(x+h)}\sqrt{x}}\right) \times \lim_{h\to 0}\left(\frac{1}{\sqrt{x} + \sqrt{(x+h)}}\right)$$

$$f'(x) = \frac{-1}{\sqrt{x}\sqrt{x}} \times \frac{1}{\sqrt{x} + \sqrt{x}}$$

$$f'(x) = \frac{-1}{x} \times \frac{1}{2\sqrt{x}}$$

$$f'(x) = -\frac{1}{2\sqrt{x^3}}$$

Ex 3.4 The Chain, Product and Quotient Rules (page 106)

1. $$\frac{dy}{dx} = -2x\left[\ln\left(1-x^2\right)+1\right]$$

2. $$\frac{dy}{dx} = \cos x \operatorname{cosec}^2 x\left[1-\ln(\sin x)\right]$$

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Ex 3.5 Implicit and Parametric Differentiation (page 110)

1. $\frac{dx}{dt} = -12\sin t\cos^2 t$

$\frac{dy}{dt} = 12\cos t\sin^2 t$

$\frac{dy}{dx} = \frac{12\cos t\sin^2 t}{-12\sin t\cos^2 t}$

$\frac{dy}{dx} = -\tan t$

$t = \frac{\pi}{4}$

$\frac{dy}{dx} = -1$

Gradient of normal $= 1$

$x = 4\cos^3\left(\frac{\pi}{4}\right) = \sqrt{2}$

$y = 4\sin^3\left(\frac{\pi}{4}\right) = \sqrt{2}$

Equation of normal:

$y = x$

2. There are numerous approaches to (a) (i) of this problem. This is only one such approach which is based on the definition of the second derivative.

(a)

(i) $\frac{dy}{dx} = \frac{dy}{dt} \div \frac{dx}{dt}$

$\frac{d^2y}{dx^2} = \frac{d\left(\frac{dy}{dx}\right)}{dx} = \frac{d\left(\frac{dy}{dx}\right)}{dx} \times \frac{dt}{dt}$

$\frac{d^2y}{dx^2} = \frac{d\left(\frac{dy}{dx}\right)}{dt} \div \frac{dx}{dt}$

$\frac{d^2y}{dx^2} = \frac{d\left(\frac{dy}{dt} \div \frac{dx}{dt}\right)}{dt} \div \frac{dx}{dt}$

Using the quotient rule:

$\frac{d^2y}{dx^2} = \left[\frac{\frac{d^2y}{dt^2} \times \frac{dx}{dt} - \frac{dy}{dt} \times \frac{d^2x}{dt^2}}{\left(\frac{dx}{dt}\right)^2}\right] \div \frac{dx}{dt}$

But $\frac{d^2y}{dx^2} = 0$ (given)

$\frac{d^2y}{dt^2} \times \frac{dx}{dt} - \frac{dy}{dt} \times \frac{d^2x}{dt^2} = 0$

$\frac{d^2y}{dt^2} \times \frac{dx}{dt} = \frac{dy}{dt} \times \frac{d^2x}{dt^2}$

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(ii) $\frac{dx}{dt} = -a\sin t - b\sin 2t$ $\qquad$ $\frac{dy}{dt} = a\cos t + b\cos 2t$

$\frac{d^2x}{dt^2} = -a\cos t - 2b\cos 2t$ $\qquad$ $\frac{d^2y}{dt^2} = -a\sin t - 2b\sin 2t$

At a point of inflection $\frac{d^2y}{dx^2} = 0$ or using the result of (i):

$(-a\sin t - b\sin 2t)\times(-a\sin t - 2b\sin 2t) = (a\cos t + b\cos 2t)\times(-a\cos t - 2b\cos 2t)$

$(a\sin t + b\sin 2t)\times(a\sin t + 2b\sin 2t) = -(a\cos t + b\cos 2t)\times(a\cos t + 2b\cos 2t)$

$a^2\sin^2 t + 3ab\sin t\sin 2t + 2b^2\sin^2 2t = -\left(a^2\cos^2 t + 3ab\cos t\cos 2t + 2b^2\cos^2 2t\right)$

$a^2\sin^2 t + a^2\cos^2 t + 3ab\sin t\sin 2t + 3ab\cos t\cos 2t + 2b^2\sin^2 2t + 2b^2\cos^2 2t = 0$

$a^2 + 3ab(\sin t\sin 2t + \cos t\cos 2t) + 2b^2 = 0$

$a^2 + 3ab\left(2\sin^2 t\cos t + \cos t\left(1 - 2\sin^2 t\right)\right) + 2b^2 = 0$

$a^2 + 3ab\cos t\left(2\sin^2 t + 1 - 2\sin^2 t\right) + 2b^2 = 0$

$a^2 + 3ab\cos t + 2b^2 = 0$

$\cos t = -\frac{a^2 + 2b^2}{3ab}$

3.

(a) $x = r\cos\theta$ $\qquad$ $y = r\sin\theta$

$x = Ae^{k\theta}\cos\theta$ $\qquad$ $y = Ae^{k\theta}\sin\theta$

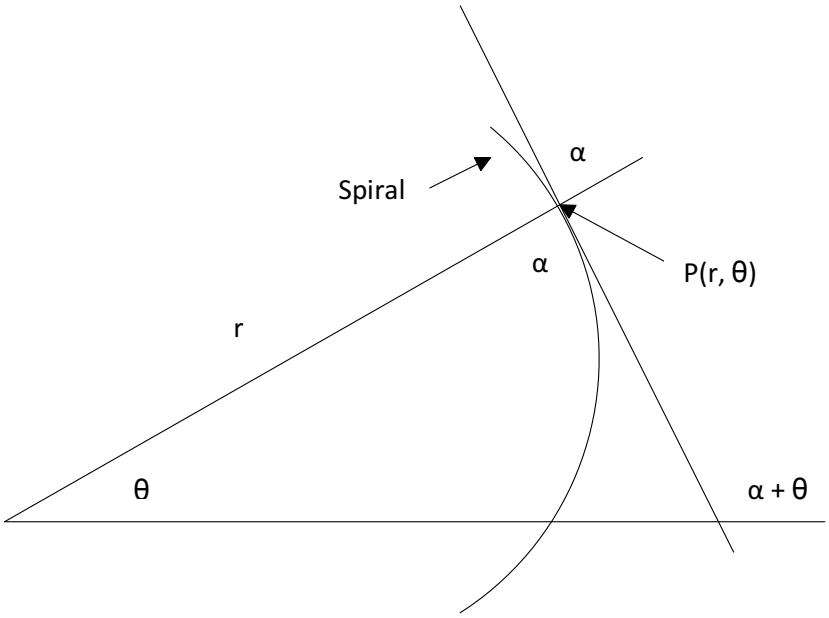
$\frac{dx}{d\theta} = A\left(ke^{k\theta}\cos\theta - e^{k\theta}\sin\theta\right)$ $\qquad$ $y = A\left(ke^{k\theta}\sin\theta + e^{k\theta}\cos\theta\right)$

$\frac{dx}{d\theta} = Ae^{k\theta}(k\cos\theta - \sin\theta)$ $\qquad$ $y = Ae^{k\theta}(k\sin\theta + \cos\theta)$

$\frac{dy}{dx} = \frac{(k\sin\theta + \cos\theta)}{(k\cos\theta - \sin\theta)}$

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(b)



A little angle geometry shows that the angle the tangent makes with the x-axis is α + θ and hence the gradient of the tangent is $\tan(\alpha+\theta)$.

$$\tan(\alpha+\theta)=\frac{\tan\alpha+\tan\theta}{1-\tan\alpha\tan\theta}$$

Hence:

$$\frac{dy}{dx}=\frac{k\sin\theta+\cos\theta}{k\cos\theta-\sin\theta}=\frac{\tan\alpha+\tan\theta}{1-\tan\alpha\tan\theta}$$

$$\frac{k\sin\theta+\cos\theta}{k\cos\theta-\sin\theta}=\frac{\tan\alpha+\tan\theta}{1-\tan\alpha\tan\theta}$$

Dividing top and bottom of the LHS by $k\cos\theta$ will express it in terms of tangent:

$$\text{LHS}=\frac{\tan\theta+\frac{1}{k}}{1-\frac{1}{k}\tan\theta}$$

Comparing with the RHS:

$$\tan\alpha=\frac{1}{k}\text{ OR }\alpha=\tan^{-1}\left(\frac{1}{k}\right)$$

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4.

(a) $h(x)=(f(x))^{g(x)}$

$\ln(h(x))=\ln\left((f(x))^{g(x)}\right)$

$\ln(h(x))=g(x)\ln(f(x))$

Differentiating implicitly:

$h'(x)\times\frac{1}{h(x)}=g'(x)\ln(f(x))+g(x)f'(x)\times\frac{1}{f(x)}$

$h'(x)=h(x)\left[g'(x)\ln(f(x))+g(x)\times\frac{f'(x)}{f(x)}\right]$

(b) $p(x)=x\ln(\ln x)$

$p'(x)=\ln(\ln x)+x.\frac{1}{x}.\frac{1}{\ln x}$

$p'(x)=\ln(\ln x)+\frac{1}{\ln x}$

$\int(\ln x)^x p'(x)dx=\int(\ln x)^x\left(\ln(\ln x)+\frac{1}{\ln x}\right)dx$

Now comparing with (a) where $f(x)=\ln x$ and $g(x)=x$:

$f'(x)=\frac{1}{x}$ and $g'(x)=1$ and hence:

$h'(x)=h(x)\left[g'(x)\ln(f(x))+g(x)\times\frac{f'(x)}{f(x)}\right]=(\ln x)^x\left[1.\ln(\ln x)+x.\frac{\frac{1}{x}}{\ln x}\right]$

$h'(x)=(\ln x)^x\left[\ln(\ln x)+\frac{1}{\ln x}\right]$, which is the expression under the integral sign.

Hence $\int(\ln x)^x p'(x)dx=\int h'(x)dx=h(x)+c=(\ln x)^x+c$

(c) If g(x) is exactly divisible by x – 1 then g(1) = 0

$-(2-a)^4+b+c=0$ OR $(2-a)^4=b+c$

$-(2x-1)^4+8x-7=0$

$a=1,\ b=8,\ c=-7$

$1=8-7$ and hence $(x-1)$ is a factor.

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Expanding:

$$-(2x-1)^4+8x-7=0$$

$$-\left(16x^4-4\times8x^3+6\times4x^2-4\times2x+1\right)+8x-7=0$$

$$-16x^4+32x^3-24x^2+8x-1+8x-7=0$$

$$-16x^4+32x^3-24x^2+16x-8=0$$

$$16x^4-32x^3+24x^2-16x+8=0$$

$$(x-1)\left(16x^3-16x^2+8x-8\right)=0$$

$$8(x-1)\left(2x^3-2x^2+x-1\right)=0$$

But substituting $x=1$ into the cubic shows that $x-1$ is a factor, hence:

$$8(x-1)(x-1)\left(2x^2+1\right)=0$$

Solving $2x^2+1=0$

$$2x^2=-1$$

$$x=\pm\frac{i}{\sqrt{2}}$$

So, $x=1,1,\pm\frac{i}{\sqrt{2}}$

Ex 3.6 Inverse Trigonometric Functions (page 119)

1, $\frac{dy}{dx}=\arcsin x+\frac{x}{\sqrt{1-x^2}}$

2. $4y+\pi x-12\ln 2=0$

3. $6y+2\sqrt{3}x-4\sqrt{3}=0$

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Ex 3.7 Properties of Curves (page 123)

1. $\frac{dy}{dx} = \frac{1.(1+x^2) - 2x.x}{(1+x^2)^2}$

$\frac{dy}{dx} = \frac{1-x^2}{(1+x^2)^2}$

$\frac{d^2y}{dx^2} = \frac{-2x(1+x^2)^2 - 4x(1+x^2)(1-x^2)}{(1+x^2)^4}$

$\frac{d^2y}{dx^2} = \frac{-2x(1+x^2)(3-x^2)}{(1+x^2)^4}$

$\frac{dy}{dx} = 0$

$x = \pm 1$

$x = 1,\ Y = \frac{1}{2}$ and $\frac{d^2y}{dx^2} = \frac{-1}{2}$

$x = -1,\ Y = \frac{-1}{2}$ and $\frac{d^2y}{dx^2} = \frac{1}{2}$

So, max at $x = 1$ and min at $x = -1$

Must test nature of TP's

$\frac{d^2y}{dx^2} = 0$

$x = 0$ or $x = \pm\sqrt{3}$

Substituting for y

Non stationary pts of inflection at $(0,0), (\sqrt{3}, \frac{\sqrt{3}}{4}), (-\sqrt{3}, \frac{-\sqrt{3}}{4})$

2. $\left(\frac{\pi}{4}, \sqrt{2}\right)$ - max, $\left(\frac{5\pi}{4}, -\sqrt{2}\right)$ - min, $\left(\frac{3\pi}{4}, 0\right), \left(\frac{7\pi}{4}, 0\right)$ - non stationary inflection (nsi)

3. $\left(2, 4 + 10\operatorname{arccot}(2)\right)$ - min, $\left(-2, -4 + 10\operatorname{arccot}(-2)\right)$ - max, $(0, 5\pi)$ - nsi

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Ex 3.8 Related Rates of Change (page 127)

1. $\frac{dh}{dt} = \frac{dV}{dt} \times \frac{dh}{dV}$

$= \frac{dV}{dt} \div \frac{dV}{dh}$

$V = \frac{1}{3}\pi r^2 h$, but $r = \frac{h}{3}$

$V = \frac{1}{3}\pi \frac{h^2}{9} \times h$

$= \frac{\pi h^3}{27}$

$\frac{dV}{dh} = \frac{\pi h^2}{9}$

$\frac{dh}{dt} = 6 \times \frac{9}{\pi h^2}$

At $h = 3$, $\frac{dh}{dt} = 6 \times \frac{9}{\pi 3^2}$

$\frac{dh}{dt} = \frac{6}{\pi}$ OR $1.91\,cm/sec\,(3sf)$

2. $1/55$ m/min or 0.1818 3. 200 km/hr

4. (a) -350 km/hr (towards each other) (b) +430 km/hr (3sf)

Ex 3.9 Optimisation Problems (page 133)

1.

$\tan\alpha = \frac{2}{x}$ and $\tan(\theta + \alpha) = \frac{3}{x}$

$\alpha = \tan^{-1}\left(\frac{2}{x}\right)$ and $\theta + \alpha = \tan^{-1}\left(\frac{3}{x}\right)$

$\theta = \tan^{-1}\left(\frac{3}{x}\right) - \tan^{-1}\left(\frac{2}{x}\right)$

$\frac{d\theta}{dx} = \frac{1}{1+\left(\frac{3}{x}\right)^2} \times -\frac{3}{x^2} - \frac{1}{1+\left(\frac{2}{x}\right)^2} \times -\frac{2}{x^2}$

$\frac{d\theta}{dx} = -\frac{3}{x^2+9} + \frac{2}{x^2+4}$

$\frac{d\theta}{dx} = 0$

$\frac{2}{x^2+4} = \frac{3}{x^2+9}$

$2x^2 + 18 = 3x^2 + 12$

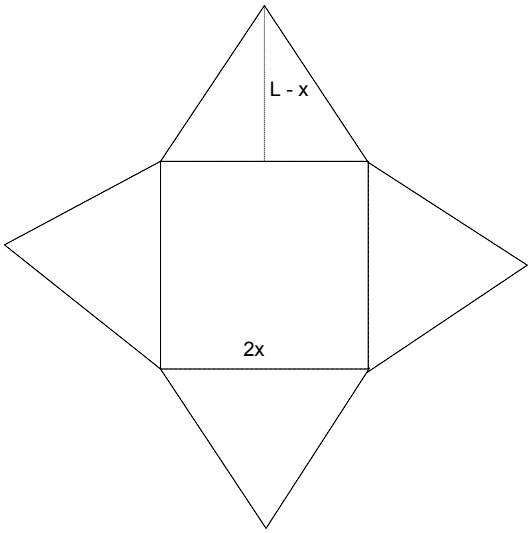
$x^2 = 6$

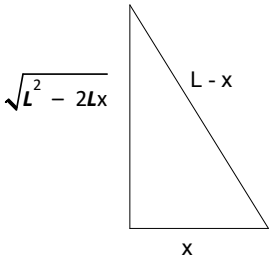
$x = \pm\sqrt{6}$

He should sit 2.4 metres from the screen.

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2. (a)





To form a pyramid the triangular side folds over such that it's vertex is over the centre of the base of the square. The height of the pyramid becomes $\sqrt{L^2 - 2Lx}$.

$$V(x) = \frac{1}{3} \times 4x^2 \times \sqrt{L^2 - 2Lx}$$

$$V(x) = \frac{4}{3} \times \sqrt{L^2x^4 - 2Lx^5}$$

Taking the x^2 term inside the square root makes it easier to differentiate using the chain rule.

$$V'(x) = \frac{4}{3} \times \frac{1}{2} \times \left(4x^3L^2 - 10Lx^4\right) \times \left(x^4L^2 - 2Lx^5\right)^{-1/2}$$

$$V'(x) = 0$$

$$4x^3L^2 - 10Lx^4 = 0$$

$$2Lx^3\left(2L - 5x\right) = 0$$

$$x = 0 \text{ OR } x = \frac{2L}{5}$$

$x = 0$ gives the minimum volume, hence $x = \frac{2L}{5}$

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(b)

$$\frac{dV}{dt}=\frac{k^2}{3L}$$

$$\text{Find } \frac{dh}{dt}=\frac{dV}{dt}\times\frac{dh}{dV}=\frac{dV}{dt}\div\frac{dV}{dh}$$

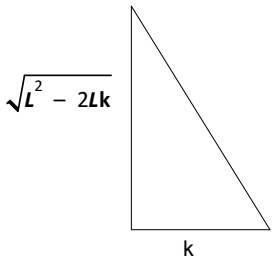


Fig 1

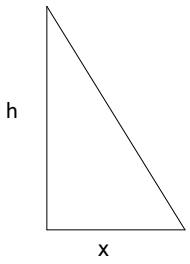


Fig 2

The triangles **Fig 1** and **Fig 2** are similar. **Fig 1** represents the whole pyramid and **Fig 2** the portion of the pyramid full of water at time t.

$$\frac{x}{h}=\frac{k}{\sqrt{L^2-2Lk}}$$

$$x^2=\frac{k^2h^2}{L(L-2k)}$$

Hence:

$$V(h)=\frac{1}{3}\times\frac{4k^2h^2}{L(L-2k)}\times h=\frac{4}{3}\times\frac{k^2h^3}{L(L-2k)}$$

$$\frac{dV}{dh}=\frac{4k^2h^2}{L(L-2k)}$$

$$\frac{dh}{dt}=\frac{k^2}{3L}\times\frac{L(L-2k)}{4k^2h^2}=\frac{(L-2k)}{12h^2}$$

Now we are given that:

$$\frac{dh}{dt}>\frac{L-2k}{h+1}$$

$$\frac{(L-2k)}{12h^2}>\frac{L-2k}{h+1}$$

$$\frac{1}{12h^2}>\frac{1}{h+1} \qquad -\text{given that } L>2x$$

$$12h^2-h-1<0$$

$$(4h+1)(3h-1)<0$$

$$0<h<\frac{1}{3} \qquad (h>0)$$

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3.

(a)

$$\text{Area of } \Delta = \frac{1}{2} \times s^2 \times \sin 60^\circ$$

$$\text{Area of } \Delta = \frac{1}{2} \times s^2 \times \frac{\sqrt{3}}{2}$$

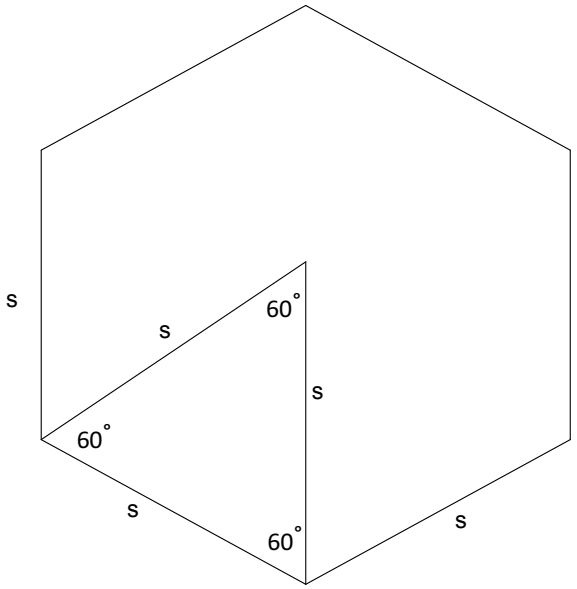
$$\text{Area of } \Delta = \frac{\sqrt{3}\, s^2}{4}$$

$$\text{Area of hexagon} = \frac{\sqrt{3}\, s^2}{4} \times 6$$

$$\text{Area of hexagon} = \frac{3\sqrt{3}\, s^2}{2}$$

$$\text{Area of stack} = 16 \times \frac{3\sqrt{3}\, s^2}{2}$$

$$\text{Area of stack} = 24\sqrt{3}\, s^2$$



(b)

$$A = 6hs + \frac{3}{2}s^2\left(-\cot\theta + \sqrt{3}\operatorname{cosec}\theta\right)$$

$$\frac{dA}{d\theta} = \frac{3}{2}s^2\left(\operatorname{cosec}^2\theta - \sqrt{3}\operatorname{cosec}\theta\cot\theta\right)$$

$$\frac{3}{2}s^2\left(\operatorname{cosec}^2\theta - \sqrt{3}\operatorname{cosec}\theta\cot\theta\right) = 0$$

$$\operatorname{cosec}\theta\left(\operatorname{cosec}\theta - \sqrt{3}\cot\theta\right) = 0$$

$$\operatorname{cosec}\theta = 0 \quad \text{OR} \quad \operatorname{cosec}\theta - \sqrt{3}\cot\theta = 0$$

Not possible, $\quad \frac{1}{\sin\theta} - \frac{\sqrt{3}\cos\theta}{\sin\theta} = 0$

$$\frac{1 - \sqrt{3}\cos\theta}{\sin\theta} = 0$$

$$\cos\theta = \frac{1}{\sqrt{3}}$$

$$\theta = 54.7^\circ$$

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(c) ***This part of the question looks like a straight forward " related rates of change question ", but the difficulty with the question is that as the Surface Area changes so does s and θ and expressing s in terms of θ is not a viable option .***

$$\frac{ds}{dt} = k\sin\theta$$

$$\text{At } \theta = \frac{\pi}{4}, \frac{ds}{dt} = \sqrt{2}$$

$$\sqrt{2} = k\frac{\sqrt{2}}{2}$$

$$k = 2$$

$$\frac{ds}{dt} = 2\sin\theta$$

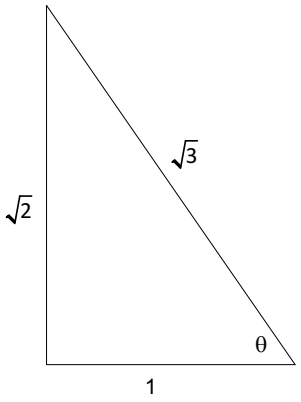
$$A = 6hs + \frac{3}{2}s^2\left(-\cot\theta + \sqrt{3}\operatorname{cosec}\theta\right)$$

$$\frac{dA}{dt} = 6h\frac{ds}{dt} + 3s\frac{ds}{dt}\left(-\cot\theta + \sqrt{3}\operatorname{cosec}\theta\right) + \frac{3}{2}s^2\left(\operatorname{cosec}^2\theta\frac{d\theta}{dt} - \sqrt{3}\operatorname{cosec}\theta\cot\theta\frac{d\theta}{dt}\right)$$

$$\frac{dA}{dt} = 6h\frac{ds}{dt} + 3s\frac{ds}{dt}\left(-\cot\theta + \sqrt{3}\operatorname{cosec}\theta\right) + \frac{3}{2}s^2\frac{d\theta}{dt}\left(\operatorname{cosec}^2\theta - \sqrt{3}\operatorname{cosec}\theta\cot\theta\right)$$

For the value of θ in (b) $\cos\theta = \frac{1}{\sqrt{3}}$.

Drawing up the appropriate right angled triangle it can then be seen that:



$$\cot\theta = \frac{1}{\sqrt{2}}, \sin\theta = \frac{\sqrt{2}}{\sqrt{3}}, \operatorname{cosec}\theta = \frac{\sqrt{3}}{\sqrt{2}}$$

and $\frac{ds}{dt} = 2\frac{\sqrt{2}}{\sqrt{3}}$.

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Continuing:

$$\frac{dA}{dt} = 6h\frac{2\sqrt{2}}{\sqrt{3}} + 3s\frac{2\sqrt{2}}{\sqrt{3}}\left(-\frac{1}{\sqrt{2}} + \sqrt{3}\frac{\sqrt{3}}{\sqrt{2}}\right) + \frac{3}{2}s^2\frac{d\theta}{dt}\left(\frac{3}{2} - \sqrt{3}\frac{\sqrt{3}}{\sqrt{2}}\frac{1}{\sqrt{2}}\right)$$

$$\frac{dA}{dt} = \frac{2\sqrt{2}}{\sqrt{3}}\left(6h + 3s\left(\frac{2}{\sqrt{2}}\right)\right) + \frac{3}{2}s^2\frac{d\theta}{dt}\left(\frac{3}{2} - \frac{3}{2}\right)$$

$$\frac{dA}{dt} = \frac{2\sqrt{6}}{3}\left(6h + 3s\left(\frac{2}{\sqrt{2}}\right)\right) + 0$$

$$\frac{dA}{dt} = 2\left(2\sqrt{6}h + 3s\frac{\sqrt{6}}{3}\left(\frac{2}{\sqrt{2}}\right)\right)$$

$$\frac{dA}{dt} = 2\left(2\sqrt{6}h + 2\sqrt{3}s\right)$$

$$\frac{dA}{dt} = 4\left(\sqrt{6}h + \sqrt{3}s\right)$$

Chapter 4 Integration

Ex 4.1 Trigonometric Substitutions (page 145)

1. (a) $\frac{2}{5}\sqrt{(x+3)^5} - 2\sqrt{(x+3)^3} + c$ (b) $12\frac{2}{3}$

   (c) $\frac{2}{3}\sqrt{\left(e^x - 1\right)^3} + c$

2. (a) $\sin^{-1}\left(\frac{x}{2}\right) + c$ (b) $\frac{9}{4}\left[\frac{2x\sqrt{9-4x^2}}{9} + \sin^{-1}\left(\frac{2x}{3}\right)\right] + c$

   (c) $\frac{-x}{9\sqrt{4x^2-9}} + c$

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3. To find the intersection of the circle and ellipse solve:

$$x^2+16(y-r)^2=x^2+y^2$$
$$16(y-r)^2=y^2$$
$$y-r=\pm\frac{y}{4}$$
$$\frac{3y}{4}=r \text{ OR } \frac{5y}{4}=r$$
$$y=\frac{4r}{3} \text{ OR } y=\frac{4r}{5}$$

But $y<r$ and hence $y=\frac{4r}{5}$

Substituting into the equation of the circle :

$$x^2+\frac{16r^2}{25}=r^2$$
$$x=\pm\frac{3r}{5}$$

Rearranging the equations of the circle and ellipse:

Circle:

$$x^2+y^2=r^2$$
$$y=\sqrt{r^2-x^2}$$

(y is +ve in the shaded area)

Ellipse:

$$x^2+16(y-r)^2=r^2$$
$$(y-r)^2=\frac{r^2-x^2}{16}$$
$$y-r=\frac{-\sqrt{r^2-x^2}}{4}$$

(y − r is negative in the shaded area)

$$y=r-\frac{\sqrt{r^2-x^2}}{4}$$

$$\text{Area}=2\int_0^{3r/5}\sqrt{r^2-x^2}+\frac{\sqrt{r^2-x^2}}{4}-r\,dx$$
$$\text{Area}=2\int_0^{3r/5}\frac{5}{4}\sqrt{r^2-x^2}\,dx-2r\int_0^{3r/5}dx$$
$$\text{Area}=\frac{5}{2}\int_0^{3r/5}\sqrt{r^2-x^2}\,dx-2r[x]_0^{3r/5}$$

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Let $x = r\sin u$

$dx = r\cos u\,du$

To evaluate the new limits of integration:

$3r/5 = r\sin u$ and $0 = r\sin u$

$u = \sin^{-1}\left(3/5\right)$ and $u = 0$

$$\text{Area} = \frac{5}{2}\int_0^{\sin^{-1}\left(3/5\right)} \sqrt{r^2 - r^2\sin^2 u}\; r\cos u\,du - \frac{6r^2}{5}$$

$$\text{Area} = \frac{5r^2}{2}\int_0^{\sin^{-1}\left(3/5\right)} \sqrt{1-\sin^2 u}\;\cos u\,du - \frac{6r^2}{5}$$

$$\text{Area} = \frac{5r^2}{2}\int_0^{\sin^{-1}\left(3/5\right)} \cos^2 u\,du - \frac{6r^2}{5}$$

$$\text{Area} = \frac{5r^2}{4}\int_0^{\sin^{-1}\left(3/5\right)} \left(\cos 2u + 1\right)du - \frac{6r^2}{5}$$

$$\text{Area} = \frac{5r^2}{4}\left[\frac{1}{2}\sin 2u + u\right]_0^{\sin^{-1}\left(3/5\right)} - \frac{6r^2}{5}$$

$$\text{Area} = \frac{5r^2}{4}\left[\sin u\cos u + u\right]_0^{\sin^{-1}\left(3/5\right)} - \frac{6r^2}{5}$$

Now, if $\sin u = 3/5$ then $\cos u = 4/5$

Hence $\sin\left(\sin^{-1}\left(3/5\right)\right) = 3/5$ and $\cos\left(\sin^{-1}\left(3/5\right)\right) = 4/5$

$$\text{Area} = \frac{5r^2}{4}\left[3/5 . 4/5 + \sin^{-1}\left(3/5\right)\right] - \frac{6r^2}{5}$$

$$\text{Area} = \frac{3r^2}{5} + \frac{5r^2}{4}.\sin^{-1}\left(3/5\right) - \frac{6r^2}{5}$$

$$\text{Area} = \frac{5r^2}{4}.\sin^{-1}\left(3/5\right) - \frac{3r^2}{5}$$

$$\text{Area} = 0.2044r^2$$

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Ex 4.2 Integration by Parts (page 153)

1. (a) $x\sin x+\cos x+c$ (b) $-2xe^{-x}-2e^{-x}+c$ (c) $-\frac{\ln x}{x}-\frac{1}{x}+c$

(d) $\frac{e^x}{5}(\cos 2x+2\sin 2x)+c$

2. $\int \ln^n x\,dx=\int 1.\ln^n x\,dx$

$v'=1 \quad u=\ln^n x$

$v=x \quad u'=n.\frac{1}{x}.\ln^{n-1} x$

$\int \ln^n x\,dx=x\ln^n x-\int n.\frac{1}{x}.\ln^{n-1} x.x\,dx$

$\int \ln^n x\,dx=x\ln^n x-n\int \ln^{n-1} x\,dx$

$\int \ln^3 x\,dx=x\ln^3 x-3x\ln^2 x+6x\ln x-6x$

Ex 4.3 Further Trigonometric Techniques (page 159)

1. (a) $-\cos x+\frac{2\cos^3 x}{3}-\frac{\cos^5 x}{5}+c$ (b) $-\ln|\cos x|+c$ OR $\ln|\sec x|+c$

(c) $\frac{\tan^2 x}{2}+\ln|\sec x|+c$ (d) $\frac{\tan^5 x}{5}+\frac{\tan^3 x}{3}+c$

(e) $-\frac{\cos^3 x}{3}+\frac{2\cos^5 x}{5}-\frac{\cos^7 x}{7}+c$

2. (i) Let $u = a - x$

$\frac{du}{dx}=-1$ and $x=a\Rightarrow u=0,\ x=0\Rightarrow u=a$

$\int_0^a f(a-x)dx$

$=-\int_a^0 f(u)du$

$=\int_0^a f(u)du$

$=\int_0^a f(x)dx$

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The final step can be made because it does not matter which letter is used for the variable.

(ii)
$$\int_0^{\frac{\pi}{2}} \frac{\sin^n x}{\sin^n x+\cos^n x}\,dx$$

$$=\int_0^{\frac{\pi}{2}} \frac{\sin^n\left(\frac{\pi}{2}-x\right)}{\sin^n\left(\frac{\pi}{2}-x\right)+\cos^n\left(\frac{\pi}{2}-x\right)}\,dx$$

$$=\int_0^{\frac{\pi}{2}} \frac{\cos^n x}{\cos^n x+\sin^n x}\,dx$$

The above step should be obvious given part (i), but what follows requires a little insight:

$$2\times\int_0^{\frac{\pi}{2}} \frac{\sin^n x}{\sin^n x+\cos^n x}\,dx=\int_0^{\frac{\pi}{2}} \frac{\sin^n x}{\sin^n x+\cos^n x}\,dx+\int_0^{\frac{\pi}{2}} \frac{\cos^n x}{\sin^n x+\cos^n x}\,dx$$

$$2\times\int_0^{\frac{\pi}{2}} \frac{\sin^n x}{\sin^n x+\cos^n x}\,dx=\int_0^{\frac{\pi}{2}} \frac{\sin^n x+\cos^n x}{\sin^n x+\cos^n x}\,dx$$

$$2\times\int_0^{\frac{\pi}{2}} \frac{\sin^n x}{\sin^n x+\cos^n x}\,dx=\int_0^{\frac{\pi}{2}} 1\,dx$$

$$2\times\int_0^{\frac{\pi}{2}} \frac{\sin^n x}{\sin^n x+\cos^n x}\,dx=[x]_0^{\frac{\pi}{2}}=\frac{\pi}{2}\Rightarrow\int_0^{\frac{\pi}{2}} \frac{\sin^n x}{\sin^n x+\cos^n x}\,dx=\frac{\pi}{4}$$

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3.

$$y = x\sin(nx) + \frac{1}{n}\cos(nx)$$

$$\frac{dy}{dx} = \sin(nx) + nx\cos(nx) - \sin(nx)$$

$$\frac{dy}{dx} = nx\cos(nx)$$

$$I_n = \int_0^{\pi}\left(\frac{\pi}{2} - x\right)\sin\left(\left(n + \frac{1}{2}\right)x\right)\operatorname{cosec}\left(\frac{1}{2}x\right)dx$$

$$I_{n-1} = \int_0^{\pi}\left(\frac{\pi}{2} - x\right)\sin\left(\left(n - \frac{1}{2}\right)x\right)\operatorname{cosec}\left(\frac{1}{2}x\right)dx$$

$$I_n - I_{n-1} = \int_0^{\pi}\left(\frac{\pi}{2} - x\right)\operatorname{cosec}\left(\frac{1}{2}x\right)\left[\sin\left(\left(n + \frac{1}{2}\right)x\right) - \sin\left(\left(n - \frac{1}{2}\right)x\right)\right]dx$$

$$I_n - I_{n-1} = \int_0^{\pi}\left(\frac{\pi}{2} - x\right)\operatorname{cosec}\left(\frac{1}{2}x\right)2\cos(nx)\sin\left(\frac{1}{2}x\right)dx$$

$$I_n - I_{n-1} = \int_0^{\pi}\left(\frac{\pi}{2} - x\right)2\cos(nx)\,dx$$

$$I_n - I_{n-1} = \int_0^{\pi}\pi\cos(nx)\,dx - 2\int_0^{\pi}x\cos(nx)\,dx$$

$$I_n - I_{n-1} = \frac{\pi}{n}\left[\sin(nx)\right]_0^{\pi} - \frac{2}{n}\int_0^{\pi}nx\cos(nx)\,dx$$

$$I_n - I_{n-1} = \frac{\pi}{n}(0-0) - \frac{2}{n}\left[x\sin(nx) + \frac{1}{n}\cos(nx)\right]_0^{\pi}$$

$$I_n - I_{n-1} = -\frac{2}{n}\left(0 + \frac{1}{n}\cos(n\pi) - \frac{1}{n}\right)$$

$$I_n - I_{n-1} = \frac{2}{n^2}\left(1 - \cos(n\pi)\right)$$

Now, the value of $\cos(n\pi)$ is +1 if n is even and -1 if n is odd. It is thus necessary to look at the result separately for n odd or even:

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n Even **n Odd**

$I_n - I_{n-1} = \frac{2}{n^2}(1-1)$ $I_n - I_{n-1} = \frac{2}{n^2}(1+1)$

$I_n - I_{n-1} = 0$ $I_n - I_{n-1} = \frac{4}{n^2}$

Given that $I_0 = 0$:

$n=1 \quad I_1 - I_0 = \frac{4}{1^2} - 0 = \frac{4}{1^2}$

$n=2 \quad I_2 - I_1 = 0 \Rightarrow I_2 = I_1 = \frac{4}{1^2}$

$n=3 \quad I_3 - I_2 = \frac{4}{3^2} \Rightarrow I_3 = \frac{4}{1^2} + \frac{4}{3^2}$

$n=4 \quad I_4 - I_3 = 0 \Rightarrow I_4 = I_3 = \frac{4}{1^2} + \frac{4}{3^2}$

$n=5 \quad I_5 - I_4 = \frac{4}{5^2} \Rightarrow I_5 = \frac{4}{1^2} + \frac{4}{3^2} + \frac{4}{5^2}$

$n=6 \quad I_6 - I_5 = 0 \Rightarrow I_6 = I_5 = \frac{4}{1^2} + \frac{4}{3^2} + \frac{4}{5^2}$

Summarising the odd and even patterns:

n Even **n Odd**

$\frac{4}{1^2} + \frac{4}{3^2} + \ldots \frac{4}{(n-1)^2}$ $\frac{4}{1^2} + \frac{4}{3^2} + \ldots \frac{4}{n^2}$

$= 4\sum_{2}^{n} \frac{1}{(i-1)^2}$ $= 4\sum_{1}^{n} \frac{1}{i^2}$

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Ex 4.4 Solids of Revolution (page 169)

1. (a) $\frac{\pi}{9}$ (b) $\frac{\pi^2}{2}$

2. $\frac{\pi}{2}$

3. (a) $\frac{\pi}{2}\left(e^2+1\right)$ - use integration by parts (b) $\frac{27\pi}{2}$ - y is negative

(c) $\pi(\pi+2)$ - use integration by parts and x is negative.

4.

(a) (i) $\frac{dy}{dx}=2x\left(1+x^2\right)^{-1}-2x^3\left(1+x^2\right)^{-2}(\ )$

$\frac{dy}{dx}=\frac{2x}{\left(1+x^2\right)}-\frac{2x^3}{\left(1+x^2\right)^2}$

$\frac{dy}{dx}=\frac{2x+2x^3-2x^3}{\left(1+x^2\right)^2}$

$\frac{dy}{dx}=\frac{2x}{\left(1+x^2\right)^2}$

$\frac{dy}{dx}=\frac{1}{2}$

$\frac{2x}{\left(1+x^2\right)^2}=\frac{1}{2}$

$1+2x^2+x^4=4x$

$x^4+2x^2-4x+1=0$

But $x=1$ is a solution:

$(x-1)\left(x^3+x^2+3x-1\right)=0$

If there is a point between $x=\frac{1}{4}$ and $x=\frac{1}{2}$ where $\frac{dy}{dx}=\frac{1}{2}$ then

for $f(x)=x^3+x^2+3x-1$

$f\left(\frac{1}{4}\right)$ and $f\left(\frac{1}{2}\right)$ must have opposite signs.

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$f\left(\frac{1}{4}\right)=-\frac{11}{64}$ and $f\left(\frac{1}{2}\right)=+\frac{7}{8}$

Hence there is a solution between $x=\frac{1}{4}$ and $x=\frac{1}{2}$ where $\frac{dy}{dx}=\frac{1}{2}$

(ii) Translating the curve by one unit to the left:

$$y=\frac{x^2}{1+x^2}$$

$$y+yx^2=x^2$$

$$y=x^2(1-y)$$

$$x^2=\frac{y}{1-y}$$

$$\text{Volume}=\pi\int_0^{1/2}\frac{y}{1-y}dy$$

Let $u=1-y$

$$\frac{du}{dy}=-1$$

Limits become $u=\frac{1}{2}$ and $u=1$

$$\text{Volume}=-\pi\int_1^{1/2}\frac{1-u}{u}\,du$$

$$\text{Volume}=\pi\int_{1/2}^{1}\frac{1-u}{u}\,du$$

$$\text{Volume}=\pi\int_{1/2}^{1}\frac{1}{u}-1\,du$$

$$\text{Volume}=\pi\left[\ln u-u\right]_{1/2}^{1}$$

$$\text{Volume}=\pi\left[-1-\ln\left(\frac{1}{2}\right)+\frac{1}{2}\right]$$

$$\text{Volume}=\pi\left[\ln(2)-\frac{1}{2}\right] \text{ OR } \frac{\pi}{2}(\ln 4-1)$$

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5.

(a) (i)

$$\text{Area} = \int_0^a f(y)\,dy$$

$$y = k + \frac{1}{m}\ln\left(\frac{k}{x}\right)$$

$$my - mk = \ln\left(\frac{k}{x}\right)$$

$$\frac{k}{x} = e^{my-mk}$$

$$x = ke^{mk-my}$$

$$\text{Area} = k\int_0^a e^{mk-my}\,dy$$

$$\text{Area} = -\frac{k}{m}\left[e^{mk-my}\right]_0^a$$

$$\text{Area} = -\frac{k}{m}\left(e^{mk-ma} - e^{mk}\right)$$

$$\text{Area} = -\frac{ke^{mk}}{m}\left(e^{-ma} - 1\right)$$

When this area is p% of the maximum:

$$-\frac{ke^{mk}}{m}\left(e^{-ma} - 1\right) = \frac{p}{100} \times \frac{k}{m}e^{mk}$$

$$\left(1 - e^{-ma}\right) = \frac{p}{100}$$

$$e^{-ma} = 1 - \frac{p}{100}$$

$$e^{-ma} = \frac{100 - p}{100}$$

$$e^{ma} = \frac{100}{100 - p}$$

$$ma = \ln\left(\frac{100}{100 - p}\right)$$

$$a = \frac{1}{m}\ln\left(\frac{100}{100 - p}\right)$$

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(ii)

$$\text{Volume} = k^2\pi\int_0^a e^{2mk-2my}\,dy$$

$$\text{Volume} = -\frac{k^2\pi}{2m}\left[e^{2mk-2my}\right]_0^a$$

$$\text{Volume} = -\frac{k^2\pi}{2m}\left[e^{2mk-2ma} - e^{2mk}\right]$$

$$\text{Volume} = -\frac{k^2\pi e^{2mk}}{2m}\left[e^{-2ma} - 1\right]$$

$$\frac{V}{A} \le e^{mk}$$

$$\frac{-\frac{k^2\pi e^{2mk}}{2m}\left[e^{-2ma}-1\right]}{-\frac{ke^{mk}}{m}\left(e^{-ma}-1\right)} \le e^{mk}$$

$$\frac{\frac{k\pi e^{mk}}{2}\left[e^{-2ma}-1\right]}{\left(e^{-ma}-1\right)} \le e^{mk}$$

$$\frac{k\pi\left[e^{-2ma}-1\right]}{2\left(e^{-ma}-1\right)} \le 1$$

$$\frac{k\pi\left[e^{-2ma}-1\right]}{\left(e^{-ma}-1\right)} \times \frac{\left(e^{-ma}+1\right)}{\left(e^{-ma}+1\right)} \le 2$$

$$\frac{k\pi\left[e^{-2ma}-1\right]\left(e^{-ma}+1\right)}{e^{-2ma}-1} \le 2$$

$$k\pi\left(e^{-ma}+1\right) \le 2$$

$$k\pi e^{-ma} + k\pi \le 2$$

$$e^{-ma} \le \frac{2-k\pi}{k\pi}$$

$$e^{ma} \ge \frac{k\pi}{2-k\pi}$$

$$a \ge \frac{1}{m}\ln\left(\frac{k\pi}{2-k\pi}\right)$$

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6. $x = 2a\cos(2y) + b$

$$\text{Volume} = \pi\int_0^{\pi} 4a^2\cos^2(2y) + 4ab\cos(2y) + b^2\,dy$$

$$\text{Volume} = \pi\int_0^{\pi} 2a^2(\cos 4y + 1) + 4ab\cos(2y) + b^2\,dy$$

$$\text{Volume} = \pi\left[2a^2\left(\frac{1}{4}\sin 4y + y\right) + 2ab\sin 2y + b^2y\right]_0^{\pi}$$

$$\text{Volume} = \pi\left[\left(2a^2\left(\frac{1}{4}\times 0 + \pi\right) + 2ab\times 0 + b^2\pi\right) - 0\right]$$

$$\text{Volume} = \pi\left[2a^2\pi + b^2\pi\right]$$

$$\text{Volume} = \pi^2\left[2a^2 + b^2\right]$$

Ex 4.5 Arc Length and Surface of Revolution (page 184)

1. (a) $\frac{128\pi}{3}\left(2\sqrt{2} - 1\right)$ (b) $\frac{54\pi}{5}$

2. (a) $\frac{13\pi}{3}$ (b) $\pi\left(e\sqrt{e^2+1} + \ln\left|e + \sqrt{e^2+1}\right| - \sqrt{2} - \ln\left|\sqrt{2}+1\right|\right)$

3. (a) $\frac{56}{27}$ (b) $\sqrt{2}\left(e^{\pi} - 1\right)$

Ex 4.6.1 Differential Equations (page 191)

1. (a) $y = 1 - 2x$ OR $y = 2x - 3$ (b) $y = \frac{1}{\ln\left|A\cot x\right|}$

(c) $2e^{y^3} - 3e^{2x} + 1 = 0$

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2. (a) $\dfrac{A}{x}+\dfrac{B}{P-x}=\dfrac{1}{x(P-x)}$

$\dfrac{A(P-x)+Bx}{x(P-x)}=\dfrac{1}{x(P-x)}$

Substituting $x=P$ on the numerator:

$BP=1$

$B=\dfrac{1}{P}$

Substituting $x=0$ on the numerator:

$AP=1$

$A=\dfrac{1}{P}$

Hence, $\dfrac{1}{x(P-x)}=\dfrac{1}{P}\left(\dfrac{1}{x}+\dfrac{1}{P-x}\right)$

(b)

$\dfrac{dN}{dt}=kN(P-N)$

$\displaystyle\int\frac{dN}{N(P-N)}=\int k\,dt$

Using the answer to (a):

$\dfrac{1}{P}\displaystyle\int\frac{1}{N}+\frac{1}{P-N}dN=\int k\,dt$

$\dfrac{1}{P}\left(\ln|N|-\ln|P-N|\right)=kt+c$

$\ln\left|\dfrac{N}{P-N}\right|=kPt+cP$

$\dfrac{N}{P-N}=e^{kPt+c'}$ – the $|\ |$ sign can be discarded as $P,\ P-N>0$

$\dfrac{N}{P-N}=Ae^{kPt}$

$N=PAe^{kPt}-NAe^{kPt}$

$N\left(Ae^{kPt}+1\right)=PAe^{kPt}$

$N=\dfrac{APe^{kPt}}{Ae^{kPt}+1}$

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(c) $N=\dfrac{APe^{kPt}}{Ae^{kPt}+1}$

$t=0,\ N=0.005P$

$0.005P=\dfrac{APe^{0}}{Ae^{0}+1}$

$0.005=\dfrac{A}{A+1}$

$0.005A+0.005=A$

$0.995A=0.005$

$A=\dfrac{1}{199}$

$N=\dfrac{Pe^{kPt}}{e^{kPt}+199}$

When $N=\dfrac{P}{5}$ then $\dfrac{dN}{dt}=0.08P$.

Reverting to the original DE:

$\dfrac{dN}{dt}=kN(P-N)$

$0.08P=k\dfrac{P}{5}\left(P-\dfrac{P}{5}\right)$

$0.08=\dfrac{k}{5}\times\dfrac{4P}{5}$

$k=\dfrac{0.5}{P}$ OR $kP=0.5$

Substituting into $N=\dfrac{Pe^{kPt}}{e^{kPt}+199}$

$N=\dfrac{Pe^{0.5t}}{e^{0.5t}+199}$

Finally solving:

$0.5P=\dfrac{Pe^{0.5t}}{e^{0.5t}+199}$

$0.5=\dfrac{e^{0.5t}}{e^{0.5t}+199}$

$0.5e^{0.5t}+0.5\times199=e^{0.5t}$

$0.5e^{0.5t}=0.5\times199$

$0.5t=\ln(199)$

$t=10.58$

Ex 4.6.2 Homogeneous Differential Equations (page 197)

1. (a) $x^2=y^2\ln\left|\dfrac{A}{x^2}\right|$ (b) $x^3-x^2y-x-y=0$ (c) $x=e^{\sin\left(y/x\right)}$

2. The gradient of the family of curves y = f(x) is given by $\dfrac{dy}{dx}=\dfrac{2y+x}{y}$

Hence the gradient of the perpendicular family y = g (x) is given by $\dfrac{dy}{dx}=-\dfrac{y}{2y+x}$

To solve $\dfrac{dy}{dx}=-\dfrac{y}{2y+x}$, let $z=\dfrac{x}{y}=xy^{-1}$

$z=xy^{-1}$

$\dfrac{dz}{dx}=\dfrac{1}{y}-\dfrac{x}{y^2}\dfrac{dy}{dx}$ (1)

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Now, dividing top and bottom of the DE by y will allow it to be expressed in terms of z:

$$\frac{dy}{dx}=-\frac{y}{2y+x}$$

$$\frac{dy}{dx}=-\frac{1}{2+\frac{x}{y}}$$

$$\frac{dy}{dx}=\frac{-1}{2+z}$$

Substituting this expression into equation (1):

$$\frac{dz}{dx}=\frac{1}{y}-\frac{x}{y^2}\times\frac{-1}{2+z}$$

$$\frac{dz}{dx}=\frac{1}{y}\left(1-\frac{x}{y}\times\frac{-1}{2+z}\right)$$

$$\frac{dz}{dx}=\frac{z}{x}\left(1-z\times\frac{-1}{2+z}\right)$$

$$x\frac{dz}{dx}=z\left(\frac{2+2z}{2+z}\right)$$

$$x\frac{dz}{dx}=2z\left(\frac{1+z}{2+z}\right)$$

$$\frac{1}{2}\int\frac{z+2}{z(z+1)}dz=\int\frac{dx}{x}$$

Using partial fractions, or the given expression:

$$\frac{1}{2}\int\frac{2}{z}-\frac{1}{z+1}dz=\int\frac{dx}{x}$$

$$2\ln|z|-\ln|z+1|=2\ln kx$$

$$\ln\left|\frac{z^2}{z+1}\right|=\ln|Ax^2|$$

$$\frac{z^2}{z+1}=Ax^2$$

Substituting back for z:

$$\frac{\frac{x^2}{y^2}}{\frac{x}{y}+1}=Ax^2$$

$$\frac{1}{y(x+y)}=A$$

$$y(x+y)=C$$

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Ex 4.6.3 Linear DE's and Integrating Factors (page 202)

1. $y=\frac{1}{3}+Ce^{-x^3}$ 2. $y=\frac{x}{2}+\frac{2}{x}$ 3. $y=\sin x+\cos x$ 4. $y=e^{-x^2}+\frac{x^2}{2}-\frac{1}{2}$

Ex 4.6.4 Reducing the Order (page 207)

1. (a) $y=\ln|x+1|$ OR $y=c$ (b) $y=\frac{2\left(e^{2x}-1\right)}{e^{2x}+1}$ OR $y=\frac{2\left(e^{2x}+1\right)}{e^{2x}-1}$ OR $y=c$

(c) $y=e^x-x$

2. $y=\sin(\ln x)$

$\frac{dy}{dx}=\frac{1}{x}\cos(\ln x)$ and $\frac{d^2y}{dx^2}=-\frac{1}{x^2}\cos(\ln x)-\frac{1}{x}\times\frac{1}{x}\sin(\ln x)$

Substituting into the DE:

$x^2\times-\frac{1}{x^2}\left(\cos(\ln x)+\sin(\ln x)\right)+x\times\frac{1}{x}\cos(\ln x)+\sin(\ln x)=k$

$-\left(\cos(\ln x)+\sin(\ln x)\right)+\cos(\ln x)+\sin(\ln x)=k$

$k=0$

To solve the following DE, use the product rule on the LHS:

$\frac{d\left(x^2\frac{dy}{dx}\right)}{dx}=x\frac{dy}{dx}-y+5$

$2x\frac{dy}{dx}+x^2\frac{d^2y}{dx^2}=x\frac{dy}{dx}-y+5$

$x^2\frac{d^2y}{dx^2}+x\frac{dy}{dx}+y=5$ (2)

This is the same as the first DE with k = 5. Hence the solution is likely to be $y=\sin(\ln x)+5$

It is easy to see that the first and second derivatives are the same as before.

The only difference on the LHS is the 5 from the y term.

Hence $y=\sin(\ln x)+5$ is a solution to equation (2).

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3. ***This question from the 2005 scholarship paper uses reducing the order of a differential equation as the first step of the solution process.***

$$\frac{dx}{dy}.\frac{d^2y}{dx^2}=k\frac{dy}{dx}$$

$$\frac{d^2y}{dx^2}=k\left(\frac{dy}{dx}\right)^2$$

Let $w=\frac{dy}{dx}\Rightarrow\frac{d^2y}{dx^2}=\frac{dw}{dx}$

$$\frac{dw}{dx}=kw^2$$

$$\int w^{-2}\,dw=\int k\,dx$$

$$-\frac{1}{w}=kx+c$$

$$\frac{dx}{dy}=-(kx+c)$$

$$\frac{dy}{dx}=-\frac{1}{kx+c}$$

At $x=0$, $\frac{dy}{dx}=1$

$$1=-\frac{1}{c}$$

$$c=-1$$

$$\frac{dy}{dx}=\frac{1}{1-kx}$$

$$\int dy=\int\frac{1}{1-kx}dx$$

$$y=-\frac{1}{k}\ln|1-kx|+c$$

At $x=0$, $y=1$

$$1=-\frac{1}{k}\ln 1+c$$

$$c=1$$

$$y=-\frac{1}{k}\ln|1-kx|+1$$

At $y=2$

$$2=-\frac{1}{k}\ln|1-kx|+1$$

$$kx=1-e^{-k}$$

$$x=\frac{1-e^{-k}}{k}$$

Substituting for k in $\frac{dy}{dx}$:

$$\frac{dy}{dx}=\frac{1}{1-kx}$$

$$\frac{dy}{dx}=\frac{1}{1-k\left(x=\frac{1-e^{-k}}{k}\right)}$$

$$\frac{dy}{dx}=\frac{1}{1-1+e^{-k}}$$

$$\frac{dy}{dx}=e^{k}$$

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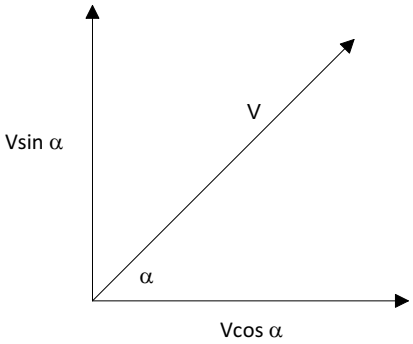
Ex 4.6.5 Mixed Integration Questions (page 212)

1.

(a)

The diagram shows the velocity split into initial horizontal and vertical components.

These are the initial velocity conditions in the solution of the differential equations.



$$\frac{d^2x}{dt^2}=0 \qquad \frac{d^2y}{dt^2}=-g$$

$$\frac{dx}{dt}=c \qquad \frac{dy}{dt}=-gt+c$$

$$\frac{dx}{dt}=V\cos\alpha \qquad \frac{dy}{dt}=-gt+V\sin\alpha$$

$$x=V\cos\alpha.t+c \qquad y=-\frac{gt^2}{2}+V\sin\alpha.t+c$$

In both cases the initial distance travelled is zero:

$$x=V\cos\alpha.t \qquad y=-\frac{gt^2}{2}+V\sin\alpha.t$$

Rearranging the x equation:

$$t=\frac{x}{V\cos\alpha}$$

Substituting in y:

$$y=-\frac{g\left(\frac{x}{V\cos\alpha}\right)^2}{2}+V\sin\alpha\frac{x}{V\cos\alpha}$$

$$y=-\frac{gx^2}{2V^2\cos^2\alpha}+x\tan\alpha$$

$$y=-\frac{gx^2\sec^2\alpha}{2V^2}+x\tan\alpha$$

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(b) Reading the question it refers to conditions on the angle α.

Therefore the equation for the path of the motion must be considered in terms of α, or specifically $\tan\alpha$. It is therefore necessary to express $\sec^2\alpha$ in terms of $\tan^2\alpha$.

(i) At $(kh,\ h)$:

$$h=-\frac{gk^2h^2\sec^2\alpha}{2V^2}+kh\tan\alpha$$

$$h=-\frac{gk^2h^2\left(\tan^2\alpha+1\right)}{2V^2}+kh\tan\alpha$$

$$2V^2=-gk^2h\left(\tan^2\alpha+1\right)+2V^2k\tan\alpha$$

$$gk^2h\tan^2\alpha-2V^2k\tan\alpha+2V^2+gk^2h=0$$

If there are two possible paths then the discriminant of the equation must be positive:

$$4k^2V^4-4gk^2h\left(gk^2h+2V^2\right)>0$$

$$V^4-gh\left(gk^2h+2V^2\right)>0$$

$$V^4-2ghV^2-g^2k^2h^2>0$$

$$V^2>\frac{2gh\pm\sqrt{4g^2h^2+4g^2h^2k^2}}{2}$$

$$V^2>\frac{2gh\pm2gh\sqrt{1+k^2}}{2}$$

$$V^2>gh\left(1\pm\sqrt{1+k^2}\right)$$

But $V^2>0$, so:

$$V^2>gh\left(1+\sqrt{1+k^2}\right)$$

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(ii) Now, the question asks that we show that $\alpha_1+\alpha_2=\tan^{-1}(-k)$ OR $\tan(\alpha_1+\alpha_2)=-k$

This suggests the use of the compound angle for tangent formula:

$$\tan(\alpha_1+\alpha_2)=\frac{\tan\alpha_1+\tan\alpha_2}{1-\tan\alpha_1\tan\alpha_2}$$

The formula contains the sum and product of the roots of a quadratic and hence these formulae should be applied to the equation:

$$gk^2h\tan^2\alpha-2V^2k\tan\alpha+2V^2+gk^2h=0$$

$$\tan\alpha_1+\tan\alpha_2=\frac{2kV^2}{gk^2h} \quad \text{and} \quad \tan\alpha_1\tan\alpha_2=\frac{gk^2h+2V^2}{gk^2h}$$

$$\tan(\alpha_1+\alpha_2)=\frac{\frac{2kV^2}{gk^2h}}{1-\frac{gk^2h+2V^2}{gk^2h}}$$

$$\tan(\alpha_1+\alpha_2)=\frac{\frac{2kV^2}{gk^2h}}{\frac{gk^2h-gk^2h-2V^2}{gk^2h}}$$

$$\tan(\alpha_1+\alpha_2)=\frac{2kV^2}{gk^2h-gk^2h-2V^2}$$

$$\tan(\alpha_1+\alpha_2)=\frac{2kV^2}{-2V^2}$$

$$\tan(\alpha_1+\alpha_2)=-k$$

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2. (a) $V = \pi\int_0^h y^2 dx$

$V = 4a\pi\int_0^h x dx$

$V = 4a\pi\left[\frac{x^2}{2}\right]_0^h$

$V = 4a\pi\frac{h^2}{2}$

$V = 2a\pi h^2$

(b) $S = 2\pi\int_0^h y\sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx$

Now,

$y^2 = 4ax$

$2y\frac{dy}{dx} = 4a$

$\frac{dy}{dx} = \frac{2a}{y} = \frac{2a}{2\sqrt{ax}} = \sqrt{\frac{a}{x}}$

Substitute in the formula above:

$S = 2\pi\int_0^h 2\sqrt{ax}\sqrt{1+\frac{a}{x}}\,dx$

$S = 4\pi\int_0^h \sqrt{ax}\sqrt{1+\frac{a}{x}}\,dx$

$S = 4\pi\int_0^h \sqrt{ax+a^2}\,dx$

$S = 4\pi\left[\frac{2}{3a}\left(ax+a^2\right)^{3/2}\right]_0^h$

$S = \frac{8\pi}{3a}\left[\left(ah+a^2\right)^{3/2} - a^3\right]$

$S = \frac{8\pi}{3a}\left[a\sqrt{a}\left(h+a\right)^{3/2} - a^3\right]$

$S = \frac{8\pi}{3}\left[\sqrt{a}\left(h+a\right)^{3/2} - a^2\right]$

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(c) $\frac{dh}{dt} = -0.26$ when $h = 8a$

$$\frac{dV}{dt} = \frac{dh}{dt} \times \frac{dV}{dh}$$

From (a):

$$V = 2\pi a h^2$$

$$\frac{dV}{dh} = 4\pi a h$$

Now we are given that $\frac{dV}{dt} \alpha$ Surface Area

$$\frac{dV}{dt} = k\frac{8\pi}{3}\left[\sqrt{a}(h+a)^{3/2} - a^2\right]$$

At $h = 8a$:

$$\frac{dV}{dt} = k\frac{8\pi}{3}\left[\sqrt{a}(9a)^{3/2} - a^2\right]$$

$$\frac{dV}{dt} = k\frac{8\pi}{3}\left[27a^2 - a^2\right]$$

$$\frac{dV}{dt} = k\frac{8\pi \times 26a^2}{3}$$

At $h = 8a$, $\frac{dh}{dt} = -0.26$

$$\frac{dh}{dt} = \frac{dV}{dt} \div \frac{dV}{dh}$$

$$-0.26 = \frac{k\frac{8\pi \times 26a^2}{3}}{4\pi a.8a}$$

$$-0.26 = k\frac{26}{12}$$

$$k = -0.12$$

To find $\frac{dV}{dt}$ at $h = 3a$:

$$\frac{dV}{dt} = \frac{-0.12 \times 8\pi}{3}\left[\sqrt{a}(4a)^{3/2} - a^2\right]$$

$$\frac{dV}{dt} = \frac{-0.12 \times 8\pi}{3}\left[7a^2\right]$$

$$\frac{dV}{dt} = -2.24\pi a^2$$

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Chapter 5 Conic Sections

Ex 5 Conic Sections (page 232)

1. (a) (i) Gradient of the parabola:

$y^2 = 4ax$

$2y\frac{dy}{dx} = 4a$

$\frac{dy}{dx} = \frac{2a}{y}$

So at (x_1, y_1):

$\frac{dy}{dx} = \frac{2a}{y_1}$

Gradient of ellipse:

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$\frac{2x}{a^2} + \frac{2y}{b^2}\frac{dy}{dx} = 0$

$\frac{dy}{dx} = -\frac{b^2x}{a^2y}$

At (x_1, y_1):

$\frac{dy}{dx} = -\frac{b^2x_1}{a^2y_1}$

But the tangents at (x_1, y_1) are perpendicular, so:

$\frac{2a}{y_1} \times -\frac{b^2x_1}{a^2y_1} = -1$

$\frac{2b^2x_1}{ay_1^2} = 1$

But $y_1^2 = 4ax_1$ and hence:

$\frac{2b^2x_1}{a4ax_1} = 1$

$2b^2 = 4a^2$

$b^2 = 2a^2$

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(ii) Distance from P to the origin $D=\sqrt{x_1^2+y_1^2}$

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$$

Using the result from (i)

$$\frac{x^2}{a^2}+\frac{y^2}{2a^2}=1$$

But $y^2=4ax$:

$$\frac{x^2}{a^2}+\frac{4ax}{2a^2}=1$$

$$x^2+2ax=a^2$$

$$x^2+2ax-a^2=0$$

$$x=\frac{-2a\pm\sqrt{4a^2-4\times-a^2}}{2}$$

$$x=\frac{-2a\pm\sqrt{8a^2}}{2}$$

$$x=\frac{-2a\pm2a\sqrt{2}}{2}$$

$$x=-a\pm\sqrt{2}a$$

But in this case x is positive, so $x=-a+\sqrt{2}a$

$$y^2=4ax$$

$$y^2=4a\left(-a+\sqrt{2}a\right)$$

$$y^2=4a^2\left(-1+\sqrt{2}\right)$$

and

$$x^2=\left(-a+\sqrt{2}a\right)^2$$

$$x^2=a^2\left(-1+\sqrt{2}\right)^2$$

Hence:

$$D=\sqrt{\left(a^2\left(-1+\sqrt{2}\right)^2+4a^2\left(-1+\sqrt{2}\right)\right)}$$

$$D=a\sqrt{\left(1-2\sqrt{2}+2-4+4\sqrt{2}\right)}$$

$$D=a\sqrt{2\sqrt{2}-1}$$

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(b) The tangent of the parabola is:

$yy_1 = 2a(x + x_1)$

But at $y = 0$:

$2a(x + x_1) = 0$

$x = -x_1$

$x = a - \sqrt{2}a$ is the x-ordinate at M

The tangent of the ellipse is:

$\frac{xx_1}{a^2} + \frac{yy_1}{2a^2} = 1$

But at $y = 0$:

$xx_1 = a^2$

$x\left(\sqrt{2}a - a\right) = a^2$

$ax\left(\sqrt{2} - 1\right) = a^2$

$x = \frac{a}{\sqrt{2} - 1} \times \frac{\sqrt{2} + 1}{\sqrt{2} + 1}$

$x = \sqrt{2}a + a$ is the x-ordinate of N

$MN = \sqrt{2}a + a - \left(a - \sqrt{2}a\right)$

$MN = 2\sqrt{2}a$

Note:

This solution is made much easier by:

(i) using the shortcut method for equation of a tangent and

(ii) substituting y = 0 as early as possible.

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2. (a) This is a hyperbola centre $(2h,h)$

The equation of the tangent at (p,q) is:

$$4px-qy-8h(x+p)+h(y+q)+15h^2-4a^2=0$$
$$(h-q)y+4x(p-2h)-8ph+hq+15h^2-4a^2=0$$
$$(h-q)y=4(2h-p)x+8ph-hq-15h^2+4a^2$$

Dividing by $(h-q)$ gives gradient $\dfrac{4(2h-p)}{h-q}$

Now we are given that the gradient is e^2-1

$$\frac{4(2h-p)}{h-q}=e^2-1$$
$$\frac{4(2h-p)}{h-q}=1+\frac{4a^2}{a^2}-1$$
$$\frac{4(2h-p)}{h-q}=4$$
$$2h-p=h-q$$
$$p-q=h$$

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(b) Translation of the hyperbola in (a) by $\begin{pmatrix} -2h \\ h \end{pmatrix}$ maps it onto the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{4a^2} = 1$

At point A, x = ae:

$$\frac{a^2e^2}{a^2} - \frac{y^2}{4a^2} = 1$$
$$e^2 - 1 = \frac{y^2}{4a^2}$$
$$4 = \frac{y^2}{4a^2}$$
$$y^2 = 16a^2$$
$$y = \pm 4a$$

So, the coordinates of A are (ae, 4a) and the coordinates of G are (-ae, 0).
Hence the gradient of AG is:

$$m = \frac{4a - 0}{ae - {}^{-}ae}$$
$$m = \frac{2}{e}$$

But $e^2 - 1 = 4$

So, $e = \sqrt{5}$ and hence the equation of AG is:

$$y - 0 = \frac{2}{\sqrt{5}}\left(x - {}^{-}a\sqrt{5}\right)$$
$$y = \frac{2}{\sqrt{5}}x + 2a$$

Now, translating the curve and line back gives:

$$y - h = \frac{2}{\sqrt{5}}(x - 2h) + 2a$$
$$\sqrt{5}y - \sqrt{5}h = 2x - 4h + 2a\sqrt{5}$$
$$\sqrt{5}y - 2x = 2a\sqrt{5} + \left(\sqrt{5} - 4\right)h$$

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(c) $\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$

$\frac{yy_1}{b^2} = \frac{xx_1}{a^2} - 1$

$y = \frac{b^2x_1x}{a^2y_1} - \frac{b^2}{y_1}$

$y = mx - \frac{b^2}{y_1}$ (1)

where $m = \frac{b^2x_1}{a^2y_1}$

rearranging:

$x_1 = \frac{ma^2y_1}{b^2}$

Substituting into the equation of the hyperbola at (x_1, y_1)

$\frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} = 1$

$\frac{m^2a^4y_1^2}{b^4a^2} - \frac{y_1^2}{b^2} = 1$

$y_1^2\left[\frac{m^2a^2}{b^4} - \frac{1}{b^2}\right] = 1$

$y_1^2\left[\frac{m^2a^2 - b^2}{b^4}\right] = 1$

$y_1 = \frac{b^2}{\pm\sqrt{m^2a^2 - b^2}}$

Substituting into (1):

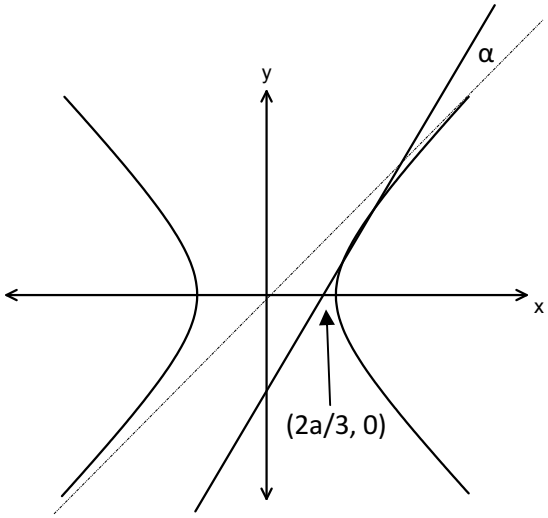
$y = mx - \frac{b^2}{\pm\sqrt{m^2a^2 - b^2}}$

$y = mx \pm \sqrt{m^2a^2 - b^2}$

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(d) Translation of the hyperbola in (a) by $\begin{pmatrix} -2h \\ -h \end{pmatrix}$ maps it onto the hyperbola

$\frac{x^2}{a^2} - \frac{y^2}{4a^2} = 1$ and the tangent required is now through point $\left(\frac{2a}{3}, 0\right)$



Using the answer from (c) and $\left(\frac{2a}{3}, 0\right)$:

$$0 = \frac{2am}{3} \pm \sqrt{m^2a^2 - 4a^2}$$

$$0 = \frac{2am}{3} \pm a\sqrt{m^2 - 4}$$

$$\frac{4a^2m^2}{9} = a^2\left(m^2 - 4\right)$$

$$\frac{4m^2}{9} = m^2 - 4$$

$$5m^2 = 36$$

$$m = \pm\frac{6}{\sqrt{5}}$$

So the gradient of the is $m_1 = \frac{6}{\sqrt{5}}$

The gradient of the positive asymptote is $m_2 = \frac{b}{2a} = \frac{2a}{a} = 2$

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Hence the angle α between the two lines is given by:

$$\tan\alpha = \frac{m_1 - m_2}{1 + m_1 m_2}$$

$$\tan\alpha = \frac{\frac{6}{\sqrt{5}} - 2}{1 + \frac{6}{\sqrt{5}} \times 2}$$

$$\tan\alpha = \frac{6 - 2\sqrt{5}}{\sqrt{5} + 12}$$

$$\tan\alpha = \frac{6 - 2\sqrt{5}}{\sqrt{5} + 12} \times \frac{\sqrt{5} - 12}{\sqrt{5} - 12}$$

$$\tan\alpha = \frac{30\sqrt{5} - 82}{-139}$$

$$\alpha = \tan^{-1}\left[\frac{82 - 30\sqrt{5}}{139}\right]$$

$$\alpha = \tan^{-1}\left[\frac{2\left(41 - 15\sqrt{5}\right)}{139}\right]$$

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3. (a)

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$$

$$\frac{2x}{a^2}+\frac{2y}{b^2}\frac{dy}{dx}=0$$

$$\frac{dy}{dx}=-\frac{b^2x}{a^2y}$$

Gradient of normal:

$$m=\frac{a^2y}{b^2x}$$

$$\text{At } \theta=\frac{\pi}{4}: \quad x=\frac{\sqrt{2}a}{2},\ y=\frac{\sqrt{2}b}{2}$$

$$m=\frac{a^2b}{b^2a}=\frac{a}{b}$$

Equation of the normal is:

$$y-\frac{\sqrt{2}b}{2}=\frac{a}{b}\left(x-\frac{\sqrt{2}a}{2}\right)$$

At $x=0$:

$$y-\frac{\sqrt{2}b}{2}=-\frac{\sqrt{2}a^2}{2b}$$

$$y=\frac{\sqrt{2}b}{2}-\frac{\sqrt{2}a^2}{2b}$$

The distance from P to where the beam cuts the y axis is:

$$D=\frac{b\sqrt{2}}{2}-\left(\frac{b\sqrt{2}}{2}-\frac{\sqrt{2}a^2}{2b}\right)$$

$$D=\frac{\sqrt{2}a^2}{2b}$$

There are two methods for part (b).

In this case we will look at both methods as:

(i) they yield different, but equivalent results and

(ii) the second, quicker method, utilises an important technique.

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(b) **Method One**

This method is the more obvious technique but looks pretty ugly:

$$\text{Gradient of OP} = \frac{b\sin\theta}{a\cos\theta}$$

$$\text{Gradient of OP} = \frac{b}{a}\tan\theta$$

$\text{OP} \perp \text{OQ}$ so:

$$\text{Gradient of OQ} = -\frac{a}{b}\cot\theta$$

Thus the equation of OQ is:

$$y = \left(-\frac{a}{b}\cot\theta\right)x$$

$$y^2 = \left(\frac{a^2}{b^2}\cot^2\theta\right)x^2 \qquad (1)$$

Substituting into the equation of the ellipse:

$$\frac{x^2}{a^2} + \frac{\frac{a^2}{b^2}\cot^2\theta.x^2}{b^2} = 1$$

$$x^2\left(\frac{1}{a^2} + \frac{a^2\cot^2\theta}{b^4}\right) = 1$$

$$x^2\left(\frac{b^4 + a^2\cot^2\theta}{a^2b^4}\right) = 1$$

$$x^2 = \frac{a^2b^4}{b^4 + a^2\cot^2\theta}$$

$$x = \pm\frac{ab^2}{\sqrt{b^4 + a^2\cot^2\theta}}$$

Substituting for into (1):

$$y^2 = \frac{a^2}{b^2}\cot^2\theta \times \frac{a^2b^4}{b^4 + a^4\cot^2\theta}$$

$$y^2 = \frac{a^4b^2\cot^2\theta}{b^4 + a^4\cot^2\theta}$$

$$y = \pm\frac{a^2b\cot\theta}{\sqrt{b^4 + a^4\cot^2\theta}}$$

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Now, as P moves around the ellipse and Q is always a quadrant ahead of it the value of θ and hence the sign of cotθ will be different. To ensure that the x and y coordinates of Q have the appropriate sign the ± will have to reverse on one of them.

Hence the coordinates of Q are: $\left(\pm \frac{ab^2}{\sqrt{b^4 + a^4\cot^2\theta}}, \mp \frac{a^2 b\cot\theta}{\sqrt{b^4 + a^4\cot^2\theta}} \right)$

Method Two

This method has two points which make it worth looking at (i) it uses a right angled triangle and Pythagoras to find required trig expressions from one given one and (ii) by labelling Q in terms of α it makes it more obvious that the values of the parameters for P and Q are always different.

Let Q be the point $(a\cos\alpha, b\sin\alpha)$

As in Method One Gradient of $OP = \frac{b}{a}\tan\theta$

But Gradient of $OQ = \frac{b}{a}\tan\alpha$

As $OP \perp OQ$:

$$\frac{b\sin\theta}{a\cos\theta} \times \frac{b\sin\alpha}{a\cos\alpha} = -1$$

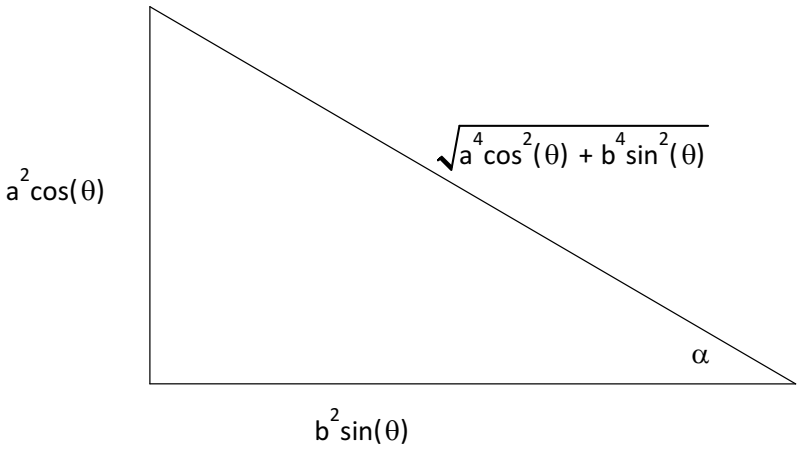
$$\frac{b\tan\alpha}{a} = -\frac{a\cos\theta}{b\sin\theta}$$

$$\tan\alpha = -\frac{a}{b} \times \frac{a\cos\theta}{b\sin\theta}$$

$$\tan\alpha = -\frac{a^2\cos\theta}{b^2\sin\theta}$$

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Now, in order to get expressions for sinα and cosα we construct a right angled triangle (we temporarily drop the negative sign above but deal with it later):



From the triangle:

$$\sin\alpha = \frac{a^2\cos\theta}{\sqrt{a^4\cos^2\theta + b^4\sin^2\theta}} \quad \text{and} \quad \cos\alpha = \frac{b^2\sin\theta}{\sqrt{a^4\cos^2\theta + b^4\sin^2\theta}}$$

Hence:

$$b\sin\alpha = \frac{a^2b\cos\theta}{\sqrt{a^4\cos^2\theta + b^4\sin^2\theta}} \quad \text{and} \quad a\cos\alpha = \frac{ab^2\sin\theta}{\sqrt{a^4\cos^2\theta + b^4\sin^2\theta}}$$

For similar reasons to Method One there needs to be a negative sign on the x-ordinate:

The coordinates of Q are $\left(-\frac{ab^2\sin\theta}{\sqrt{a^4\cos^2\theta + b^4\sin^2\theta}}, \frac{a^2b\cos\theta}{\sqrt{a^4\cos^2\theta + b^4\sin^2\theta}} \right)$

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4. (a) Use the parametric equations to eliminate parameter s:

$y = 2at$ and $y = \sqrt{2}as$

$2at = \sqrt{2}as$

$2t = \sqrt{2}s$

$4t^2 = 2s^2$

$s^2 = 2t^2$

$x = at^2 + b$ and $x = 3b - as^2$

$at^2 + b = 3b - as^2$

Substituting for s^2 :

$at^2 + b = 3b - a2t^2$

$3at^2 = 2b$

$t^2 = \frac{2b}{3a}$

Substituting into $x = at^2 + b$:

$x = a\frac{2b}{3a} + b$

$x = \frac{2b}{3} + b = \frac{5b}{3}$

Substituting into $y - 2at$:

$y = \pm 2a\sqrt{\frac{2b}{3a}}$

$y = \pm 2\sqrt{\frac{2ba^2}{3a}}$

$y = \pm 2\sqrt{\frac{2ab}{3}}$

So the points of intersection are $\left(\frac{5b}{3}, \pm 2\sqrt{\frac{2ab}{3}}\right)$

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(b) There are several methods outlined in the assessment schedule for this question. Two are included here:

Method One – Parametric Approach

This approach is best if a parametric approach has already been used in (a) but it does leave a complicated index simplification at the end. It is worth looking at to observe how a Cartesian integral is turned into a parametric one.

Area 1:

$\int y\,dx = \int 2at\frac{dx}{dt}dt$

$x = at^2 + b$

$\frac{dx}{dt} = 2at$

$\text{Area } 1 = \int 2at \times 2at\,dt$

$\text{Area } 1 = 4a^2\int t^2 dt$

$\text{Area } 1 = 4a^2\int t^2 dt$

Area 2:

$\int y\,dx = \int \sqrt{2}as\frac{dx}{ds}ds$

$x = 3b - as^2$

$\frac{dx}{ds} = -2as$

$\text{Area } 2 = \int \sqrt{2}as \times -2as\,ds$

$\text{Area } 2 = -2\sqrt{2}a^2\int s^2 ds$

$\text{Area } 2 = -2\sqrt{2}a^2\int s^2 ds$

We need to sort out limits for the integration:

Curve 1:

At the intersection of the curves $t^2 = \frac{2b}{3a}$ OR $t = \sqrt{\frac{2b}{3a}}$ and on the x-axis $t = 0$

Curve 2:

At the intersection of the curves $s^2 = \frac{4b}{3a}$ OR $s = \sqrt{\frac{4b}{3a}}$ and on the x-axis $s = 0$

$\text{Area } 1 = 4a^2\int_0^{\sqrt{\frac{2b}{3a}}} t^2 dt$

$\text{Area } 1 = 4a^2\left[\frac{t^3}{3}\right]_0^{\sqrt{\frac{2b}{3a}}}$

$\text{Area } 1 = \frac{4a^2}{3}\left(\frac{2b}{3a}\sqrt{\frac{2b}{3a}}\right)$

$\text{Area } 2 = -2\sqrt{2}a^2\int_{\sqrt{\frac{4b}{3a}}}^0 s^2 ds$

$\text{Area } 2 = -2\sqrt{2}a^2\left[\frac{s^3}{3}\right]_{\sqrt{\frac{4b}{3a}}}^0$

$\text{Area } 2 = \frac{-2\sqrt{2}a^2}{3}\left(0 - \frac{4b}{3a}\sqrt{\frac{4b}{3a}}\right)$

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$$\text{Area1}:\text{Area 2}=\frac{\frac{4a^2}{3}\left(\frac{2b}{3a}\sqrt{\frac{2b}{3a}}\right)}{\frac{2\sqrt{2}a^2}{3}\left(\frac{4b}{3a}\sqrt{\frac{4b}{3a}}\right)}$$

$$\text{Area1}:\text{Area 2}=\frac{1}{\sqrt{2}}\times\frac{\sqrt{2}}{\sqrt{4}}=\frac{1}{2}$$

Method Two – Cartesian Approach

It is first necessary to work out the Cartesian equations.
This method is definitely easier if a Cartesian approach was used in (a).

Curve 1 :

$$t=\frac{y}{2a}$$

$$x=a\left(\frac{y}{2a}\right)^2+b$$

$$x-b=\frac{y^2}{4a}$$

$$y^2=4a(x-b)$$

Curve 2 :

$$s=\frac{y}{\sqrt{2}a}$$

$$x=3b-a\left(\frac{y}{\sqrt{2}a}\right)^2$$

$$x-3b=-\frac{y^2}{2a}$$

$$y^2=-2a(x-3b)$$

The vertices (the limits of integration along with $x=\frac{5b}{3}$ from (a)):

$\text{Curve}1: x=b$ $\quad$ $\text{Curve}2: x=3b$

So, the ratio of the areas is given by:

$$\text{Area 1}:\text{Area 2}=\frac{2\int_{b}^{5b/3}2\sqrt{a}(x-b)^{\frac{1}{2}}\,dx}{2\int_{5b/3}^{3b}\sqrt{2}\sqrt{a}(3b-x)^{\frac{1}{2}}\,dx}$$

$$\text{Area 1}:\text{Area 2}=\frac{2\left[2(x-b)^{3/2}\right]_{b}^{5b/3}}{\sqrt{2}\left[-2(3b-x)\right]_{5b/3}^{3b}}$$

$$\text{Area 1}:\text{Area 2}=-\frac{2\left(\left(5b/3-b\right)^{3/2}-(b-b)^{3/2}\right)}{\sqrt{2}\left((3b-3b)^{3/2}-\left(3b-5b/3\right)^{3/2}\right)}$$

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$$\text{Area 1 : Area 2} = \frac{2\left(\frac{2b}{3}\right)^{3/2}}{\sqrt{2}\left(\frac{4b}{3}\right)^{3/2}}$$

$$\text{Area 1 : Area 2} = \frac{2}{\sqrt{2}} \times \left(\frac{2}{4}\right)^{3/2}$$

$$\text{Area 1 : Area 2} = \frac{2}{\sqrt{2}} \times \frac{1}{2} \times \frac{1}{\sqrt{2}}$$

$$\text{Area 1 : Area 2} = \frac{1}{2}$$

(c) At P:

$$\frac{dx}{ds} = -2as \qquad \frac{dy}{ds} = \sqrt{2}a$$

$$\frac{dy}{dx} = \frac{\sqrt{2}a}{-2as}$$

$$\frac{dy}{dx} = \frac{\sqrt{2}}{-2s}$$

Gradient of the normal is $m = \frac{2s}{\sqrt{2}} = \sqrt{2}s$

Equationof the normal is:

$$y - \sqrt{2}as = \sqrt{2}s\left(x - 3b + as^2\right)$$

At $y = 0$:

$$-\sqrt{2}as = \sqrt{2}s\left(x - 3b + as^2\right)$$

$$-a = x - 3b + as^2$$

$$x = 3b - as^2 - a$$

At the midpoint M:

$$y = \frac{0 + \sqrt{2}as}{2} = \frac{\sqrt{2}as}{2}$$

$$x = \frac{3b - as^2 - a + 3b - as^2}{2} = \frac{6b - 2as^2 - a}{2}$$

Rearranging y and substituting into x:

$$s = \frac{2y}{\sqrt{2}a} \text{ OR } s^2 = \frac{4y^2}{2a^2} = \frac{2y^2}{a^2}$$

$$x = \frac{6b - 2a\frac{2y^2}{a^2} - a}{2}$$

$$2x = 6b - \frac{4y^2}{a} - a$$

$$\frac{4y^2}{a} = -2x + 6b - a$$

$$y^2 = \frac{a}{4}\left(-2x + 6b - a\right)$$

$$y^2 = -\frac{a}{2}\left(x - 3b + \frac{a}{2}\right)$$

So $k = -\frac{1}{2}$, $m = \frac{1}{2}$, $n = -3$

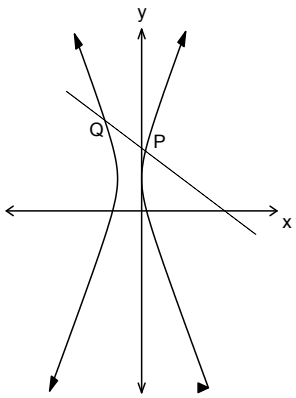
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5. (a) $x+1=\sec\theta,\ \dfrac{y-2}{2}=\tan\theta$

$\sec^2\theta-\tan^2\theta=1$

$(x+1)^2-\dfrac{(y-2)^2}{4}=1$

This is a hyperbola centre (-1, 2):



(b) In this case we shall use parametric differentiation (differentiation of the Cartesian equation is equally valid);

$\dfrac{dx}{d\theta}=\sec\theta\tan\theta,\ \dfrac{dy}{d\theta}=2\sec^2\theta$

$\dfrac{dy}{dx}=\dfrac{2\sec^2\theta}{\sec\theta\tan\theta}$

$\dfrac{dy}{dx}=\dfrac{2\sec\theta}{\tan\theta}$

$\dfrac{dy}{dx}=\dfrac{2}{\sin\theta}$

At $\theta=\dfrac{\pi}{4}$:

$\dfrac{dy}{dx}=\dfrac{2}{\frac{1}{\sqrt{2}}}$

$\dfrac{dy}{dx}=2\sqrt{2}$

So, the gradient of the normal is:

$m=-\dfrac{1}{2\sqrt{2}}$

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At $\theta = \frac{\pi}{4}$:

$$x = \frac{1}{\cos\left(\frac{\pi}{4}\right)} - 1 = \sqrt{2} - 1$$

$$y = 2\tan\left(\frac{\pi}{4}\right) + 2 = 4$$

So, the equation of the normal is:

$$y - 4 = -\frac{1}{2\sqrt{2}}\left(x - \sqrt{2} + 1\right)$$

$$2\sqrt{2}y - 8\sqrt{2} = -x - 1 + \sqrt{2}$$

$$x + 1 = 9\sqrt{2} - 2\sqrt{2}y$$

Because the equation of the hyperbola has $(x+1)^2$ it is best to leave the x and 1 together.

$$(x+1)^2 = 2(9 - 2y)^2$$

Substituting into the equation of the hyperbola:

$$2(9 - 2y)^2 - \frac{(y-2)^2}{4} = 1$$

$$8\left(81 - 36y + 4y^2\right) - \left(y^2 - 4y + 4\right) = 4$$

$$648 - 288y + 32y^2 - y^2 + 4y - 4 = 4$$

$$31y^2 - 284y + 640 = 0$$

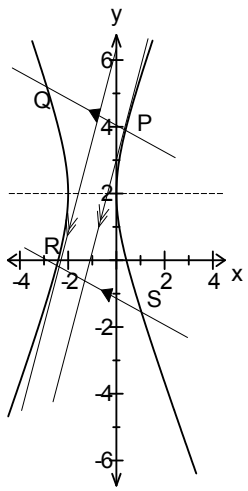
But we already know that y = 4 is a solution, so:

$$(y-4)(31y - 160) = 0$$

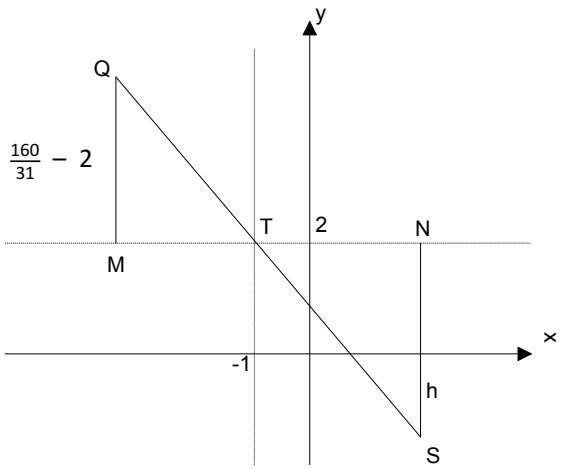
So $y = \frac{160}{31}$ is the required solution.

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(c) The diagram immediately below shows the situation but is somewhat cluttered.



Repeating the critical parts of the diagram:



T is (-1, 2), the centre of the hyperbola.

By symmetry triangles QMT and SNT are congruent and hence:

$$\frac{160}{31} - 2 = h + 2$$

$$h = \frac{160}{31} - 4$$

$$h = \frac{36}{31} \quad \text{or the y-ordinate of is } -\frac{36}{31}$$

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A second (and probably more obvious) approach is to work out the equation of normal RS and find its intersection with the curve:

At P the gradient of the curve is given by:

$$\frac{2}{\sin\theta} = 2\sqrt{2}$$

$$\sin\theta = \frac{1}{\sqrt{2}}$$

So, at R

$$\sin\theta = \frac{1}{\sqrt{2}}$$

$$\theta = \frac{3\pi}{4}$$

$$x = \sec\left(\frac{3\pi}{4}\right) - 1 \qquad y = 2\tan\left(\frac{3\pi}{4}\right) + 2$$

$$x = -\sqrt{2} - 1 \qquad y = 2 \times -1 + 2 = 0$$

Equation of the normal at R:

$$y - 0 = -\frac{1}{2\sqrt{2}}\left(x + \sqrt{2} + 1\right)$$

$$\sqrt{2} + 2\sqrt{2}y = -(x+1)$$

$$(x+1)^2 = 2(1+2y)^2$$

Substituting back into the equation of the curve:

$$2(1+2y)^2 - \frac{(y-2)^2}{4} = 1$$

$$8\left(1 + 4y + 4y^2\right) - \left(y^2 - 4y + 4\right) = 4$$

$$8 + 32y + 32y^2 - y^2 + 4y - 4 = 4$$

$$31y^2 + 36y = 0$$

$$y(31y + 36) = 0$$

$$y = -\frac{36}{31}$$